

The canonical problem of time as a category error:
 an empirically forced cosmic time, deparametrization, the dissolution of frozen
 dynamics,
 and a graviton quantization the de Sitter horizon closes without a free parameter—the
 seam where $\hbar$ enters gravity—in Cosmological Relativity

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Abstract

In the canonical formulation of general relativity the Hamiltonian is a sum of constraints that vanish on the physical phase space: no preferred time survives, and the generator of evolution annihilates physical states rather than evolving them. This is the canonical “problem of time.” We argue that it is not a technical defect awaiting a technical repair, but the canonical symptom of a category error—treating the four-dimensional manifold of *occurrences* as the thing that *exists*, the block universe.

The companion framework paper [3] introduced, as an ontological commitment of Cosmological Relativity (CR), a global cosmic time that general relativity’s formalism admits but does not supply. Here we show what that commitment does to the canonical formalism. The decisive premise is not a new calculation but a selection: the bare Arnowitt–Deser–Misner (ADM) machinery propagates data along an *arbitrary* hypersurface and settles nothing, whereas CR selects a preferred cosmic foliation—forced empirically by the isotropy of the cosmic microwave background (CMB) and structurally by the horizon geometry of collapse [4]. On that foliation the lapse recovers its meaning as the metrical rate of the existing layer’s advance, $d\tau = N dt$; the constraint, deparametrized along cosmic time, is solved for a *true* Hamiltonian that generates the advance; and quantum evolution in cosmic time is unitary. The frozen constraint and the true Hamiltonian are the same canonical content under two readings, differing only in whether the manifold is granted existence.

The deparametrization is, as a technique, entirely standard; what CR adds is not a mechanism but a *warrant*. The multiple-choice problem—that many candidate internal times exist, that inequivalent quantizations follow from different choices, and that the formalism prefers none—has force exactly when the clock is chosen for formal convenience internal to the theory, and loses it when the clock is fixed from outside the formalism by ontology and observation. It is dissolved, not solved within the space of formal clocks.

We exhibit the structure first in flat Friedmann–Lemaître minisuperspace, where the deparametrized Hamiltonian reproduces the flat- Λ CDM expansion—the $\sinh^{2/3}$ law established in [3]—and then lift it to the closed- S^3 layer’s propagating sector: the transverse-traceless^{L20} graviton modes form a discrete tower that deparametrizes to a unitary evolution in cosmic time, and project into a fundamental observer’s frame as the flat- Λ CDM graviton, with matter density appearing only as the bend of the slicing rather than as ontological content. The external clock is no convenience: an internal area clock separates the dynamics at $\Lambda = 0$ and would serve, but at $\Lambda > 0$ it no longer does, and the absolute cosmic time is forced.

In that sector the quantization moreover closes *without a free parameter*. The scale-factor Hamiltonian carries deficiency indices $(1, 1)$ independently of operator ordering—the inverse-square coefficient at the origin attaining $\gamma = \frac{1}{4}$ across the natural ordering family, strictly below the essential-self-adjointness threshold $\frac{3}{4}$ —so a single boundary condition at $a = 0$ remains; and that point is not a generic half-line endpoint but the de Sitter cosmological horizon, whose Hartle–Hawking thermal state, at the surface gravity $\kappa = 1/\alpha$ the substrate itself fixes, selects the Friedrichs extension. The deficiency freedom is thereby not merely

parametrized but closed, and closed without a free parameter: $\hbar$ enters the gravitational sector at the seam, scaled by Λ alone.

With the tower coupled, the boundary coefficient is promoted to an operator straddling the threshold and the same thermal regularity supplies the condition fibre by fibre, so that what remains open is not the boundary condition but the *definition of the interacting tower*. *That is a different thing from a residual freedom in the quantization, and its shared character with every interacting field theory does not settle it*: the divergence is measured here rather than characterised—quartic at a leading constant fixed by the propagating-component count rather than assumed; logarithmic at a coefficient the tower’s own spectrum settles exactly; and, where the shear breaks the degeneracy among the quadratic invariants, one further counterterm whose Weyl-squared coefficient is twice a real scalar’s—and it is carried as this programme’s own open frontier.

The problem of time is thereby *dissolved* as a category error, not solved by an addition to the formalism, in exactly the way the same ontological correction dissolves the inference that gravitational collapse completes in finite cosmic time [4].

1 Introduction: the problem of time, as standardly posed

The canonical quantization of general relativity confronts a difficulty that has resisted resolution for half a century. In the ADM decomposition [16], spacetime is foliated by spatial hypersurfaces and the Einstein–Hilbert action is recast in Hamiltonian form. The result is a fully constrained system in the sense of Dirac [17]: the total Hamiltonian is a sum of the scalar (Hamiltonian) constraint $\mathcal{H}$ and the momentum constraint $\mathcal{H}_i$, each weighted by the lapse N and shift N^i , and each vanishing on the physical phase space. The lapse and shift are not dynamical; they are Lagrange multipliers enforcing the constraints. Promoting the scalar constraint to an operator yields the Wheeler–DeWitt equation $\hat{\mathcal{H}}\Psi = 0$ [18]: the physical state is annihilated by the generator that should evolve it. The wavefunction of the universe does not appear to depend on any time parameter at all.

This is the problem of time [19, 20]. Its sharpest statement is that the formalism contains no preferred time variable, that every spatial foliation of a given spacetime is related to every other by a gauge transformation generated by the constraints, and that “evolution” in the coordinate label t is therefore pure gauge—a re-slicing of one fixed four-dimensional object, carrying no physical content. The many proposed responses—identifying an internal time among the canonical variables before quantization, coupling a material clock, treating observables as relational “evolving constants,” or abandoning a fundamental time altogether—each carry well-known costs [19, 20, 23, 24], to which we return in Section 8.

Our claim is that the difficulty is conceptual before it is technical, and that the conceptual diagnosis is precise rather than vague. The frozen formalism is the canonical expression of a single metaphysical commitment: that the four-dimensional manifold is the object that exists. We will argue, following the ontological foundation of CR [13, 3] and the necessity assembled in Section 2, that this commitment is a category error—the confusion of what *happens* with what *exists*—and that it is empirically false, the universe having furnished, in the CMB, direct evidence of the very cosmic time the formalism declines to recognize. When that error is declined, the same canonical equations describe a real cosmic time all along.

We stress at the outset the register of this paper. We do not propose a modification of general relativity, a new term in the action, or an additional structure bolted onto the canonical formalism. The equations are those of Einstein and of ADM, unchanged. What changes is the reading. The frozen constraint and a true Hamiltonian generating cosmic-time evolution are, as we will show, the same content under two readings, differing only in whether the manifold of occurrences is granted the kind of existence that belongs to the evolving three-dimensional world. The problem of time is not solved here. It is dissolved.

The paper proceeds as follows. Section 2 earns the premise the argument needs—that a preferred cosmic time is real—on two independent grounds, one conceptual and one empirical.

Section 3 reads the unchanged ADM formalism under the two ontologies, and Section 4 isolates why the move is a *selection* rather than a construction: the bare machinery settles nothing, and everything turns on whether a foliation is distinguished to be selected. Section 5 carries out the deparametrization, structurally and then in flat minisuperspace, marking the boundary between the two. Section 6 lifts it to the closed- S^3 layer’s propagating sector—the transverse-traceless graviton tower, advanced unitarily in cosmic time—and there the quantization closes: the scale-factor Hamiltonian’s deficiency- $(1, 1)$ freedom, ordering-independent, is fixed by the de Sitter horizon’s own thermal state, without a free parameter, leaving open not the boundary condition but the definition of the interacting tower. Section 7 states the verdict negatively, and Section 8 places the argument against the internal-time, relational, and timeless programmes, where the multiple-choice objection is met not with a mechanism but with a warrant.

2 The necessity: a cosmic time that is real, not posited

The canonical move of this paper rests on one premise: that there is a preferred cosmic time, and that it is a real feature of the world rather than a convention chosen for convenience. This premise is not assumed. It is forced, on two independent grounds—one conceptual, one empirical—which we summarize here; the full argument is developed in [13, 14, 15] and stated as the necessary-and-sufficient augmentation of general relativity in [26].

The conceptual ground: occurrence is not existence. To exist is to endure—to persist through an interval. The three-dimensional world exists in this sense, and so do the things in it. An *event*, by contrast, is something that *happens*: it occurs at a place and an instant and is gone. A worldline is the continuous succession of events associated with an enduring thing, and four-dimensional spacetime is the manifold of all such events—the totality of what happens. The persistent confusion about time, traceable from Heraclitus and Parmenides through to modern eternalism, is the treatment of occurrences as though they were existents: the reification of the event-manifold into a thing that exists [13]. Once spacetime is imagined to exist, every moment along a worldline is imagined to exist on equal footing, the distinction between past, present, and future is declared an illusion, and time is said not to pass. But that conclusion was smuggled into the premise. Granting existence to the manifold of occurrences is not a discovery about time; it is a misuse of the word “exist.” The coherent picture is the one in which an evolving three-dimensional world exists and advances, and events happen in the course of that advance, mapping to a four-dimensional spacetime that represents them without being a further existent of the same kind.

In CR this distinction is structural. What exists is the layer $\mathcal{S}_t$ —a real three-sphere evolving in a real de Sitter substrate [3, 8]. The four-manifold M is the representational record of the events that happen as the layer advances. The layer’s existence and advance are not optional gloss on a fundamentally four-dimensional object; they are the ontology, and M is its shadow.

The empirical ground: the CMB measures the cosmic rest frame. That a preferred cosmic time might be real but undetectable is the situation relativity leaves open: local experiments cannot distinguish a true state of rest, because the relativity of synchrony—a general kinematic consequence of one’s chosen rest convention, independent of the light postulate—and the round-trip symmetry of local signalling together hide one’s motion. Einstein’s identification of simultaneity with frame-relative synchrony was, on this much, a philosophical choice among two empirically equivalent options, not a deduction; treating the local inaccessibility of a cosmic rest frame as evidence of its nonexistence is a modal fallacy [14].

The universe, however, did furnish the missing evidence. The cosmological redshift of the CMB is isotropic to high precision, and the primordial temperature anisotropies at the level of

one part in 10^5 have been preserved across the entire history of expansion [25]. Had the expansion rate varied locally with the inhomogeneous matter distribution, the accumulated redshift along different lines of sight would differ by far more than is observed—by some three orders of magnitude, as we establish quantitatively in the companion paper [26]—and the isotropy therefore certifies that cosmological space has expanded uniformly throughout cosmic history. But to speak of a three-dimensional space expanding uniformly *through time* already presupposes a cosmic time, a cosmic rest frame, and an associated simultaneity. The same data fix our own velocity relative to that frame through the CMB dipole. The cosmic rest frame is thus not a large-scale average or a coordinate convenience but one of the better-measured structures in cosmology. Synchrony is relative; simultaneity is absolute [14, 15].

The two grounds are one correction. These are not two arguments but two faces of one. The relativity-of-simultaneity premise is precisely what licenses the reification of the block: if there is no objective “now,” the four-dimensional manifold is all there coherently is, and the layer’s privileged advance has no referent. Conversely, an empirically real cosmic “now” restores the layer as the existent and demotes the manifold to its record. Declining the block (the conceptual correction) and admitting the cosmic rest frame (the empirical correction) are the same move. The equations of relativity are untouched by it; only the interpretation shifts—from “if two events appear simultaneous they must be simultaneous” to “apparent simultaneity is a local matter with no bearing on the global cosmic simultaneity that cosmology establishes empirically” [15].

This is the premise the canonical argument needs, and it has been earned rather than assumed: there is a preferred cosmic foliation, and it is real.

3 The canonical formalism, read two ways

We now read the standard canonical formalism under the two ontologies. Nothing in this section alters the equations.

In the ADM form, a spacetime (M, g) is foliated by spacelike hypersurfaces Σ_t with induced metric h_{ij} and conjugate momentum π^{ij} , and the line element is

$$ds^2 = -N^2 dt^2 + h_{ij} (dx^i + N^i dt)(dx^j + N^j dt), \quad (1)$$

with lapse N and shift N^i . The action becomes $S = \int dt d^3x (\pi^{ij} \dot{h}_{ij} - N\mathcal{H} - N^i \mathcal{H}_i)$, and variation of N and N^i enforces the constraints $\mathcal{H} = 0$, $\mathcal{H}_i = 0$. The total Hamiltonian $H = \int d^3x (N\mathcal{H} + N^i \mathcal{H}_i)$ is a sum of constraints; on the constraint surface it vanishes.

The frozen reading. Suppose the existing object is the four-dimensional manifold itself. Then a choice of foliation is a choice of how to slice one already-existing thing into a stack of three-dimensional pictures, and a different foliation is a different slicing of the same thing. No slicing is distinguished; the relation between any two is a gauge transformation generated by the constraints; and since the generator of t -evolution is a constraint, it moves a physical state nowhere. Quantum-mechanically, $\hat{\mathcal{H}}\Psi = 0$: the state is timeless. This reading is internally consistent. Its content is exactly the content of the premise: *if* the block is what exists, time does not pass, and the formalism correctly reports that there is nothing for a Hamiltonian to do. The frozen formalism is not a mistake of calculation. It is the faithful canonical image of the block.

The reading CR adopts. Suppose instead, with CR, that what exists is the evolving layer S_t , and that M is the record of the events that happen as it advances (Section 2). Then a foliation is not a slicing of a pre-existing block; it is a choice of how to coordinate the advance

of the existing world, and one such foliation—the cosmic one—tracks that advance rather than cutting across it. On this reading the lapse is not a bookkeeping multiplier but the metrical amount by which the existing layer advances per unit of the time label,

$$d\tau = N dt, \tag{2}$$

with the shift N^i carrying the foliation’s non-orthogonality to the layer. The constraint $\mathcal{H} = 0$ is still an equation; but now it is read as the relation that, once solved for the momentum conjugate to the cosmic time, yields the generator of the layer’s advance—a true Hamiltonian. The same symbols, the same constraint surface, the same dynamics; a different answer to the single question of what exists. This lapse and shift are the very levels the cosmological development computes on: the lapse is the foliation stacking rate the observable expansion rides—set by the geometry, the density the content that set rate carries, read off the clock rather than sourcing it—and the shift is the synchronization through which that expansion is *observed* as distance and redshift, with the leaf’s bend the matter it carries [9, 33]. That a plasma process rides the one rate or the other is then fixed by what the process is—a process running in the content reads the leaf’s rate, a comoving separation read across leaves the stacking rate [34]—not chosen: the same selection of the existent’s foliation that turns the frozen constraint into a true Hamiltonian here fixes, one level down, which rate a recombination-era or a nucleosynthesis-era computation must read—and reifying the shift’s synchronous slicing as the existent is the block error, met at the level of a rate.

4 The selection is the whole of the move

It is worth isolating why the foregoing is a recognition rather than a new construction, because this is the crux of the paper.

The ADM machinery propagates canonical data along whatever foliation one supplies. It is indifferent to which foliation that is; it has no preferred time of its own, and in this exact sense the bare formalism contains no resolution of the problem of time and no obstruction to one. Everything turns on what is fed to it. If nothing distinguishes a foliation, the constraint is all there is and the state is frozen. If a foliation is distinguished, the constraint deparametrizes along it and a true Hamiltonian appears. The canonical formalism is the same in both cases; what differs is whether a preferred foliation exists to be selected.

CR supplies exactly that, and supplies it from outside the bare formalism, on the grounds of Section 2: the cosmic foliation is real, forced empirically by the CMB and conceptually by the occurrence/existence distinction. It is moreover the foliation that the rest of the programme independently singles out. The metric-singularity structure of the event horizon determines a unique limiting causal orientation in collapse, generically non-orthogonal to any spacelike slice, and this is the orientation CR reassigns as cosmic time [4, 3]; the representational freedom in choosing among Lorentzian metrics on the fixed manifold is a genuine gauge symmetry, organized by the description groupoid [12], but the cosmic foliation is not one of its orbits to be quotiented away—it is fixed by the ontology the gauge freedom is defined over.

The consequence is methodological as well as physical. There is no separate “canonical machinery” result to be proved here; the deparametrization that follows is standard once a clock is in hand. The content of this paper is that CR’s cosmic time is the physically correct clock, on grounds that are ontological and empirical rather than formal, and that the canonical face of the existing physics is therefore a true Hamiltonian rather than a frozen constraint. The move is the selection. Everything else is reading the textbook on it.

The same selection settles what a mass charge could be. The ADM mass is defined relative to an asymptotic time translation, and exists only where the geometry supplies one—which asymptotically-de Sitter spacetimes do not, there being no global timelike Killing vector, so that no conserved charge is well defined there at all [8]. *On the reading taken here that absence*

is not a deficiency of the geometry but a misplacement of the question: the time this construction runs on is not recovered from an asymptotic symmetry but selected and *measured*, so a quantity whose definition waits on an asymptotic Killing vector is waiting on the wrong thing. It is the same move as the paper’s own—what the formalism cannot supply from inside, the selection supplies from outside, and the question that presupposed the missing structure lapses with it.

5 Deparametrization and the true Hamiltonian

We now carry out the reading. The structural statement is independent of any model; the explicit minisuperspace realization that follows is an illustration of it, and we are careful to mark the boundary between the two.

The structural statement. Let the cosmic time of Section 2 be the distinguished foliation parameter, and let p_τ denote the momentum conjugate to it. The scalar constraint, linear in p_τ after the cosmic clock is isolated, takes the deparametrized form

$$p_\tau + H_{\text{phys}}(q^A, p_A) = 0, \quad (3)$$

where (q^A, p_A) are the remaining (true, gauge-invariant) degrees of freedom of the layer and H_{phys} depends on the cosmic time only through them. Equation (3) is the same constraint that, read with no preferred foliation, gives $\hat{\mathcal{H}}\Psi = 0$. Read with the cosmic clock selected, it is solved for the generator of advance: classically $dq^A/d\tau = \partial H_{\text{phys}}/\partial p_A$, and on quantization

$$i \partial_\tau \Psi(q^A, \tau) = \hat{H}_{\text{phys}} \Psi(q^A, \tau). \quad (4)$$

This is a genuine time-dependent Schrödinger equation generating unitary evolution in cosmic time, not the frozen Wheeler–DeWitt constraint. The foliation is fixed—by the ontology, not by a gauge choice—so there is a true time and a true Hamiltonian. The frozen constraint and the true Hamiltonian are (3) read two ways; they are the same content, differing only in whether the manifold is granted existence. Deparametrization here is not a trick that restores a time that was really absent; it is the formal face of a time that was never lost, because the cosmic foliation was real all along.

A zero-duration boundary, and what it does to the evolution. The cosmic time of Section 2 is the real part of the companion framework’s complex cosmic time, and that framework’s closed contour contains one segment—the *lift*, joining the collapse’s comoving turnaround to the branch point at which a collapse becomes a cosmology—along which *the real part does not advance* [3]. The consequence for (4) is immediate and worth stating plainly, because it is unusual: cosmogenesis occupies *no cosmic time at all*, so the evolution operator across it is $U(\Delta\tau = 0) = \mathbb{K}$. The true Hamiltonian has no interval in which to act there. The geometry nonetheless changes across the segment—the areal radius climbs from the turnaround to zero and the expansion rate is carried from zero to divergent—so the change is not generated by H_{phys} but is the segment’s own analytic content, the contour read at another point of itself. *The beginning is therefore not a first instant of the evolution this section constructs; it is a boundary of zero duration within it*, and that is why nothing can be lost across it: no evolution acts, so none can be non-unitary.

We mark the limit of the claim. What is established is the classical statement—the foliation parameter does not advance on that segment, so no interval of τ separates the two legs—together with its immediate consequence for the evolution operator. *What the deparametrization does not supply is a quantization of the Euclidean segment itself.* The state on either side is related by the identity in cosmic time; whether the imaginary-time stretch admits a quantum treatment in its own right, and what that treatment would say about the $\{q^A, p_A\}$ across it, is

not addressed by this deparametrization and is not claimed by it. *The gap is named rather than managed, and its shape is definite. The silence is structural, not a matter of refinement*—an evolution parametrized by cosmic time cannot describe a segment of zero cosmic duration, so no sharpening of $\hat{H}_{\text{phys}}$ reaches it. What the deparametrization cannot supply is an account of an imaginary-time stretch across which the classical configuration changes while the foliation parameter does not, and that is the natural point of contact between this construction and Euclidean approaches to the quantum beginning. *The classical account of the beginning is complete, and the quantum account is not the deparametrization's to give*: the step below gives its kernel, and the framework carries the segment's quantum reading on from there [3].

The propagator across the lift is Euclidean. The evolution (4) is generated in the framework's complex cosmic time, and on the lift that time runs along the imaginary direction: $\tilde{\tau} = \text{const} + is$, so $\partial_s = i\partial_{\tilde{\tau}}$ and (4) becomes $\partial_s \Psi = -\hat{H}_{\text{phys}} \Psi$. The kernel carrying the state across is therefore not the unitary $U = e^{-i\hat{H}_{\text{phys}}\Delta\tau}$ but

$$K = e^{-\hat{H}_{\text{phys}}|\Delta\eta|}, \quad (5)$$

a Euclidean kernel—real, contracting, and not unitary—with $|\Delta\eta|$ the lift's imaginary conformal-time interval. This is consistent with $U(\Delta\tau = 0) = \mathbb{I}$ rather than in tension with it: *no cosmic time elapses, and the kernel is not an evolution in cosmic time.*

Two things follow, and the first is a check the construction did not have to pass. A mode of frequency ω is damped by $e^{-\omega|\Delta\eta|}$ under (5); the framework's independent classical reading of the same segment damps a mode carrying $e^{ikc_s\eta}$ by $e^{-kc_s|\Delta\eta|}$ [3]. With $\omega = kc_s$ these are the same expression, term for term. *The classical selection rule—frozen content crosses, oscillating content does not—is the Euclidean kernel, read classically.* Second, and this is the content the classical reading could not supply: a Euclidean kernel run for a long imaginary interval *projects onto the ground state*, suppressing each excitation by $e^{-(E_n - E_0)|\Delta\eta|}$. *The scope of that statement should be fixed carefully, because the two sectors it separates are not symmetric.* The kernel acts on *oscillatory* content; a frozen, zero-frequency mode is a fixed point of it and passes unchanged. And at the crossing no mode is oscillating: on the contracting leg aH grows without bound as $r \rightarrow 0$, so the comoving horizon shrinks to zero and every mode exits it and freezes before the branch point [34]. *So the projection does not select the frozen content that crosses—it has nothing to act on there.* What it acts on is the oscillatory content of modes while they were sub-horizon, which it annihilates; the ground-state projection is the statement that this content, and not the frozen amplitude, is what the segment removes.

The projection is not exact, and where it fails is a scale rather than a caveat. At $|\Delta\eta| \simeq 1.7 \times 10^4 \text{ Mpc}$ the damping reaches e^{-2} only at $k \sim 2 \times 10^{-4} \text{ Mpc}^{-1}$, so modes below that wavenumber are not strongly suppressed and cross with residual excitation. That wavenumber subtends $\ell \approx 3$ on the last-scattering sphere, which is where the framework's independent transmission calculation places the boundary of the filter's reach [3]. The two routes agree on the scale as well as on the factor.

The obvious objection is that a Euclidean kernel needs a Hamiltonian bounded below, and that Euclidean gravity notoriously lacks one—the conformal factor entering the Einstein–Hilbert action with the opposite sign, so that the Euclidean action can be driven arbitrarily negative. That objection does not reach this construction, and the reason is structural rather than fortunate. The conformal-factor problem arises when the path integral ranges over the conformal factor of the metric. Here it does not. The substrate's scale is α and is fixed—not chosen, but required, as the unique maximally symmetric structure carrying no unforced modulus [27, 8]—so there is no conformal mode to integrate over; and the propagating gravitational degree of freedom of the layer is the *transverse-traceless* shear of Section 6, which is precisely the sector the conformal mode is absent from. Mode by mode that sector is (12), a harmonic oscillator, whose Hamiltonian is bounded below. The areal radius along the lift is likewise confined,

running between the comoving turnaround and the branch point on an interval fixed by the progenitor mass, with the substrate curvature finite throughout [3, 35]: there is no runaway direction for the kernel to diverge along.

And the state the kernel projects onto is not a new posit but one this paper has already selected on independent grounds. Section 6 finds the scale-factor Hamiltonian to have deficiency indices $(1, 1)$ —a one-parameter family of candidate quantizations—and closes it, without stipulation, by observing that the boundary at $a = 0$ is the de Sitter horizon, that its surface gravity is $\kappa = 1/\alpha$, and that the regular Euclidean state at that κ leaves nothing to choose. *That is the same condition the kernel (5) enforces: regularity in the Euclidean continuation. And that regularity is available only because of what the causality companion establishes about the boundary.* A regular Euclidean state exists at a horizon only if the Euclidean continuation is smooth there—free of a conical defect, curvature finite. The horizon at $a = 0$ is a *metric* singularity, at which the spatial measure collapses while the curvature stays finite, and not a curvature singularity [4]; had it been the latter there would be no smooth Euclidean section, no regular state to select, and the extension would remain unfixed. *The distinction that dissolves the information paradox is the same distinction that makes this quantization unique.* The extension-fixing condition and the lift’s kernel are one requirement met twice, once at the horizon and once across the beginning.

One approximation is carried and its size is bounded below rather than left open. The kernel’s frequency is not constant along the segment—the layer’s scale varies with the path parameter—so the projection is adiabatic rather than exact, and the residual at the largest scales quantified above is one face of that. *The ceiling on what it can transmit is computed in the adiabatic correction below,* and it is small enough that nothing above turns on it. *The segment’s variational treatment is given below.* The relation between the configuration variables $\{q^A, p_A\}$ on the two sides is described through the kernel’s action on the modes, and the segment is exhibited there as a solution of a variational principle in its own right.

The segment rides the marginal congruence $(dr/d\tilde{\tau})^2 + f = 1$ —the $\gamma = 1$, $J = 0$ case of the effective-potential form $(dr/d\tau)^2 = \gamma^2 - V_{\text{eff}}$ with $V_{\text{eff}} = f$ derived in [1, Eq. (4.24)]—which is $\frac{1}{2}\dot{r}^2 + V = \frac{1}{2}$ with $V(r) = \frac{1}{2}f(r)$. Continuing $\tilde{\tau} = is$ sends $\dot{r} \mapsto -i r'$ and the Lorentzian action $S = \int d\tilde{\tau} [\frac{1}{2}\dot{r}^2 - V]$ to $S = -iS_E$ with

$$S_E = \int ds \left[\frac{1}{2}r'^2 + V(r) \right], \quad (6)$$

whose stationary points obey $r'' = +V'(r)$ —motion in the *inverted* potential, the standard signature of a Euclidean solution—with first integral $\frac{1}{2}r'^2 - V = -\frac{1}{2}$. The lift’s closed form $r(s) = -(2M\alpha^2)^{1/3}|\sin(3s/2\alpha)|^{2/3}$ satisfies both: the first integral is $-\frac{1}{2}$ identically along the segment, and r'' agrees with $V'(r)$ to the precision of the check^{P10R11}. *So the imaginary-time segment is not a formal device but a solution of a variational principle—an instanton connecting the Lorentzian turning point to the branch point.*

Its action converges. Near the branch point $V \simeq -M/r$ while $r \propto s^{2/3}$, so the integrand of (6) grows as $s^{-2/3}$ and is integrable; numerically $S_E \rightarrow 1.487$ in units $\alpha = 1$ on the forced member. *The segment therefore carries a finite weight rather than a divergent one,* which is what a saddle-point treatment of the beginning would require of it. *Two normalisations are in play, and the relation between them can be settled rather than flagged.* The functional (6) carries a unit kinetic term; the gravitational action carries the metric factor and the prefactor implied by (9). They share a solution—the contour’s law $(dr/d\tilde{\tau})^2 = 2M/r + r^2/\alpha^2$ is the Friedmann equation with $a = r$, the identification $2M \leftrightarrow (8\pi/3)\rho_d$ being just $M = (4\pi/3)\rho_d$ —so the instanton is one trajectory read in two measures, and only its *action value* differs.

That value follows without further input. On shell the reduced action is $S = \int p_a da$, and inverting (9) gives $p_a = (3/4\pi)a\dot{a}$; continuing $\tilde{\tau} = is$ yields $S = -iS_E$ with

$$S_E^{\text{grav}} = \frac{3}{4\pi} \int ds r r'^2 = \frac{3}{4\pi} \int ds r [f(r) - 1], \quad (7)$$

using $r'^2 = f - 1$ on the segment. The integral converges rapidly and gives $S_E^{\text{grav}} = -0.0481 \alpha^2 / G$ on the forced member^{P10R10}.

Its sign is the informative part, and it is not a pathology. A negative Euclidean gravitational action is the Hartle–Hawking sign: the no-boundary de Sitter action is likewise negative, $-\alpha^2/8G$ in this convention, yielding an enhanced rather than suppressed weight [30]. The present value is the same sign and the same order, smaller by a factor of order two, which is what a segment of the contour rather than a full hemisphere should give.

The sign traces to the segment lying on the $r < 0$ branch, where $p_a < 0$ while r increases. *So the beginning's weight is of the no-boundary type, obtained here from a contour the construction already possessed rather than from a boundary condition imposed on the path integral. One register point must be made here, because the vocabulary invites a reading the construction refuses.* The segment's parametrisation runs imaginary, and the kernel and action computed above are the kernel and action of that parametrisation; *they are not the Euclidean objects of a Wick-rotated spacetime.* The geometric-core paper states the guard generally: the construction reaches a real manifold through imaginary instruments, and the continuations it uses—the embedding coordinate, the equatorial seam, the cosmogenesis reassignment—are *real analytic continuations* on a spacetime that is Lorentzian throughout, not Wick rotations [7]. The de Sitter horizon's Gibbons–Hawking state, by contrast, *is* a Euclidean continuation, and that paper marks it as distinct in kind from these. *So the two must not be conflated:* the thermal register in which $\hbar$ is fixed by the period $\beta = 2\pi\alpha$ is not the register the segment's action belongs to, and dividing that action by that $\hbar$ would be joining two continuations the construction keeps apart. What the action is, is a real integral along a real curve of a real Lorentzian geometry, whose parameter happens to run imaginary over one stretch; what it is not is an exponent awaiting a quantum of action. *Its comparison with the no-boundary sign above is a comparison of signs and magnitudes, not an identification of frameworks.*

The adiabatic correction, and where it fails. The kernel projects exactly only if $\hat{H}_{\text{phys}}$ is constant along the segment, and it is not: the tensor tower of Section 6 carries frequencies $\omega_n = \mu_n/a$, and a varies along the lift, diverging in ω as the branch point is approached. The exact suppression exponent is therefore $\int \omega_n ds$ rather than $\omega_n |\Delta\eta|$, and the first question is whether it converges at all. *Two different questions are in play here and it is worth keeping them apart, because only one of them is what the word “adiabatic” names.* An adiabatic expansion is controlled by $|d\omega/ds|/\omega^2$, and along this segment that quantity is $\propto s^{-1/3}$ and *diverges* at the branch point—so the adiabatic invariant E/ω , which is what an adiabatic approximation conserves, is not conserved through it. *The suppression is nonetheless finite, and for a different reason:* the exponent $\int \omega_n ds$ converges because $\omega \propto s^{-2/3}$ is integrable, which is a statement about the action integral and not about slow variation. *So the correction here is semiclassical rather than adiabatic*—the companion cosmology paper names the same object correctly, as a WKB form whose adiabaticity parameter is of order unity [34]^{L14}. It does: near the branch point $|r| \propto s^{2/3}$, so the integrand grows only as $s^{-2/3}$ and

$$\int_0^{\pi\alpha/3} \frac{ds}{|r(s)|} = 3.3387 \quad (8)$$

on the forced member^{P10R9}. *The correction is not small and it runs the conservative way:* the constant-frequency estimate using the turnaround value gives 1.4396, so the exact exponent is larger by a factor 2.32 and the suppression is *stronger* than the naive reading of (5) suggests.

Whether the projection is adiabatic is a separate question with a definite answer. The adiabaticity parameter $|d\omega/ds|/\omega^2$ is C/μ_n with C running from 1.72 near the branch point to 0.16 near the turnaround, so the approximation is controlled by the harmonic index alone. With $\mu_n^2 = n(n+2) - 2$, $n \geq 2$, this gives 0.70 at $n = 2$, 0.48 at $n = 3$, and 0.16 by $n = 10$. *The projection is therefore adiabatic for all but the lowest few harmonics, and degrades to order*

unity only at $n = 2$ and $n = 3$. That the approximation should fail exactly where the tower is coarsest is expected rather than surprising—there are no modes below $n = 2$ on S^3 —and it locates the correction rather than leaving it as a caveat. *We record, without claiming it, that the harmonic indices at which the treatment loses control are the lowest ones.*

What that loss of control costs is bounded, and the bound is one-sided because of the tower’s own floor. The suppression falls monotonically with the harmonic index— 2.8×10^{-4} in amplitude at $n = 2$, 5.9×10^{-6} at $n = 3$, and by roughly two orders of magnitude per step thereafter—so the mixing the adiabatic projection neglects can only carry amplitude into more-suppressed modes. And there is no mode below $n = 2$ to carry amplitude the other way: the least-suppressed mode has no source beneath it, so the bound is structural rather than estimated. Taking the whole $O(\epsilon)$ fraction out of $n = 2$ at the worst case leaves that mode more suppressed, not less, and the mode it enriches does not reach the unmixed ceiling. *The ceiling on transmitted power is therefore the unmixed $n = 2$ value, 7.9×10^{-8} —and below 10^{-3} even on the naive exponent, which is the weaker estimate. So the residual is located, bounded, and reaches nothing observable*^{P10R14}. The relation between S^3 tensor harmonics and observed multipoles is developed in the companion framework [3], whose perturbation-spectrum treatment carries the source spectrum of a closed S^3 , discrete by degree, projected to the sky through the flat distance slicing rather than a closed one—the flat/closed decoupling carried by the non-synchrony. *That projection is that paper’s; this section’s own contribution to it is the harmonic-side adiabaticity above, controlled by the harmonic index alone—so the two meet at the lowest harmonics, where that paper’s lowest physical mode and this section’s loss of control coincide.*

An explicit illustration, and its limits. To show the structure is not empty we exhibit it in the simplest cosmological model, flat Friedmann–Lemaître minisuperspace. With scale factor a , conjugate momentum p_a , cosmological constant Λ , and a pressureless (dust) reference fluid serving as the cosmic clock in the manner of Brown and Kuchař [21], the deparametrized gravitational Hamiltonian reduced from the Einstein–Hilbert action (in units $G = 1$, fiducial comoving volume unity) is

$$H_{\text{phys}}(a, p_a) = \frac{2\pi}{3} \frac{p_a^2}{a} - \frac{\Lambda}{8\pi} a^3, \quad (9)$$

equal on shell to the conserved comoving dust energy^{P10R1}. Hamilton’s equation $\dot{a} = \partial H_{\text{phys}} / \partial p_a = 4\pi p_a / 3a$, together with $H_{\text{phys}} = \rho_d$ on shell, gives

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi}{3} \frac{\rho_d}{a^3} + \frac{\Lambda}{3}, \quad (10)$$

the flat- Λ CDM Friedmann equation. Its solution is the $\sinh^{2/3}$ law

$$a(\tau) = A \sinh^{2/3}\left(\frac{3\tau}{2\alpha}\right), \quad \alpha = \sqrt{3/\Lambda}, \quad (11)$$

with the single scale α of the de Sitter substrate; one verifies directly that this $a(\tau)$ satisfies (10) identically. This is precisely the expansion history that the companion framework paper derives, by a wholly independent route, as the SdS expansion law of the layered construction [3]: the deparametrized generator reproduces, in cosmic time, the same flat- Λ CDM evolution that the null-boundary correspondence delivers geometrically. Promoting (9) to an operator on L^2 of the half-line and applying (4) gives unitary evolution of the cosmological wavefunction in cosmic time.

Two qualifications attach to this illustration. First, it is a minisuperspace toy: a homogeneous truncation with a dust clock, not the full Schwarzschild–de Sitter object of CR, and the operator ordering of p_a^2/a and the self-adjoint extension on the half-line are genuine technical questions not settled by the reduction. The illustration establishes that the deparametrized

structure is consistent and reproduces the right cosmology; it does not by itself constitute the canonical quantization of the full theory. That next object—the lift to the closed- S^3 layer and its propagating degrees of freedom—is carried out in Section 6. Second—and this is a feature rather than a gap—the CR beginning is the branch point, a divergence of the geometry read over the areal chart on a substrate whose curvature is finite there, which the bead’s contour passes through rather than ending at [4, 3]. The usual motivation for quantizing minisuperspace, to resolve an initial singularity, therefore does not arise. The point of (4) is not singularity avoidance but the demonstration that, on the selected foliation, the canonical generator evolves rather than annihilates.

6 The closed- S^3 lift: the graviton sector on the layer

The minisuperspace illustration of Section 5 truncates the layer to its scale factor alone; it carries no propagating degree of freedom and so cannot exhibit what the true Hamiltonian generates beyond the background expansion. The full layer does carry such a degree of freedom—the transverse-traceless shear of its spatial geometry, the graviton—and the lift of the deparametrization to it is the substantive canonical content. We carry out that lift here. Throughout, the foliation parameter is the same absolute cosmic time of Section 2: the comoving congruence of the layer, which the companion framework paper identifies with the cosmic rest frame the CMB measures and with the S^3 family of the null-boundary correspondence [3]. It is one congruence, carrying the flat- Λ CDM and the closed- S^3 slicings as two synchronizations of itself, not two frames.

The layer is best represented by the background geometry on which the cosmology is built—the reassigned de Sitter geometry, distinct (as a Lorentzian metric) from the Schwarzschild-de Sitter cosmology it underlies but sharing its manifold and foliation [3]. Its closed synchronous slicing is the evolving round three-sphere of radius $a(T) = \alpha \cosh(T/\alpha)$ in cosmic time T . A transverse-traceless perturbation of the spatial metric decomposes into the tensor harmonics of S^3 , $h_{ij}(T, x) = \sum_n \phi_n(T) Y_{ij}^{(n)}(x)$, with the $Y_{ij}^{(n)}$ the standard transverse-traceless rank-two harmonics of the unit three-sphere and Laplace eigenvalues $\mu_n^2 = n(n+2) - 2$, $n \geq 2$. The second-order Einstein–Hilbert action in the transverse-traceless sector reduces, mode by mode, to that of a harmonic oscillator with time-dependent mass a^3 and frequency μ_n/a ,

$$S_n = \frac{1}{2} \int dT a^3 \left[\dot{\phi}_n^2 - \frac{\mu_n^2}{a^2} \phi_n^2 \right], \quad \pi_n = a^3 \dot{\phi}_n. \quad (12)$$

Read on the absolute foliation exactly as in Section 5, the constraint deparametrizes and (4) becomes

$$i \partial_T \Psi = H_{\text{phys}} \Psi, \quad H_{\text{phys}} = \sum_n \left[\frac{\hat{\pi}_n^2}{2a(T)^3} + \frac{1}{2} a(T) \mu_n^2 \hat{\phi}_n^2 \right], \quad (13)$$

a unitary evolution in cosmic time of a tower of time-dependent oscillators^{P10R2}, one per tensor harmonic. This is the lift the toy lacked: the abstract gauge-invariant degrees of freedom (q^A, p_A) of Section 5 are here concrete—the graviton modes ϕ_n —and the deparametrized generator advances them rather than annihilating them. Each mode is a Schrödinger oscillator on $L^2(\mathbb{R})$ and is manifestly self-adjoint; the closed topology of the layer enters as the *discreteness* of the tower, in contrast to the continuous spectra of the flat minisuperspace and planar reductions.

The same tower has a second reading, in the frame of a fundamental observer. Under the null-boundary reassignment the observer’s worldline is one of the reassigned null rulings, and the layer’s expansion read along it is the areal radius $a(\tau) = (2M\alpha^2)^{1/3} \sinh^{2/3}(3\tau/2\alpha)$ in the observer’s proper time τ —the flat- Λ CDM scale factor of the framework paper, the expansion leg of the *cosmogenetic bead* the framework proves one closed curve [3]. The cosmic time

deparametrized here is the very parameter along that bead: real on the expansion leg the observer inhabits, it is what continues off the real axis, through a bounded contour, along the bead's lap into the antecedent collapse. The tower projected into this frame is (13) with $a(T)$ replaced by $a(\tau)$, the discrete index passing to the continuous wavenumber of the observer's locally flat slices. What distinguishes the two readings is the bend: the observer's Friedmann readout is

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{2M}{a^3} + \frac{1}{\alpha^2}, \quad (14)$$

carrying a pressureless matter term $2M/a^3$ that is absent from the background geometry's pure- Λ expansion, $(\dot{a}/a)^2 \rightarrow \alpha^{-2}$. The matter the observer attributes to the graviton's background is thus the bend of the slicing—a feature of the projection, not of the layer—exactly the reclassification the framework makes for matter content generally [3, 9]. The graviton sector therefore exhibits, in one object, both faces of the layered reading: the discrete unitary tower on the ontological background, and the continuous flat- Λ CDM spectrum a real observer measures, with the matter density appearing only in the latter.

This realizes concretely the loop drawn in Section 7 with the dynamical content. The propagating degree of freedom is the transverse-traceless shear of the evolving layer, advanced by a true Hamiltonian on the absolute foliation—the same structure the companion dynamics paper exhibits for the confined polarized wave [11], here for the cosmological sector and the full tensor tower. The necessity of the external clock is sharpest here: an internal (area) clock is available at $\Lambda = 0$ and would serve, but at $\Lambda > 0$ it no longer separates the dynamics, and the absolute cosmic time is forced. Of the two background-sector questions, the first is settled in structure. The operator ordering and the half-line self-adjoint extension of the scale factor (Section 5) are one coupled question: in the geodesic coordinate $x \propto a^{3/2}$ the kinetic operator carries an inverse-square term γ/x^2 at the origin whose coefficient, across the natural ordering family, attains a maximum $\gamma = \frac{1}{4}$ —strictly below the essential-self-adjointness threshold $\gamma = \frac{3}{4}$ —so the operator is limit-circle at $a = 0$ for every ordering, while the $-\frac{\Lambda}{8\pi}a^3$ term maps to the marginal $-x^2$ and is limit-point at $a \rightarrow \infty$, forcing no condition there. The scale-factor Hamiltonian therefore has deficiency indices $(1, 1)$ independently of the ordering [31, 32]: a one-parameter family of self-adjoint extensions fixed by a single boundary condition at $a = 0$. The ordering ambiguity cannot render the quantization unique; the residual reduces to the physical choice of that boundary condition. That choice is fixed, for the free scale-factor sector, by the geometry of the boundary itself. It is worth naming what kind of closure that is, because the corpus has a criterion for exactly this and this is its sharpest instance. A one-parameter family of self-adjoint extensions^{L10} is a one-parameter family of *candidate quantizations*—distinct physical predictions indexed by a λ that the formalism does not pin. That is the structure the epistemic reading excludes: a structure carrying an unforced parameter is not a single world but a family, and answers “what is the world?” with a family rather than a world [27]. What follows below is not a stipulation that removes the family but the substrate *requiring* its own configuration: the boundary at $a = 0$ is the de Sitter horizon, its surface gravity is $\kappa = 1/\alpha$, and the regular Euclidean state at that κ leaves nothing to choose. *The distinction that makes this the criterion's instance and not merely an analogy is worth keeping: λ indexes distinct worlds, so it is an ontological family and inadmissible as it stands, whereas a parameter that merely records an unknown datum of one world—the progenitor's mass, say—is an epistemic gap and not a family at all.* The point $a = 0$ is not a generic half-line endpoint but the de Sitter cosmological horizon—the smooth, finite-curvature metric singularity at which the comoving congruence's past null ruling closes, the substrate curvature staying finite while the comoving ruler collapses [4, 9, 3]. A Killing horizon of surface gravity κ carries a thermal (Hartle–Hawking) state [29], the unique state regular in the Euclidean continuation and free of a conical defect at the horizon; for the de Sitter cosmological horizon $f = 1 - r^2/\alpha^2$ gives $\kappa = \frac{1}{2}|f'(\alpha)| = 1/\alpha$, regularity period $\beta = 2\pi/\kappa = 2\pi\alpha$, and the Gibbons–Hawking temperature $T = \hbar\kappa/2\pi k_B = \hbar/2\pi\alpha k_B$ [28]. Imposed on the scale-factor

wavefunction, Euclidean regularity is exactly the condition that retains the regular indicial branch $x^{1/2+\nu}$, $\nu = \sqrt{\gamma + \frac{1}{4}} < 1$, and excludes the irregular admixture $x^{1/2-\nu}$, whose presence is a conical defect—an off-temperature state. This is the Friedrichs extension, and it is a single boundary condition: the deficiency- $(1, 1)$ freedom is thereby not merely parametrized but *closed*, and closed without a free parameter, since the de Sitter horizon fixes one temperature. The physical choice the ordering ambiguity left open is the horizon’s own thermal state, and $\hbar$ enters the gravitational sector here, at the seam, scaled by Λ alone—the quantum instance of the substrate’s constant ledger, in which the fundamental constants enter as unit gauges over the single scale and the one place a free quantum parameter could sit is closed by the horizon’s own thermal state [7]. *And that continuation is not a device brought in to close the freedom; it is the substrate’s other real form.* The global Wick rotation $x_0 \mapsto ix_0$ carries $dS_5 = SO(5, 1)/SO(4, 1)$ to the compact $S^5 = SO(6)/SO(5)$, and the two are the real forms of the one complex group $SO(6, \mathbb{C})$ [2, 7]. *So the register in which $\hbar$ enters is the same substrate read on its compact face,* and the horizon—where the period is $\beta = 2\pi\alpha$ —is where the two faces meet: the extension is fixed not by an extra assumption but by the geometry the Lorentzian reading already carries, seen from its other side. This determines the free scale-factor sector; the coupled sector below is where the open structure remains. *And what remains there is one of three parts, the other two being settled.* The boundary operator is bounded below and its spectrum is computable branch by branch, both shown below; what is not settled is the ultraviolet definition of the mode sums. *That the remainder is that part and not another is worth stating, because it bounds what any calculation on the free spectrum can reach:* a deformation of the free frequencies is answerable exactly—no multiplicative renormalisation moves the logarithmic coefficient, and the one mass-like displacement that removes it is the curvature offset itself—but *whether the coupling acts on the sum that way cannot be settled ahead of the sum’s own definition*, which is the open item. *So the free-spectrum method reaches exactly as far as it reaches*, and the remainder is not a finer calculation of the same kind^{P10R7}. The second background-sector question—the couplings of the tower to the scale-factor sector beyond quadratic order—sharpens this structure rather than threatening it. Deparametrizability is not at risk: the cosmic clock is a Brown–Kuchař dust fluid (Section 5), whose momentum enters the scalar constraint linearly whatever the graviton content, so the constraint solves for a true Hamiltonian at every order of the coupling—which enters H_{phys} as interaction terms between a and the modes, not as an obstruction. The free tower above evolves on $a(T)$ as a fixed classical background, which is why each mode is manifestly self-adjoint on $L^2(\mathbb{R})$; the coupling question is what happens once the scale factor is itself quantized and back-reacts, and there it acts at the $a = 0$ boundary of the scale-factor half-line. Each mode’s kinetic term $\pi_n^2/2a^3$ is, in the geodesic coordinate, the inverse-square operator $\pi_n^2/2x^2$ at the origin, so the boundary coefficient is promoted from the c-number $\gamma \leq \frac{1}{4}$ of the free scale factor to an operator $\hat{\Gamma}$ on the tower whose spectrum straddles the $\frac{3}{4}$ threshold. *At leading order that spectrum is bounded below by γ and unbounded above—the operator is a sum of squares, displayed below—but the decomposition that follows uses only that both sides of the threshold are occupied, and not that the spectrum has a floor; whether the complete $\hat{\Gamma}$ is bounded below is not assumed here*^{P10R12}, *though it does in fact follow: the full inverse-square coefficient is positive on non-degenerate metrics, so $\hat{\Gamma} = \gamma + c \sum_n \hat{\pi}_n^2$ retains its floor at γ beyond leading order, and the question of a spectrum below $-\frac{1}{4}$ —where $\nu = \sqrt{\Gamma + \frac{1}{4}}$ would leave the reals and no regular branch would exist—has no object. The straddle itself is a computed fact*^{P10R13}: *the spectrum does occupy both sides of $\frac{3}{4}$, since $\hat{\Gamma} = \gamma + c \sum_n \hat{\pi}_n^2$ with $c > 0$ has $\text{spec } \hat{\Gamma} = [\gamma, \infty)$, and $\gamma \leq \frac{1}{4} < \frac{3}{4}$ places spectrum strictly below the threshold while the unboundedness above places spectrum strictly above it—the negation requiring $\gamma \geq \frac{3}{4}$, three times its own bound. That threshold is a different one from $-\frac{1}{4}$, deciding a different property: $\frac{3}{4}$ separates limit-point from limit-circle and so decides whether a boundary condition must be chosen at all, while $-\frac{1}{4}$ decides whether a regular branch exists to choose. How the straddle falls does not bear on the closure*

below, which is supplied fibre by fibre and so cannot be broken by the size of the sub-threshold set⁹. As $\hat{\Gamma}$ commutes with the radial part, $-\partial_x^2 + \hat{\Gamma}/x^2$ decomposes as a direct integral over its spectrum: essentially self-adjoint where $\hat{\Gamma} \geq \frac{3}{4}$ —the graviton momentum makes the origin limit-point and removes the boundary freedom—and limit-circle where $\hat{\Gamma} < \frac{3}{4}$. The single boundary condition of the free scale factor is thereby replaced, in the coupled theory, by a self-adjoint condition supported on the sub-threshold graviton subspace, its deficiency that subspace rather than a line. At leading order $\hat{\Gamma} = \gamma + c \sum_n \hat{\pi}_n^2$; but the cubic and higher self-interactions enter at the same inverse-square order at the origin ($\pi_n^2 \phi_m / a^3$ in kind)—the potential sector’s own cubic being regular there, and so reaching the trace without reaching the boundary coefficient—so the complete boundary coefficient is the singular part of the interacting graviton Hamiltonian, and the *ultraviolet definition* of the tower sums is the open frontier—the interacting tower as a defined theory, met here at its boundary face. *The spectrum is not open with it*: it is computed branch by branch below, and which branch is a question the construction has been shown not to answer internally. *Two things can be said about that spectrum without settling the frontier, and both bear on the decomposition. First, it is computable branch by branch: on the natural ordering family the minimum of $\hat{\Gamma}$ is $\frac{1}{4}$ under normal ordering and $\frac{3}{4}$ under symmetric ordering with one mode occupied, and the difference between the two is exactly the zero-point quantum of that mode—which is why the threshold sits at $\frac{3}{4} = \frac{1}{4} + \frac{1}{2}$, the free boundary coefficient plus one such quantum, rather than at a value the construction chose. Second, the two orderings answer the paper’s own question oppositely: under normal ordering the origin remains limit-circle and the boundary freedom survives, while under symmetric ordering it becomes limit-point from the first occupied mode upward. The decomposition below is untouched by the choice, since it uses only that both sides of the threshold are occupied and both orderings occupy both sides; but the physical content is not, and the choice is not one this construction can make internally. Asking which ordering is asking whether the graviton tower’s zero-point energy gravitates at the horizon—the cosmological-constant problem in local dress, reached from inside the boundary coefficient rather than imported. That the construction makes no such choice is an exhaustion rather than an omission. The ordering is inert in the bulk—normal and symmetric ordering differ by the c-number zero-point $\frac{1}{2}\hbar\omega_n$, a global phase under the deparametrized evolution—and becomes physical only in the boundary coefficient, so any selector would have to act there; and each candidate the construction offers fails to, for a structural reason: the horizon’s thermal state and the seam’s characteristic structure act *downstream*, closing the extension given $\hat{\Gamma}$ (the regular branch $x^{1/2+\nu}$ exists for either ordering); the substrate isometry, positivity, the single-scale ledger and covariance under configuration redefinition each respect *both* orderings; and the deparametrization, which supplies a preferred vacuum and so makes normal ordering available, does not make it mandatory—and, by solving the constraint rather than imposing it, removes the anomaly-freedom lever a Wheeler–DeWitt quantization would have used. What is left is not a residual freedom the quantization failed to close but a single physical datum—whether vacuum energy gravitates—localized precisely as one computable quantity: an epistemic gap of the kind the construction admits, not an ontological family of the kind it excludes^{P10R21}. The boundary condition on that sub-threshold subspace, however, is supplied by the same principle that fixed the free sector. Thermal (Hartle–Hawking) regularity is Euclidean smoothness at the horizon mode by mode: across the direct integral it imposes the regular branch $x^{1/2+\nu}$, $\nu = \sqrt{\hat{\Gamma} + \frac{1}{4}}$, on each sub-threshold fibre at the one horizon period $\beta = 2\pi\alpha$ (the surface gravity $\kappa = 1/\alpha$ belongs to the background horizon, not to the graviton content, and so is common to every fibre), and asks nothing of the limit-point fibres $\hat{\Gamma} \geq \frac{3}{4}$. The self-adjoint condition on the sub-threshold subspace is therefore the de Sitter horizon’s thermal state at every order of the coupling, exactly as in the free sector; what remains open is accordingly not the boundary condition but the *definition of the interacting tower*—the spectrum of $\hat{\Gamma}$ and the ultraviolet definition of the tower sums—the first of which admits a verdict here, and it is affirmative^{P10R15}. *The DeWitt form carries exactly one negative direction and it is the conformal**

one, which is why the Wheeler–DeWitt equation is hyperbolic; on transverse-traceless momenta that term drops and the form is $\text{tr}((g\pi)^2)$, equal to $\text{tr}(S^2)$ for the symmetric $S = L^\top \pi L$ with L the Cholesky factor of g , hence positive definite for every non-degenerate spatial metric. *So the cubic term’s apparent unboundedness is an artefact of truncation:* $\pi^2(1 - \lambda\phi + \dots)$ is the expansion of $\pi^2/(1 + \lambda\phi)$, whose full coefficient is positive wherever the metric is non-degenerate.⁹ *The truncated form is easily misread, because a truncation of a positive function reads as an operator that has lost its floor:* the first-order term $-\lambda\phi\pi^2$ sampled alone does go arbitrarily negative, while the resummed coefficient on the same points never falls below γ . *The deparametrization that yields a true Hamiltonian is the same move that yields a bounded-below one, both being the removal of the conformal direction*—though the positivity speaks only of the interior, the degenerate boundary being what the thermal condition above is for. What remains genuinely open is accordingly the ultraviolet definition of the tower sums—a different thing from a residual freedom in the quantization at the boundary, and not settled by being shared with every interacting field theory. *And what remains open is narrow, because the coefficient’s response to the coupling is settled even though the coupled sector is not.* The logarithmic coefficient is the $1/m$ term of $d(m)\mu(m)$, hence an exact functional of the spectrum, so the deformations that could discharge it can be enumerated without solving the coupling: an overall rescale $\mu^2 \rightarrow (1 + \epsilon)(m^2 - 3)$ returns $L = (39/4)\sqrt{1 + \epsilon}$, which vanishes only where every frequency in the tower does. *So no multiplicative renormalisation of the frequencies discharges the log*—and that is the class the leading back-reaction lives in, the coupling acting through a and through $\hat{\Gamma}$, whose instantaneous expectation is the same sum rescaled. A mass-like shift $\mu^2 \rightarrow m^2 - 3 + \delta$ returns $L = -(\delta - 3)(\delta + 13)/4$, so the one such discharge is $\delta = 3$: there $\mu_n = n + 1$ and $d(m)\mu(m) = 2m^3 - 8m$ is a polynomial, the $1/m$ term not cancelling but *not existing*—and $+3$ is exactly the curvature offset that makes these frequencies non-integer.^{P10R17} *That is a characterisation, and a measurement and a structural remark sharpen it.*⁹ *The measurement first.* The tower is a discrete sum rather than a momentum integral—the harmonics run from $n \geq 2$ with no zero mode and no soft region—so the infrared is regulated by the compactness of the layer and needs nothing. The ultraviolet is untouched by that: with $\mu_n \sim n$, $\langle \hat{\pi}_n^2 \rangle \sim n$ in the instantaneous ground state, and a degeneracy growing as n^2 , the shell contribution goes as n^3 and the sum *diverges quartically*—the ordinary zero-point degree of a field in four dimensions. *The third of those inputs is a formula and not only a scaling:* since $S^3 = SU(2)$ is parallelizable, a frame-indexed field of frame-spin s is $L^2(SU(2)) \otimes V_s$, Peter–Weyl returns level- j totals of 1, 3 and 5 times $(2j + 1)^2$ for $s = 0, 1, 2$ —the component counts—and the transverse-traceless part is the two extreme summands, exactly two fifths of the symmetric-tracefree total. Organised by eigenvalue, *the degeneracy is $2(n - 1)(n + 3)$* , ten at the floor $n = 2$, so the shell contribution is $2n^3$ and the leading constant is settled rather than assumed⁹. *And the constant is the propagating-component count rather than a universal:* the scalar tower’s counting function grows with leading coefficient $\frac{1}{3}$ against the tensor tower’s $\frac{2}{3}$, so the growth is n^2 per propagating component and not per tensor rank. *So “standard” is the right word and it is a number. And the leading term’s counterterm is one this framework already carries.* The quartic piece of the one-loop vacuum energy is independent of mass and of field—it survives at $m = 0$ and is the same for every species—so it is a constant vacuum energy, and *the counterterm a constant vacuum energy requires is a cosmological-constant term.* That is the framework’s single dimensionful constant, absorbed into the one observed curvature with no bare- Λ -versus-vacuum split [7]; and since ℓ_P is a gauge-combination rather than a second physical length, the cutoff is not smuggling a scale in either.

The subleading term is measured on the same spectrum rather than estimated. Written in the labels $m = n + 1$ that carry $\mu^2 = m^2 - 3$ and $d = 2(m^2 - 4)$, the spectral zeta at the origin is $\zeta(0) = 10$ *exactly*—the binomial expansion terminating there, so the value is an identity and not an asymptotic estimate—and the zero-point sum’s logarithmic coefficient is $\frac{39}{4}$, the $1/m$ term of $d\mu = 2m^3 - 11m + \frac{39}{4}m^{-1} + \dots$, which a hard cutoff carrying no zeta function reproduces.

And the scale factor does not enter it: the free tower's frequencies are μ_n/a , so a factors out of the sum identically and the coefficient's independence of the expansion is exact rather than adiabatic—which is also why the coupled sector, and not a time-dependent free one, is where what remains lives^{P10R3}.

The structural remark is about the terms after it, and it is sharper than the leading case because their counterterms are not cosmological-constant terms. Only the quadratic and logarithmic successors carry the mass, which is to say only they renormalise masses and couplings; in the standard curved-space expansion the first goes with the Ricci scalar and the second with curvature-squared invariants, and a curvature-squared coupling is not an entry in this framework's ledger. The reason that is not fatal here is a property of the backgrounds this construction admits rather than of the divergences. On a maximally symmetric geometry of radius α every curvature invariant is a pure power of $1/\alpha^2$ [7], so $\int \sqrt{g} R^2$, $\int \sqrt{g} R_{\mu\nu} R^{\mu\nu}$, $\int \sqrt{g} R_{\mu\nu\rho\sigma} R^{\mu\nu\rho\sigma}$ and $\int \sqrt{g}$ are not four independent functionals but four multiples of one; and the substrates the construction admits are a one-parameter family, the α -foliation, every descent step above the last changing the scale and nothing else [7, 5]. So the counterterm basis is one-dimensional on the admitted family, and a divergence of any degree requires exactly one counterterm where a generic theory requires three.

And that reduction does not in fact need the family, which matters because the family's own flatness does not survive second order in the shear. In four dimensions $C^2 = E_4 + 2(R_{\mu\nu} R^{\mu\nu} - R^2/3)$ is an algebraic identity on any metric, with E_4 the Gauss–Bonnet density: the curvature-squared invariants are dependent because the dimension is four and not because the background is symmetric, so no order in the shear can separate them. What the admitted backgrounds then supply is the rest of the step rather than the whole of it: on an exact vacuum- Λ space the Ricci terms are constants and $C^2 = E_4 - 8\Lambda^2/3$ exactly, the Gauss–Bonnet part shifting the on-shell action by a topology-fixed constant and the Ricci remainder being what the field equations trade for the stress tensor. So a change of subtraction point is a cosmological-term renormalisation, reached from an identity rather than from a property a deformation could remove^{P10R23}. The restriction is the on-shell one and it is the whole restriction: off shell C^2 is an independent invariant and nothing here bounds it.

What that argument does not reach is stated here rather than left for a reader to find, because it is the whole of what remains. A degeneracy among counterterms is a property of the background class: the functionals are distinguished by their response to varying the background, and on a one-parameter family they cannot be told apart. The first thing to check is therefore which background the tower is defined on, and it is this section's own: the closed synchronous slicing $a(T) = \alpha \cosh(T/\alpha)$ of §5, whose Ricci scalar is the constant $12/\alpha^2$. That geometry is exactly de Sitter, so the degeneracy holds on the very background the free tower uses, and the argument above is not an appeal to a nearby object.

Where it genuinely stops is elsewhere, and the section has already named the place. The free tower evolves on $a(T)$ as a fixed classical background; the coupled sector is what happens once the scale factor is itself quantized and back-reacts, and that is precisely the regime the boundary operator $\hat{\Gamma}$ belongs to. The natural worry there is that a counterterm basis is a statement about a class of fixed backgrounds, so that once the scale factor is quantized there is no fixed background left to state it on. That worry does not survive being checked, and what replaces it is sharper.

The collapse of the three quadratic invariants is not a consequence of maximal symmetry: in four dimensions their deficit $\text{Riem}^2 - 2\text{Ric}^2 + \frac{1}{3}R^2$ is exactly $C_{\mu\nu\rho\sigma}C^{\mu\nu\rho\sigma}$, every Friedmann geometry is conformally flat for every scale factor, and on such a geometry the Gauss–Bonnet^{L17} combination is an exact total derivative, $\sqrt{g}(R^2 - 4\text{Ric}^2 + \text{Riem}^2) = 24 d(\dot{a}^3/3 + \dot{a})/dT$.

So $\int \sqrt{g} \text{Ric}^2$ and $\int \sqrt{g} \text{Riem}^2$ are both fixed by $\int \sqrt{g} R^2$ up to a boundary term whatever a does: no scale factor breaks it, because no scale factor can make an FRW geometry anything but conformally flat⁹. And because that is an identity holding pointwise in $a(\cdot)$ rather than an evaluation on a chosen class, it survives superposition and descends to the quantized sector as

an operator relation: the coupled sector never required a class of fixed backgrounds, because the statement was never made by evaluating on one. *What maximal symmetry did buy is a weaker and different thing, and that half does not survive*: on a constant-curvature background terms of *different* dimension— $\int \sqrt{g}$, $\int \sqrt{g} R$, $\int \sqrt{g} R^2$ —are proportional as well. *What that half actually needs is weaker than a constant-curvature background and weaker than the cosh*: it is exactly that R be $\sqrt{g}$ -almost-everywhere constant, and two things that are not the cosh keep it so. A cosmological term with radiation gives $R = 4\Lambda$ exactly, radiation being traceless; and *the free tower's own bare zero-point energy is exactly radiation-like*—its energy is $\mathcal{S}/a$ with $\mathcal{S}$ free of a , so the density goes as a^{-4} , the pressure is a third of it, and the trace vanishes identically. *So the naive back-reaction of the quantized tower breaks nothing*, and both cases serve as null controls that measure the numerical floor rather than assuming it.

What does break it is the logarithm, which is to say the very constant at issue. The spectral $Z(s)$ has a pole at $s = -1$ of residue $\frac{39}{4}$, so the renormalized energy carries a $\ln a$ —and a $\ln a$ spoils the exact a^{-4} that made the bare sum traceless. The trace is then $\Theta = \rho - 3p \propto \frac{39}{4}a^{-4}$, *independent of the renormalization scale*: it is scheme-independent, which is what an anomaly is, and is why the split between the constant and the logarithm being a convention does not reach it. So R acquires an a^{-4} piece and is constant *if and only if* that residue vanishes. *The three integrals then span three dimensions rather than two*, at a ratio thirty-five decades above the floor the null controls measure, with the departure signed by the identity $I_2 I_0 - I_1^2 = I_0^2 \text{Var}_{\sqrt{g}}(R) \geq 0$ and growing as the square of the residue: *the logarithm's counterterm is a genuinely new entry, and the one ultraviolet constant becomes observable once the scale factor is quantized*^{P10R24}.

The shape of that is worth stating plainly, because it is not an accident of this tower. On a fixed background the one constant is unobservable *because* of a degeneracy—and the degeneracy is switched off by exactly the constant it was hiding. *Were there no logarithm, R would stay constant and the counterterm would stay invisible; but then there would be no constant to hide*. The saving mechanism and the thing it saves us from are the same number, so the sums cannot be carried into the coupled sector free of charge by that route: *the counterterm basis is one-dimensional at fixed background and two-dimensional once the scale factor is quantized. And it does not rest on the substrate scale being indefinite*, which is the reading the statement invites and which fails twice over. A superselection rule forbids *coherence* between sectors and not an *ensemble* over them, and the three integrals are read off the ensemble linearly, so an ensemble over distinct curvature radii spans three dimensions inside a superselected theory and with no anomaly at all; and the question has no object as put, since α is a coefficient of the physical Hamiltonian rather than an observable of it, so the corresponding operator is a multiple of the identity with a one-point spectrum, and a central operator with one point decomposes nothing. *Neither route is the one taken here*: the rank is three already at a single curvature radius, which is where the a^{-4} anomaly puts it, and the operator the Ricci scalar depends on commutes with nothing in the coupled algebra. *Centrality is a relation to an algebra*, and the theory's one genuine superselection label is the boundary operator—central in the radial algebra, labelling the boundary condition and not the background scale^{P10R25}. *The back-reaction here is semiclassical and the interacting theory is not built*; what higher orders could do is narrow, since restoring the degeneracy would need the total trace exactly constant—a pure cosmological term—and an a^{-4} anomaly is not one. *And the mode in which the constant is observable is fixed, because no state resolves the curvature it is read off*. At this order the excitation trace vanishes on the *diagonal* of the adiabatic basis, which is the harmonic virial theorem and not an operator identity: with $\omega_n = \mu_n/a$ the excitation energy is $\mathcal{S}/a$ with $\mathcal{S}$ free of a *at fixed occupation*, so the pressure is a third of the density in every stationary state and in every mixture of them. *Written in the field operators themselves the trace carries two terms and not none*— π_n^2/a^6 and $\mu_n^2 \varphi_n^2/a^2$, the second a potential-like term from the tower's own gradient energy, with $\mu_n^2 \geq 6$ for every mode so that it is neither zero nor small—and their combination is $a^{-4}(a_n^2 + a_n^{\dagger 2})$: *purely off-diagonal, non-zero on any squeezed state, and untouched by normal ordering*, which

has no contraction to take. *The conclusion the diagonal statement was carrying survives and no longer rests on it.* That combination is an *inverted* oscillator, and an inverted oscillator has no normalizable eigenvector at any real eigenvalue: in the momentum representation the null equation is solved by $\sqrt{p} J_{\pm 1/4}(p^2/2\mu a^2)$, both solutions regular at the origin and neither square-integrable at infinity, the modulus squared falling as $1/p$ with the envelope exponent holding at $-1/2$ exactly at every real eigenvalue. *So the curvature has no point spectrum, and it has none without any term being absent*—which is the stronger warrant, since it does not turn on an absence^{P10R32}. *The spectrum is purely continuous and two-sided:* the same operator puts spectrum below 4Λ as well as above it.

And the obstruction is the anomaly coefficient itself, which is this result's shape for the third time and from a third direction. The commutator of the curvature with the momentum conjugate to the scale factor gives $\text{Var}(R) \text{Var}(p_a) \geq \frac{\hbar^2}{4} \langle R' \rangle^2$, saturated to a third of a per cent by narrow admissible states, and *the right-hand side vanishes exactly when the residue does*—which is the case in which the curvature is constant and there is nothing to resolve. *So the same number that makes the counterterm observable is what forbids any state from sharpening the curvature the counterterm is read off,* and the constant is observable *distributionally* rather than as an ordinary sharp-valued observable. That is unconditional rather than weakened: there are no special states to except, the ordinary reading being excluded by a theorem and not by a missing calculation. *One consistency there was not put in by hand:* the expectation of a^{-4} converges at the origin only above a threshold in the boundary index, and the horizon's own thermal condition clears it on either ordering—the *boundary condition that closes the deficiency is the same one that makes the curvature's expectation exist*^{P10R26}. *What the coupling is genuinely needed for is smaller than that, and it is smaller than the order of the promotion suggests.* At higher order the cubic and above put genuine tower operators into the trace, and neither the bound nor the level-set argument turns on their absence. *The commutator is an algebraic identity and not a dynamical one:* $[f(a) \otimes T, p_a] = i\hbar f'(a) \otimes T$ for *any* tower operator T , commuting or not and ordered any way, since p_a differentiates the scale-factor function alone and the tower factor rides through—so the inequality holds verbatim in form at every order, and its form is blind to the ordering. *And pointwise cancellation is excluded by the trace rather than found to be absent:* the trace of an energy term $h(a) \otimes T$ is $(h + ah')T/2\pi^2 a^3$ exactly, so a term $ca^{-n} \ln^m a$ contributes $2\pi^2 \Theta = ca^{-(n+3)} [(1-n) \ln^m a + m \ln^{m-1} a]$, and the counterterm's own a^{-4} is the member of that family at $n = 1, m = 1$. *The whole family is closed under the operation rather than annihilated by it, and closure in one direction only is what settles the question.* From a term at $n = 1$ the highest power surviving in $\partial_a \Theta$ is $\ln^{m-1} a$ with coefficient exactly $-4mc$, while a term one logarithm higher reaches both $\ln^m a$ and $\ln^{m-1} a$: so the terms chain *downward* in the logarithm and never upward, the coefficient system over log powers is lower triangular with diagonal $-4m$, and $\partial_a \Theta \equiv 0$ therefore forces every coefficient to vanish. *The leading logarithm has no partner,* and that is why nothing cancels the residue at any (n, m) —the pure-power case, where $n = 1$ kills the coefficient outright, is this statement at $m = 0$. *What a tuned coefficient would do is relocate the obstruction rather than remove it*—at exactly twice the residue a second-logarithm term annihilates the constant coefficient the counterterm sits in and leaves the next logarithm up at sixteen times the residue, with the curvature's variance at that same state eleven decades above the measured floor^{P10R28}—*but that relocation is a fact about the trace formula's closure and not about this theory, because the term it needs does not exist here.* The relocation must sit at the power of a the counterterm sits at, since that is what lets it reach the constant coefficient at all. *And in the reduced theory the only scales are a and the Planck length,* so the vacuum energy is $a^{-1} \sum_j f_j (\ell_P/a)^j$ against a polynomial in $\ln(a/\ell_P)$: the power is $1 + j$, one vertex pair costs ℓ_P^2 , and the interaction therefore enters at $j \geq 2$. *So the counterterm's own power is populated at $j = 0$ alone*—by the free tower, whose large-label weight $2(m^2 - 4)\sqrt{m^2 - 3}$ is a pure Laurent series and carries no logarithm of the label, so the pole there is simple and the logarithm is capped at the first power. *The second logarithm is*

therefore unavailable at every order of the coupling and not merely at the lowest, and asking the two families to cancel each other instead has no solution, one coefficient being unable to annihilate two different powers of a at once. *The cubic's own sum carries no logarithm either:* the three-harmonic overlap on the spatial section obeys an exact multiplet sum rule, weight $n_1 n_2 n_3$ on the triples satisfying the triangle inequality with odd perimeter, and every term of the soft expansion has the form $j(j^2 - 3)^p$ —an odd polynomial terminating above the first power when p is an integer, an even-power series otherwise—so the inverse first power, which is what a harmonic number would occupy, has coefficient exactly zero at every order^{P10R29}. The level-set argument likewise survives on every fibre that exists: wherever the cubic's tower operators are simultaneously diagonal, the curvature is multiplication by $4\Lambda + \sum_j c_j a^{-n_j}$, whose derivative has no zero while $r > 0$, whose level sets clear to a polynomial of degree six and are therefore finite, and a direct sum over such fibres manufactures no point spectrum from parts that have none^{P10R27}. *A tuned coefficient can make the right-hand side vanish, and that is not sharpness:* the bound goes vacuous while the curvature's variance stays almost six decades above the measured floor, and two distinct powers of a agree at isolated points rather than on a set of positive measure. *And the correlated case is a question about one operator family rather than about states.* Because the trace equation is *algebraic* in the matter trace— $R = 4\Lambda + \kappa\Theta$ carries no derivative of the scale factor—the curvature commutes with it exactly, so the curvature is a direct integral of $4\Lambda + \sum_k \kappa_k a^{-m_k} T_k$, one operator on the excitation factor at each scale. *A constant eigenvalue branch then forces its own value:* every power is negative, so the family tends to 4Λ and a branch constant on a set of positive measure can only be 4Λ itself—leaving the condition that $\sum_k \kappa_k a^{-m_k} T_k$ be singular for almost every scale. *At finite truncation that is exactly the singularity of the excitation operators' own pencil,* a condition carrying neither the scale nor the couplings, because with distinct powers no two monomials of the determinant can meet on the curve the scale traces; and the pencil is non-singular at every truncation computed, in exact arithmetic, for the two non-commuting operators the cubic supplies and for three at the powers where they collide^{P10R30}. *The one place the truncation speaks for itself runs toward the affirmative rather than away from it:* at odd truncation a ladder matrix is singular and the coupling system does acquire solutions, but every one of them switches off at least two couplings, and the operators the tower actually carries have purely continuous spectrum and so no null vector at all. *And in the tower the criterion is not a limit of determinants at all but an ordinary differential equation:* in the momentum representation the excitation operators the cubic's kinetic vertex supplies are p^2 and $i p (d/dp)(p \cdot)$, one power of the field and so first order, and the null equation integrates to $C e^{i c(a)p} / p$. *The exponential is a pure phase, so the obstruction is the inverse power and it is the symmetrisation's own inner derivative:* the modulus squared falls as p^{-2} , integrable at infinity and divergent at the origin, on either side of it. *The operator is not essentially self-adjoint there—the deficiency indices are (1, 1) and the extensions form a circle—and the criterion is settled uniformly over them rather than by choosing one:* a self-adjoint realisation is a restriction of the adjoint, so its eigenvectors solve the same equation and must be normalizable, and a boundary condition selects among square-integrable solutions where here there are none to select from. *So zero lies in no realisation's point spectrum, and the extension choice never has to be made.* The same verdict comes from the position representation, where the operator is Sturm–Liouville with a constant and a logarithm for its null solutions and neither decays^{P10R31}. *The curvature does carry a second power of the field alongside the first,* from the tower's own gradient energy and at quadratic rather than cubic order, so the obstruction at the momentum origin is gone—both solutions are regular there—and the second-order case is the one that obtains. *It does not end in an extension choice either, because the point at which the leading coefficient vanishes belongs to one representation,* and one gauge of unit modulus—so unitary, and every spectral count carries across—removes the first-order term exactly and leaves a plain Schrödinger operator on the whole line: *an inverted quartic*, its quartic coefficient negative whatever the sign of the first-order vertex. *What the*

singular point of the other representation is not is a locus: the origin of the scale factor is an endpoint of the configuration space and a physical surface with its own surface gravity, where the extension freedom is genuine and the horizon's thermal state closes it, while a zero of a leading coefficient in one chart carries no locus for anything to act at. *At every real eigenvalue both solutions of the inverted quartic fall as an inverse first power*, so the modulus squared is integrable at each end, both ends are limit-circle, the deficiency indices are $(2, 2)$, and a null solution does exist—the dichotomy against the quadratic content being exact rather than continuous, since the same integral converges over a quartic turning point and diverges over a quadratic one. *So the argument that closed the quadratic case has no purchase here*: there a boundary condition had no square-integrable solution to select among, and here it has two.

What closes it instead is the same criterion one level up, and it closes for every realisation held fixed. Rescaled, the operator is a one-parameter family whose parameter is a fixed power of the scale factor—constant only at a trace power the construction does not give it—and at limit-circle ends every realisation has compact resolvent and therefore purely discrete spectrum, while each eigenvalue rises *strictly* with the parameter, the derivative being the expectation of a positive operator and so positive for every realisation alike. *Each branch therefore crosses zero at most once*, so the scale factors at which zero is attained form a discrete set, so the condition cannot hold on a set of positive measure, and there is no eigenvector^{P10R33}. *The one freedom that argument does not fix is a realisation chosen fibre by fibre, and it is not a new open thing*: the extension lives at unbounded momentum, which is the tower's ultraviolet—the same ultraviolet definition of the mode sums this section already carries as its open frontier. So the two extension questions this sector meets are one closed and one open at opposite ends of the same tower: the physical boundary at the origin of the scale factor, closed by the horizon's own thermal state, and the ultraviolet, which is the frontier itself.

And the spatial curvature term's own cubic does reach the trace, by an identity rather than a computation: the cube's trace is three times the determinant for any traceless symmetric matrix, the first-order curvature perturbation vanishes for transverse traceless data on a maximally symmetric section, and what survives at cubic order is the volume element's own term against the unperturbed curvature, needing no perturbation of the curvature at all. *What is open is whether it reaches the single-mode problem*: for one wave vector transversality is algebraic and makes that vector a null eigenvector of the polarisation, so the determinant—and with it the whole term—vanishes identically across the flat polarisation space, lifting only for two wave vectors together. *The term is therefore intrinsically many-moded at flat order, and what would put a third power of one excitation into the equation sits exactly on the degenerate set*; on the closed section transversality is differential rather than algebraic and does not force that degeneracy, so the question is the value of one overlap integral, which is not settled here.

And there is a second place the degeneracy ends, at fixed background, which is the shear. The deficit is C^2 ; for an anisotropic shear over an isotropic expansion the invariant is $C^2 = 2 \sum_i (\dot{\sigma}_i + H \sigma_i)^2$ at second order in the amplitude^{P10R4}—so it carries two derivatives more than the shear itself, and for a mode of frequency ω it grows as $\omega^2 \sigma^2$, which is what makes it ultraviolet-sensitive at all; and the propagating content of this tower is the transverse-traceless shear. *So what remains is the tower's own shear, which is a calculation and not a question about meaning*—entering at second order in the mode amplitude, with the sub-leading heat-kernel coefficients as its instrument. *That calculation is run, and its answer is smaller than the question sounds*: for a transverse-traceless perturbation the first variation of the Ricci scalar vanishes identically and $\sqrt{g}$ is unchanged, so R^2 at second order in the amplitude is $2\bar{R}$ times R at that order—*pointwise, with no integration by parts*—and $\int \sqrt{g} R^2$ is accordingly a multiple of the Einstein–Hilbert term rather than a new structure.

With the Gauss–Bonnet combination contributing no field equation, the three quadratic invariants leave $\int \sqrt{g} R^2$ and $\int \sqrt{g} C^2$, of which only the second is new: *the shear therefore costs exactly one new counterterm, and it is the Weyl-squared one*⁹.

And its coefficient follows from what the tower is, without a second-order spectrum. Each mode is a harmonic oscillator of mass a^3 and frequency μ_n/a , which is term for term the reduction of a minimally coupled massless scalar on this background, so the tower is *two* such degrees of freedom carrying the transverse-traceless spectrum. The C^2 coefficient is a property of the field content rather than of the background, which is why it is fixed without one: for a single real scalar the entire C^2 content of the second heat-kernel coefficient [6] sits in $2\text{Riem}^2 - 2\text{Ric}^2$ —independently of the curvature coupling, which is why the answer needs none—and that combination is $3C^2 - E_4$, giving $1/120$ in units of $(4\pi)^{-2}$ and therefore $\frac{1}{60}$ for the tower^{P10R8}: *exactly twice a real scalar's. The count is of physical modes, and that is the reading on which the number stands*: the constraints here are solved rather than gauge-fixed, so no ghost sector enters and no covariant spin-2 tabulation applies. The overall normalisation by which the coefficient enters a divergence is convention-dependent; the ratio is not.

Two things about that entry are settled and belong beside it. First, the two helicities are not identified by any symmetry the layer connectedly carries: on $S^3 = \text{SU}(2)$ the connected isometry group is $(\text{SU}(2)_L \times \text{SU}(2)_R)/\mathbb{Z}_2$, transverse-traceless rank two carries $|j_L - j_R| = 2$, and each level is $(j, j+2) \oplus (j+2, j)$ with $2j+1 = n-1$ —whose dimensions sum to exactly the $2(n-1)(n+3)$ derived above^{P10R5}. *They are inequivalent irreducibles, and the map exchanging them is the exchange of the two factors, an orientation-reversing isometry*—so a parity-odd term is not excluded by the geometry, and whatever excludes it does so through the state. *Second, a topological term contributes no field equation, so a coefficient multiplying one is not read off any dynamics*; its only remaining home is the horizon entropy, on which this construction is deliberately uncommitted [7]. *So whether such a term is an entry in this ledger is not a separate question from that one.*

One caveat belongs in the statement rather than after it: the dimension-four scalars also include the parity-odd Pontryagin density, which is not a curvature-squared invariant of the kind counted above and is a total derivative, but which can be non-zero at second order in the metric perturbation and reverses with the handedness—*so the zero is a property of the mode chosen and not of the geometry* [11]. *Which modes give the zero is a question about the configuration and not a general rule*: on a plane-wave pair the density goes with the relative phase of the two channels, vanishing where they are a quarter-cycle apart and largest where they are in step, so a prescription reaching for one handedness may be reaching for the case that vanishes.

And what that scopes, first of all, is this computation. A counterterm answers to a parity-odd divergence of the effective action, which is a property of the theory and the state; *it is not a property of a basis mode*. The two helicities span the same space in either polarisation basis, and on a parity-symmetric state the two contributions cancel whatever an individual basis element returns. *So a mode-level evaluation cannot fix what counterterm the theory needs, in either direction* — the caveat bounds its own instrument rather than deferring the question to a choice of polarisation.

Two things would have to be settled before a parity-odd term could be scored at all, and they are independent. Whether the tower is chiral on the corpus's own criterion — which is not that a reflection exists but that *no connected action completes it*, a condition applied to the radiating sector and not to this one. *And whether a total derivative is a ledger entry.* The density above contributes no field equation, so it is not a coupling in the sense the constant-counting audit means, and the ledger counts no counterterm coefficient [7].

The propagating sector itself is a clean, unitary, deparametrized evolution—a true Hamiltonian generating the graviton on the layer, in cosmic time.

7 Dissolution, not solution

The result of Sections 4–5 is best stated negatively. We have added nothing to the formalism. We have not introduced a new time variable into general relativity, nor a matter field whose role is to be a clock, nor a modification of the constraint algebra. We have identified, on external grounds, which foliation is physically real, and read the existing constraint on it. The problem of time, on this account, was never a problem internal to the formalism; it was the formalism faithfully reporting the consequences of an ontological premise—the block—that is both a category error and empirically false.

This is why the appropriate verdict is dissolution rather than solution, and the distinction is not merely rhetorical. A solution would add structure to make a defective formalism work. A dissolution removes a mistaken premise and finds the formalism was working all along. The four epistemic rules that govern such corrections [15, 27]—resist the naïve identification of appearance with reality; prefer a reading on which the observed structure is a structural necessity rather than a tuned permission; consolidate phenomena under one principle; treat accumulating patches as a red flag—each favour the CR reading over the frozen one. The cosmic time is a structural necessity of the ontology and an empirical datum, not a tuned addition. The reading consolidates: the same correction that turns the frozen constraint into a true Hamiltonian also dissolves the inference that gravitational collapse completes in finite cosmic time, since both inferences—the timeless wavefunction and the completed horizon—rest on granting existence to a manifold of occurrences that no observer’s present ever contains [4, 15]. One ontological correction, two canonical payoffs.

The reading also closes a loop with the dynamical content of the programme. The true Hamiltonian generates the advance of the layer $\mathcal{S}_t$; the geometry of that advance is what the operator and range papers read directly, where the matter content of a layer is the bend of its spatial cut off the vacuum profile and gravitational radiation is the transverse-traceless shear of the evolving layer, up to the wall at which a symmetric layer can no longer carry the wave and ordinary inhomogeneous evolution begins [9, 10]. The canonical face (a Hamiltonian that advances the layer) and the geometric face (matter and radiation as features of the advancing layer’s intrinsic geometry) are two descriptions of the one evolving three-dimensional world. The cosmogenesis re-expansion is one such advance: the companion cosmogenesis paper [33] runs the collapse-and-re-expansion thermal history—the layer’s advance along cosmic time, generated by this true Hamiltonian rather than frozen—through a nuclear network to the primordial light-element abundances. The companion foundation roots these faces in a single augmentation of general relativity [26]; the present paper supplies the canonical one, and with it the recognition that CR’s distinctive contribution to quantum gravity is not a new dynamics but an absolute cosmic time—real, forced, and already implicit in the formalism—that turns the constraint into a generator. This dissolution is the canonical face of the substrate’s maximal symmetry gathered in the geometric core [7]: the problem of time is identified in the constraint-algebroid construction as the base-dependence of the substrate’s coset metric [5], and the selection of the forced foliation is what deparametrizes it away—the structural and the canonical sides of one dissolution.

8 Relation to existing approaches

The deparametrization carried out in Section 5 is, as a technique, entirely standard, and it is important to be clear about what is and is not new here. Kuchař’s survey [19] classifies responses to the problem of time into those that locate an internal time before quantization, those that quantize first and seek time afterwards, and those that abandon time outright; Isham’s review [20] maps the same terrain. The construction of Section 5 sits squarely in the first class, and its minisuperspace realization uses a dust clock of exactly the Brown–Kuchař

type [21]. We claim no novelty in the mechanism.

What CR adds is not a mechanism but a warrant. The standard and well-known objection to internal-time approaches is the multiple-choice problem: many candidate internal times exist, inequivalent quantizations follow from different choices, and the formalism provides no principle to prefer one [19, 20]. This objection has force precisely when the choice of clock is a matter of formal convenience internal to the theory. It loses that force when the clock is fixed from outside the formalism by ontology and observation. CR’s cosmic time is not selected because it makes the equations tractable; it is selected because the occurrence/existence distinction identifies the evolving layer as what exists, and because the CMB measures the corresponding cosmic rest frame (Section 2). The multiple-choice problem is thus not solved within the space of formal clocks but dissolved by the recognition that one clock is physically real and the others are not. This is the same pattern as the rest of the paper: a difficulty that is genuine so long as a mistaken premise is held, and that disappears with the premise.

The timeless and relational programmes—Page and Wootters’ conditional probabilities [22], Rovelli’s relational quantum mechanics and evolving constants of motion [23], Barbour’s denial of fundamental time [24]—take the opposite path with full consistency: they accept that no preferred foliation exists and reconstruct the appearance of time from correlations within a fundamentally timeless structure. We do not claim these programmes are internally flawed. We claim that the premise they share—that there is no objective cosmic now—is the very premise the CMB refutes [26], and that once a real cosmic time is admitted, the elaborate reconstruction of temporal appearance from timeless correlations is solving a problem that no longer arises. The choice between the programmes is, in the end, the choice of Section 2: whether to grant the manifold of occurrences the existence that the evidence assigns to the evolving layer.

9 Conclusion

The canonical problem of time is the block universe in Hamiltonian dress. Read the four-dimensional manifold as the existent, and the constraint correctly reports that nothing evolves; read the evolving three-dimensional world as the existent and its record as the manifold, and the same constraint, deparametrized along the cosmic time that the universe has let us measure, is a true Hamiltonian generating a unitary advance that reproduces the observed flat- Λ CDM expansion. Lifted beyond the background to the layer’s transverse-traceless sector, that same Hamiltonian advances the graviton tower of the closed- S^3 layer unitarily—the propagating degree of freedom a fundamental observer reads as the flat- Λ CDM graviton, with matter the bend of its slicing rather than an ontological content. The equations never changed. What changed was the answer to a single question—what exists—and that answer is not a matter of taste but one the occurrence/existence distinction and the isotropy of the CMB settle together. The problem of time was the canonical symptom of a category error; read without it, the formalism was describing a real cosmic time all along. Read so, the dissolution is one of a family: the same distinction that turns the frozen constraint into a true Hamiltonian is shown, in the companion framework paper’s first synthesis, to dissolve a wider set of general relativity’s standing problems together—the problem of time among them, and the unitary evolution established here underwriting the resolution of the black-hole information paradox on the realised, globally hyperbolic spacetime [3]. That places the canonical face within the framework’s one synthesis: the maximally symmetric de Sitter substrate the framework reads at once as general relativity’s solution space [10], as the discrete *CPT* and charge structure, and as the gauge sector on its conjugate real form, is read here on its *canonical* constraint—where the deparametrized generator is the layer’s true Hamiltonian, and the quantum of action enters the gravitational sector at the seam, scaled by Λ alone [7].

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Appendix R Computational Receipts

Each computation marked ^R in the text is verified by a runnable receipt in the `receipts/` directory that ships with this work; status OK = verified (traced, run, output matches the claim). This appendix is generated from the receipt index, not maintained by hand.

- P10R1** `P10_minisuperspace_friedmann` [OK]
sec:deparam (deparametrized minisuperspace $H_{\text{phys}} = (2\pi/3)p_a^2/a - (L/8\pi)a^3$ generates Friedmann $(\text{adot}/a)^2 = (8\pi/3)\rho_d/a^3 + L/3$ via Hamilton's equations). *Computes:* $\text{adot} = 4\pi p_a/3a$; H_{phys} autonomous=conserved; eliminate p_a ; Friedmann; dust a^{-3} ; Lambda-sign control *Bound:* the true-Hamiltonian structure is non-empty *Run:* `python3 receipts/P10_canonical_time/P10_minisuperspace`
- P10R2** `P10_graviton_lift` [OK]
sec:lock (closed- S^3 graviton lift: tensor-harmonic action $S_n = 1/2 \int dT a^3 [\text{phidot}^2 - (\mu^2/a^2)\text{phi}^2]$ Legendre-transforms to $H_{\text{phys},n} = \pi^2/2a^3 + 1/2 a \mu^2 \text{phi}^2$; mode eq $\text{phiddot} + 3(\text{adot}/a)\text{phidot} + (\mu^2/a^2)\text{phi}$ *Computes:* $\pi = a^3 \text{phidot}$; Legendre H; EL mode eq; Hamilton eqns combine to it; flat- a oscillator control *Bound:* unitary cosmic-time evolution of the graviton tower *Run:* `python3 receipts/P10_canonical_time/P10_graviton_lift.py`
- P10R3** `P10_the_scale_factor_factors_out_of_the_free_tower` [OK]
'PO-23'; 'sec:lock' (for the FREE tower $\sum d_n \omega_n = \sum d_n \mu_n/a$, so $\sum d_n \omega_n = (1/a) \sum d_n \mu_n$ and ***the scale factor factors out of the whole sum identically, for every a *** — so the leading coefficient's time-independence is exact FACTORISATION rather than insensitivity, and ***a non-adiabatic FREE computation cannot move it***; the row's discharge must be the coupled one. Both of the row's numbers are also made identities: $\zeta(0) = 10$ by a terminating expansion, and the log coefficient $39/4$ as the $1/m$ term of $d(m) \mu(m) = 2m^3 - 11m + (39/4)/m + \dots$). *Computes:* the label map to the standard TT-on- S^3 form; $Z(0) = 5/2$, $Z(-1) = 5/2$; the summand's expansion; a cancellation-free hard cutoff returning 9.74998 with no zeta function in it; and $R = 12/\alpha^2$ constant, so $\int R^2$ and $\int R$ are both $\proptopto V$ *Bound:* bounded: ***the coupled and non-adiabatic case is untouched*** — what is shown is that the free case cannot be the route to it, and that on this background the one counterterm is degenerate with Einstein–Hilbert so the constant is not OBSERVABLE there *Run:* `python3 receipts/P10_canonical_time/P10_the_scale_factor_factors_out_of_the_free_tower.py`
- P10R4** `P10_the_weyl_invariant_at_second_order_carries_two_more_derivatives` [OK]
'PO-43'; 'sec:lock's shear clause (the Weyl-squared invariant at second order in the shear is $C^2 = 2 \sum_i (\dot{\sigma}_i + H \sigma_i)^2$ — ***two more derivatives than the stated σ^2 , and the right dimension $1/L^4$ *** — so for a mode of frequency ω it goes as $\omega^2 \sigma^2$, which is what makes the counterterm ultraviolet-sensitive at all). *Computes:* the full contraction from the metric with nothing assumed, and the closed form ***read off one computation and verified against another*** — ratio to the exact C^2 converging to 1 linearly in the amplitude (1.0404 , 1.0040 , 1.00040); plus the separating case, a constant comoving shear on a static background, giving $C^2 = 0$ where σ^2 would not *Bound:* bounded: ***the COEFFICIENT is not computed*** — that is 'PO-43's object and needs the sub-leading heat-kernel coefficients for the tower's own operator. This fixes what they multiply *Run:* `python3 receipts/P10_canonical_time/P10_the_weyl_invariant_at_second_order_carries_two`
- P10R5** `P10_the_two_replacement_questions_are_one_answered_one_coupled` [OK]
'PO-43' 'r4545's two replacement questions (***neither survives as stated***: the connected-isometry question is answered (the tower's degeneracy $2(n-1)(n+3)$ is exactly $(j,j+2) \oplus (j+2,j)$ with $2j+1=n-1$, so the helicities are ***inequivalent*** irreducibles of $\mathfrak{so}(4) = (\mathfrak{su}(2) \oplus \mathfrak{su}(2))$ and the exchange is the factor swap, orientation-reversing); and the topological-terms question ***is not independent*** — a topological term contributes no field equation, so its coefficient's only home is the horizon entropy, where the corpus is uncommitted). *Computes:* the branching identity checked at every level $n=2$ – 11 against 'P10's Peter–Weyl degeneracy, and the two families shown distinct; and the enumeration of where a topological coefficient could be observable *Bound:* bounded: ***Q2 is identified, not answered*** — it closes only when the de Sitter horizon entropy does, and 'P17' declines that number rather than asserting it *Run:* `python3 receipts/P10_canonical_time/P10_the_two_replacement_questions_are_one_answered_one_coupled.py`

P10R6 P10_the_thermal_condition_is_helicity_blind_at_the_mode_functions_and_the_parity_odd_entry_is_not [OK]

‘PO-43’*’s one unrun check; ‘sec:lock’*’s fibre-by-fibre condition, and ‘P17’ ‘sec:ledger’** (**THE CHECK COMES OUT FOR THE CLAIM, AND NOT BY WORDING: THE TWO HELICITY TOWERS ARE POPULATED ALIKE, SO THE PARITY-ODD ENTRY IS NOT OWED AND THE ROW CLOSES ABOVE THE DECLINED ENTROPY QUESTION.** The condition’s entire dependence on the tower is $\hat{\Gamma} = \gamma + c \sum_n \hat{\pi}_n^2$, a sum of squares over a mode set the helicity swap PERMUTES, pairing modes of *equal frequency* — so $\hat{\Gamma}, P = 0$ exactly, and with it $\nu = \sqrt{\hat{\Gamma} + \frac{1}{4}}$, the sub-threshold projector and the retained branch $x^{1/2 + \nu}$; and $\beta = 2\pi/\alpha$ carries no mode index at all, $\kappa = 1/\alpha$ being the *background* horizon’s. **AND THE EQUAL FREQUENCIES ARE FORCED BY AN ISOMETRY RATHER THAN BY ARITHMETIC.** $\sigma: g \mapsto g^{-1}$ on $S^3 = \mathrm{SU}(2)$ is $\mathrm{diag}(1, -1, -1, -1)$: in $O(4)$ with $\det = -1$, *orientation-reversing*, conjugating left translations into right ones so it exchanges (j_L, j_R) , commuting with the Laplacian and *anti-commuting* with the curl — one map that preserves every frequency and reverses every handedness. Read off the Casimirs, $\mu^2 = 2(C_L + C_R) - 6$ is manifestly symmetric and reproduces ‘P10’*’s own* $\mu_n^2 = n(n+2) - 2$ at every level, with the degeneracy splitting $50/50$ ($(n-1)(n+3)$ each) and the floor a self-dual and an anti-self-dual *five*. $\Rightarrow$ *The three attacks the order named: (a) and (b) FAIL; (c), the Euclidean continuation, is ANSWERED* — the rotation $x_0 \mapsto ix_0$ and σ act on disjoint coordinate blocks and commute, and while a Riemannian four-section does admit a real self-dual split a Lorentzian one does not, *weighting the towers apart needs a parity-odd exponent*, and both candidates are absent: the de Sitter cosmological horizon has $\Omega = 0$, and assuming a θ -term is circular). *Computes:* the degeneracy split symbolically and to eight levels; $\mu^2 = 2(C_L + C_R) - 6$ against ‘P10’*’s own eigenvalue; σ ’s orthogonality, determinant and the left-to-right conjugation on 200 random quaternion pairs; $\hat{\Gamma}, P$, $\nu(\hat{\Gamma}), P$ and $\Pi_{\Gamma, 3/4}, P$ on a truncated Fock space; κ and β DERIVED off $f = 1 - r^2/\alpha^2$; the parity-odd expectation in the Fock space and in closed form level by level; and the Wick rotation against σ — *28 gates, with THREE controls that break as they must*: a helicity-weighted $\hat{\Gamma}$, unequal degeneracies or frequencies, and a rotating horizon returning an Ω -linear parity-odd expectation *Bound:* *not that the tower is achiral* — ‘r4547’ stands, it IS chirally capable and what is shown is that the STATE does not use the capacity; nothing about the interacting tower’s *ultraviolet definition*, ‘P10’*’s own open frontier, the reach here being that a parity-even bare action and a parity-even state generate no parity-odd divergence under a parity-even regulator (a parity-breaking one is a scheme choice, and the anomaly route needs unbalanced chiral FERMIONS, which the row’s own index obstruction excludes — that last step is a READING of a corpus result, not computed); not the declined $S = A/4$ question; and the floor-as-subtraction-point candidate is deliberately not built*, per the order *Run:* python3 receipts/P10_canonical.time/P10_the_thermal_condition_is_helicity_blind_at_the_mode_functions_and.t*

P10R7 P10_the_remainder_is_the_last_of_three_and_is_not_reachable_by_refining_the_method [OK]

‘PO-23’; ‘P07’*’s three parts, and the ‘r6445’ bound* (**the ‘r6445’ premise cannot be established ahead of the row itself: it asks what the coupling does to the sum, and *the open item IS the definition of that sum* — so ‘60’*’s table is the reach of the free-spectrum method and the remainder is not reachable by refining it. And ‘PO-23’ is the last of three named parts*, ‘P07’ saying two can now be said (the operator is bounded below; its spectrum is computable branch by branch at $\frac{1}{4}$ and $\frac{3}{4}$). *Computes:* the triple enumerated with two said and one open; that the premise and the open item name the same object; and that the tempting re-scoping is the one already reversed *Bound:* *NOT a re-scoping to “not CR-specific”* — that was done at ‘r1243’, found wrong, *restored at ‘r1291’ and Daryl-flagged*, and is not re-made. The remainder is a genuine CR open and room (1) does not close *Run:* python3 receipts/P10_canonical.time/P10_the_remainder_is_the_last_of_three_and_is_not_reachable_by_refining

P10R8 P10_the_weyl_squared_coefficient_is_twice_a_scalar [OK]

‘PO-43’; ‘sec:lock’ (**the Weyl-squared coefficient is $1/60$ in units of $1/(4\pi)^2$ — exactly TWICE a real minimally coupled scalar’s — because ‘P10’*’s own description of the tower*

(mass a^3 , frequency μ_n/a) matches two such scalars term for term, and because the C^2 coefficient is a property of the field content rather than the background). *Computes:* the bridging identity $2\{\mathrm{Riem}\}^2 - 2\{\mathrm{Ric}\}^2 = 3C^2 - E_4$ verified ****algebraically on 200 random Riemann tensors**** to 2×10^{-15} , with the Weyl trace checked to vanish; and a cross-check from the standard anomaly formula $c = (N_0 + 6N_{1/2} + 12N_1)/120$ at $N_0 = 2$, sharing the physics and none of the algebra *Bound:* ****IMPORTED:** Gilkey's a_2 , cited not derived. **SCOPED:** this is the PHYSICAL-MODE count ****** — a covariant spin-2 computation gauge-fixes and carries ghosts and is a different object; ‘P10’ solves the constraints rather than gauge-fixing them, which is an argument from what the construction does and not a theorem, and is where this is attackable. The overall normalisation is convention-dependent; the convention-free content is the **RATIO**, exactly two *Run:* `python3 receipts/P10_canonical_time/P10_the_weyl_squared_coef`

P10R9 `LIFT_adiabatic_correction` [OK]

eq:adiabatic-exponent (THE ADIABATIC CORRECTION. Exact Euclidean suppression for mode n is $\exp(-\int \omega_n ds)$, with $\omega_n = \mu_n / a$ (P10 sec:lock) and $a = |r|$ along the lift. ω_n DIVERGES at the branch point $(a, 0)$. Does the exponent converge? And is the evolution adiabatic $(|dw/ds|/w^2 \ll 1)$?). *Computes:* runs clean; registered from its own docstring and a live run *Bound:* description is the receipt's own wording, condensed; not independently re-derived here *Run:* `python3 receipts/P10_canonical_time/LIFT_adiabatic_correction.py`

P10R10 `LIFT_gravitational_action` [OK]

eq:grav-action (THE CALIBRATION. On-shell $S = \int p_a da$ with $p_a = (3/4\pi) a \dot{a}$ (from P10's H). Continuing $\tau = i s$: $S = -i S_E$ with $S_E^{\text{grav}} = (3/4\pi) \int r'^2 ds$ along the SAME trajectory. Compare with the contour action $S_E^c = \int [(1/2) r'^2 + f/2] ds = 1.489$ ($\alpha=1$).). *Computes:* runs clean; registered from its own docstring and a live run *Bound:* description is the receipt's own wording, condensed; not independently re-derived here *Run:* `python3 receipts/P10_canonical_time/LIFT_gravitational_action.py`

P10R11 `LIFT_instanton_action` [OK]

eq:euclidean-action (FAMILY 10, final item: does the lift have a variational principle of its own? Claim: $S_E = \int ds [(1/2) r'^2 + V(r)]$, $V = f/2$, with Euclidean first integral $(1/2)r'^2 - V = -1/2$. Test the closed form $r(s) = -(2 M a^2)^{1/3} |\sin(3s/2a)|^{2/3}$ against BOTH.). *Computes:* runs clean; registered from its own docstring and a live run *Bound:* description is the receipt's own wording, condensed; not independently re-derived here *Run:* `python3 receipts/P10_canonical_time/LIFT_instanton`

P10R12 `P10_the_straddle_does_not_need_a_floor` [OK]

sec:lock (****THE STRADDLE DOES NOT NEED A FLOOR, AND THE PARAGRAPH'S TWO STATEMENTS ABOUT $\hat{\Gamma}$ ARE ABOUT DIFFERENT OPERATORS.**** The coupled sector promotes the boundary coefficient from the c-number $\gamma \mapsto 1/4$ to an operator $\hat{\Gamma}$ on the tower. ****The paragraph asserted it 'bounded below by γ ' where it builds the straddle, and listed 'whether it is bounded below' as OPEN four hundred characters later.**** Both are right of different operators: the LEADING-ORDER $\hat{\Gamma} = \gamma + c \sum_n \pi_n^2$ is a sum of squares over a positive coefficient, hence spectrum in $[\gamma, \infty)$; the COMPLETE one carries the cubic and higher self-interactions, which enter at the same inverse-square order and are not sign-definite. **** AND THE REPAIR COSTS THE ARGUMENT NOTHING, WHICH IS THE POINT:** the direct-integral decomposition of $-d^2/dx^2 + \hat{\Gamma}/x^2$ uses only that the spectrum MEETS BOTH SIDES of $3/4$ – limit point at or above, limit circle below – and a spectral projection onto $\{\hat{\Gamma} \leq 3/4\}$ exists whether or not the spectrum has a floor.** Exhibited on a two-channel model in which the cubic takes one channel: the union is unbounded below and both spectral subspaces stay non-trivial.). *Computes:* the leading-order infimum asserted equal to γ and the operator asserted monotone in k ; the complete model asserted to break the floor and asserted driven down at large momentum; both spectral subspaces asserted non-trivial in both cases *Bound:* ****bound:** the model is a diagonal two-channel stand-in for the singular part of the interacting graviton Hamiltonian, not that Hamiltonian. It establishes what the decomposition NEEDS, not what the spectrum IS – and the spectrum is exactly the frontier the paragraph names ('L-165') *Run:* `python3 receipts/P10_canonical_time/P10_the_straddle_does_not_need_a_floor.py`

P10R13 `P10_the_straddle_is_computed` [ALL PASS]

sec:lock (THE STRADDLE AS A COMPUTED FACT: $\text{spec}(\hat{\Gamma}) = [\gamma, \infty)$ from

the displayed form, and $\gamma=1/4;3/4$ puts spectrum strictly below the threshold while unboundedness puts it strictly above, so both sides are occupied for every admissible γ ; the $3/4$ exhibited by solving the indicial equation rather than quoted). *Computes:* r3803 *Bound:* 59 *Run:* python3 receipts/P10.canonical_time/P10.the_straddle_is_computed.py

P10R14 P10_the_adiabatic_residual_at_low_n_is_bounded_by_the_towers_own_floor [ALL PASS]

sec:dissolution (**THE ADIABATIC RESIDUAL AT LOW n IS BOUNDED BY THE TOWER'S OWN FLOOR, AND IT REACHES NOTHING OBSERVABLE.** The suppression falls monotonically with harmonic index (2.8e-4 at $n=2$, 5.9e-6 at $n=3$, ~ 2 orders per step after), so the mixing the adiabatic projection neglects can only carry amplitude into MORE-suppressed modes; and the S^3 tensor tower has no mode below $n=2$, so the least-suppressed mode has no source beneath it. **The bound is therefore one-sided by construction rather than by estimate.** Worst-case transfer of the whole $O(\epsilon)$ fraction out of $n=2$ leaves that mode MORE suppressed and does not raise the ceiling. Ceiling on transmitted power = unmixed $n=2 = 7.9e-8$, and below $1e-3$ even on the naive exponent.). *Computes:* r4231 *Bound:* 61 *Run:* python3 receipts/P10.canonical_time/P10.the_adiabatic_residual_at_low_n_is_bounded_by_the_towers_own_floor.

P10R15 P10_gamma_hat_is_bounded_below [OK]

sec:lock (**THE COMPLETE BOUNDARY OPERATOR IS BOUNDED BELOW, AND THE REASON IS THE DEPARAMETRIZATION ITSELF: THE SCALE FACTOR IS THE SUPERMETRIC'S ONE NEGATIVE DIRECTION.** P10 lists 'the spectrum of Gamma-hat, including whether it is bounded below' among what is open in the interacting tower; that clause is separable and is settled here. **The DeWitt form $\text{tr}((g \pi)^2) - \lambda \text{tr} g \pi^2$ has exactly ONE negative eigenvalue on the round metric, realised by the identity direction — the conformal one, which is the whole indefiniteness of superspace and the reason Wheeler-DeWitt is hyperbolic.** On g -TRACELESS momenta the trace term drops and the form is exactly $\text{tr}((g \pi)^2)$, which for L the Cholesky factor of g equals $\text{tr}(S^2)$ with $S = L^T \pi L$ SYMMETRIC — a sum of squares of real eigenvalues, zero only at $\pi = 0$. **Positive definite for every non-degenerate spatial metric, verified as an IDENTITY on 400 random pairs rather than as a sample.** **So the cubic term's apparent unboundedness is a TRUNCATION ARTEFACT: $\pi^2(1 - \lambda \phi + \dots)$ is the expansion of $\pi^2/(1 + \lambda \phi)$, and the full coefficient is positive wherever the spatial metric is non-degenerate.** The deparametrization that produces a true Hamiltonian is the same move that produces a bounded-below one — both are the removal of the conformal direction, and P10 argues the first without noticing it had the second.). *Computes:* the signature asserted one negative; the Cholesky identity and positivity asserted on 400 valid (g, π) pairs; min Q asserted positive on three metrics including a 625:1 anisotropy; the truncation asserted to go negative where the full expression does not *Bound:* **bound: positivity holds on NON-DEGENERATE spatial metrics. At the degenerate boundary of superspace the inverse metric blows up and the argument says nothing — which is exactly the $a \rightarrow 0$ boundary, handled instead by the thermal condition. Two different jobs, neither substituting. NOT TOUCHED: the UV definition of the tower sums, and the closed-form nonlinear solution** *Run:* python3 receipts/P10.canonical_time/P10.gamma_hat_is_bounded_below.py

P10R16 P10_the_towers_zeta_at_zero_is_ten_and_the_claim_needs_its_scoping [PASS]

sec:lock (**THE TENSOR TOWER'S SPECTRAL ZETA(0) IS 10, SO THE MODE SUMS SPEND ONE DIMENSIONLESS CONSTANT AND THE NO-FREE-CONSTANT CLAIM NEEDS THE SCOPING ITS OWN SENTENCE NAMES.** CR_synthesis sec:ledger turned the claim into a prediction about a computable number — 'that coefficient vanishes' — and it does not. On P10 sec:lock's own spectrum ($\mu_n^2 = n(n+2)-2$, $n_0=2$, degeneracy $2(n-1)(n+3)$) the free static tower gives $\zeta(0) = 10$ EXACTLY, and the zero-point sum's log-scale coefficient $\text{Res}_{\{s=-1\}} \zeta = 39/4$. Non-zero on either reading of which spectral quantity sec:ledger meant, so the verdict does not turn on that ambiguity. AND THE PAPER NAMES THE COST ITSELF: the quartic leader is absorbed into Λ , but the logarithmic successor goes with curvature-squared invariants and 'a curvature-squared coupling is not an entry in this framework's ledger'). *Computes:* the spectral zeta of the deparametrized TT graviton tower: $\zeta(0)$ exactly (the asymptotic expansion terminates at $s=0$, so the continuation has no remainder), the residue at $s=-1$ by two independent routes, and a hard-cutoff cross-check with no zeta function in it *Bound:* FREE and STATIC: the tower's couplings to the scale-factor sector are not in it and the instantaneous spectrum is used, so a coupled or non-adiabatic computation could still return zero — what is excluded is the

claim holding trivially, on the most favourable case (physical TT modes only, no ghosts, constraints already solved). Does not touch essential self-adjointness of the coupled system, and does not disturb the geometric ledger: c , G , $\hbar$ as unit gauges over Λ are untouched; what is measured is that the QUANTUM sector's mode sums spend one constant. *Run:* `python3 receipts/P10_canonical.time/P10_the_towers.zeta.at.zero.is.ten.and.the.claim.needs.its.scoping.py`

P10R17 `P10_no.rescaling.discharges.the.log.and.the.one.mass.shift.that.does.is.the.curvature.offset` [PASS]

sec:lock (**THE COUPLED ATTEMPT RETURNS 39/4 UNCHANGED IN KIND: NO RESCALING OF THE TOWER'S FREQUENCIES CAN DISCHARGE THE LOG, AND THE ONE MASS-LIKE SHIFT THAT DOES IS EXACTLY THE THREE-SPHERE CURVATURE OFFSET.** The log coefficient is the $1/m$ term of $d(m)\mu(m)$, an exact functional of the spectrum, so which deformations can move it is answerable without perturbation theory in a coupling nobody has. Overall rescale gives $L = (39/4)\sqrt{1+e}$, zero only where every frequency is; mass shift gives $L = -(d-3)(d+13)/4$, zero at $d = 3$ and -13 ; a $1/m^2$ tail at $e = -39/4$; linear-in- m at $e = +2\sqrt{65}/5$. NO RESCALING DISCHARGES IT, which generalises r6411 from the scale factor to any multiplicative renormalisation – the class the leading back-reaction lives in, since the coupling acts through a and through $\Gamma\text{-hat} = \gamma + c \sum(\pi_n^2)$ whose expectation is $(a^2/2)\sum(d_n \mu_n)$, the same sum rescaled. At $d = 3$ the summand is $2m^3 - 8m$, a POLYNOMIAL: the $1/m$ term does not cancel, it does not exist – and $+3$ is exactly the curvature offset that makes the frequencies non-integer.). *Computes:* the exact $1/m$ coefficient of $d(m)\mu(m)$ over four one-parameter deformations of the spectrum, symbolically, plus a cancellation-free numerical confirmation summed term by term *Bound:* NOT a solution of the coupled sector and does not claim to be: it states which deformations COULD discharge the row and observes that the leading back-reaction is not among them. Does not compute the coupling's induced shift; if a higher-order effect supplies a mass-like delta this says what it would have to equal, not that it cannot. The non-adiabatic half is untouched. Does not disturb r6435's counterterm reading: on this conformally flat background the log's counterterm is degenerate with Einstein-Hilbert and the cosmological term, so a non-vanishing L is not a new ledger entry here – the result narrows the row rather than reopening it. *Run:* `python3 receipts/P10_canonical.time/P10_no.rescaling.discharges.the.log.and.the.one.mass.shift.that.does.is`

P10R18 `P10_the_floor.is.forced.as.a.mode.but.the.subtraction.point.is.a.convention.and.the.residue.is.t` [OK]

'PO-51'; 'sec:lock's log, and 'P17' 'sec:ledger' (**THE FLOOR IS FORCED AS A MODE AND THE SUBTRACTION POINT IS A CONVENTION — AND THOSE ARE DIFFERENT STATEMENTS THE ROW'S PREMISE RUNS TOGETHER.** *Forced:* $m=3$ is the lowest transverse-traceless harmonic the three-sphere admits — $j_R=(m-3)/2$ vanishes exactly there and is negative below, the degeneracy $2(m^2-4)$ being zero at $m=2$ — and the tower is **gapped at $\mu_3=\sqrt{6}$ with no zero mode and no soft region**, so the infrared is regulated by the geometry. *And the log is real and scheme-independent*: $d\mu=2m^3-11m+\frac{39}{4}m+\frac{45}{8}m^3+\dots$, and the Dirichlet series $\sum d(m)\mu(m)m^{-s}$ has a **simple pole at $s=0$ with residue exactly $39/4$ ** (with $s=4$ and $s=2$ carrying the quartic and quadratic at 2 and -11), ***so zeta regularisation does not evade a subtraction point***. $\Rightarrow$ $39/4$ does NOT move with that point — identical across $m_0=3,5,10,50$ to the summation's own rounding, 25 orders below the truncation error — while the finite part shifts by **exactly $\frac{39}{4}\ln(m_0'/m_0)$ ** and nothing else. ***BUT THE DECIDING TEST COMES OUT AGAINST, IN TWO STEPS.*** *First, **a mode number is dimensionless**, so $\ln(M/m_0)$ introduces no length for ANY m_0 and the one-scale claim — the row's whole motivation — protects every choice equally and **cannot single out the floor**. Second, on the admitted one-parameter family 'sec:lock's own argument makes the counterterm basis **one-dimensional**, so the log's counterterm is degenerate with the cosmological term on the tower's own background ($R=12/\alpha^2$, verified), and 'P17' absorbs a constant vacuum energy into the one observed curvature with no bare-versus-vacuum split — **so no observable depends on m_0 **.* $\Rightarrow$ *And the floor fixes the constant to nothing: subtracting there leaves $C=-8.51485690643\dots$, neither zero nor named.* ***THE ANSWER TO THE BURDEN IS "CONVENIENT", AND THE ROW CLOSES***). *Computes:* the degeneracy and eigenvalue against 'sec:lock's own; $j_R\geq 0$ at the floor; the asymptotic expansion and $L=39/4$ symbolically; the analytic continuation's three poles and residues

to 10^{-5} ; $\mathcal{L}$ extracted at four subtraction points and three cuts; the finite-part shift to 10^{-12} ; the Ricci scalar of $\alpha \cosh(T/\alpha)$ derived as $12/\alpha^2$; and $\mathcal{C}$ to 16 digits — **19** gates with **THREE** controls that break as they must: a mass shift **MOVES** $\mathcal{L}$ (so the invariance is not a tautology), a soft floor diverges in the infrared (so the gap is this spectrum's), and at $\mu^2 = m^2$ the product is a polynomial with **no** $1/m$ term at all (so the machinery detects the log rather than manufacturing it) *Bound*: **not** that the one-scale claim fails — the opposite for the part that matters, a discrete sum needing no second **LENGTH** for any subtraction point; what fails is the further step that the floor is thereby determined. **Nothing** here uses the coupled tower: every number is the free spectrum's, 'PO-23's ultraviolet definition of the sums is not reached, and a subtraction point for a log is not a definition of the sum. Where the shear breaks the counterterm degeneracy the Weyl-squared coefficient is a separate entry whose own condition is 'PO-43's uncosted one — **named**, not **claimed**. And $39/4$ is not discharged; 'r6436' stands *Run*: `python3 receipts/P10_canonical_time/P10_the_floor_is_forced_as_a_mode_but_the_subtraction_point_is_a_conven`

P10R19 `I9_the_adiabatic_parameter_diverges_where_the_action_integral_converges` [OK]
sec:lock (I9 — THE ADIABATIC PARAMETER DIVERGES WHERE THE ACTION INTEGRAL CONVERGES. Integrable-systems field bake, probe I9. On P10's own near-branch-point scaling $\text{abs}(r) \sim s^{2/3}$, the adiabaticity parameter $\text{abs}(dw/ds)/w^2$ goes as $s^{-1/3}$ and DIVERGES, so the adiabatic invariant E/ω is not conserved through the branch point; the suppression exponent $\int \omega ds$ nevertheless converges as $S^{1/3}$, because $\omega \sim s^{-2/3}$ is integrable. Two different powers, two different questions: the finiteness P10 verifies comes from the action integral and not from slow variation, so the correction is semiclassical — which is what P15 calls the same object. Two adversarial controls, one for each half.). *Computes*: runs clean; eight verdicts including controls that fire in opposite directions *Bound*: **COMPUTES**: the scaling $\text{abs}(r) \sim s^{2/3}$ is P10's own; A and μ carried symbolically, no value pinned *Run*: `python3 receipts/P10_canonical_time/I9_the_adiabatic_parameter_diverges_where_the_action_integral_converge`

P10R20 `P10_gamma_is_one_quarter_and_is_the_maximum` [OK]
sec:lock ($\gamma = 1/4$ is the **MAXIMUM** of the natural ordering family, so (1,1) holds for every ordering). *Computes*: reduces T_s exactly, shows the first-derivative term vanishes, and gets $\Gamma(s)$ with its maximum; 11 checks *Bound*: **NOT-A-PAPER-CLAIM** — verification of a standing claim; node 17 withdrew its own caveat *Run*: `python3 receipts/P10_canonical_time/P10_gamma_is-on`

D1 `the_boundary_is_per_fibre_and_the_UV_is_over_fibres` [OK]
sec:lock (PO-6 re-weighted not closed: the boundary closure is **PER-FIBRE** and the UV is **OVER-FIBRES**, and the clause c54.129 answered is the one the argument does not need). *Computes*: asserts the direct-integral structure at source, computes the per-fibre deficiency across the threshold, and checks the straddle under an unbounded-below operator (18 checks) *Bound*: **NOT-A-PAPER-CLAIM** — a bounded re-statement of a **PROTECTED** row; PO-6 stays open *Run*: `python3 receipts/L165_interacting_tower/D1_the_boundary_is_per_fibre_and_the_UV_is_over_fibres.py`

D2 `the_UV_degree_is_quartic_and_the_IR_is_free` [OK]
sec:lock (PO-6s UV clause measured: the tower sum diverges **QUARTICALLY**, the ordinary zero-point degree, and compactness buys the IR rather than the UV). *Computes*: takes the tower and mode action from P10 at source, derives the linear growth of μ_n and $j_{pi,n}^2$, and tracks the partial sums against $N^{4/4}$ (13 checks) *Bound*: **NOT-A-PAPER-CLAIM** — a measurement of one PO-6 clause; PO-6 stays open *Run*: `python3 receipts/L165_interacting_tower/D2_the_UV_degree_is_quartic_and`

S1 `the_subleading_log_divergence_is_gauss_bonnet_and_the_ledger_survives` [OK]
— (PO-6 fixed-background half: the **SUB-LEADING** (log) one-loop divergence on the tower's own de Sitter slicing is the Gauss-Bonnet term (topological in $D=4$), so it forces **NO** curvature-squared coupling outside the one-constant ledger — extending L-809's quartic result to the log order; on de Sitter $\text{Weyl}^2=0$, and the on-shell graviton divergence reduces to GB (t Hooft-Veltman/Christensen-Duff)). *Computes*: verifies $R^2/\text{Ric}^2/\text{Riem}^2=144/36/24$ per α^4 (S50) and $\text{Weyl}^2=0$, that $\text{GB}=24/\alpha^4$ is the topological Euler density, the on-shell $R=4\Lambda$, and the ledger-survival conclusion (4 checks) *Bound*: **NOT-A-PAPER-CLAIM** — the fixed-background half of PO-6; extends L-809; does **NOT** touch the coupled-sector dark half (54's/definitional) *Run*: `python3 receipts/L816_po6_subleading_survives/S1_the_subleading_log_divergence_is_gauss_bonnet_and`

S1.the_one_constant_ledger_survives_on_the_running_layer_not_only_fixed_de_sitter [OK]

sec:lock (PO-6 RUNNING-LAYER half (the residue S50 named: does the one-constant basis survive on a background whose curvature RUNS?): on $a(T)=\sinh^{\{2/3\}}(3T/2\alpha)$, where R runs ($R\sim 12/\alpha^2$ only asymptotically), the pure-graviton log divergence reduces to {Gauss-Bonnet (topological) + Lambda-renorm + Newton/ell_P-renorm + field-equation-proportional (field-redef removable)}) with NO irreducible curvature-squared coupling outside {Lambda,G} — for ANY one-loop coefficients. Extends L-816 off fixed de Sitter, via background-independent algebraic identities (not $R=\text{const}$)). *Computes:* confirms $\text{Weyl}^2=0$ and R runs on the $\sinh^{\{2/3\}}$ layer, the topological $(53/90)E_{\text{GB}}$ split, the $\text{Lambda}=0$ identity $(7/20)\text{Ric}^2+(1/120)R^2=E_{\text{mn}}f^{\text{mn}}$ exactly, the coefficient-independent reduction of $a_1\text{Riem}^2+a_2\text{Ric}^2+a_3R^2$ to $\{E_{\text{GB}},\text{Lam}^2,\text{Lam}R,\text{EOM}\}$ with remainder 0, and the vacuum-dS consistency with L-816 (6 checks) *Bound:* NOT-A-PAPER-CLAIM — the running-layer half of PO-6; F5 reserves the verdict; pure graviton (matter as bend), does NOT enter the quantised-background sector or claim one-loop finiteness *Run:* python3 receipts/L818_po6_running_layer/S1.the_one_constant_ledger_survives_on_the_running_layer_not_only_f

S1.the_ledgers_one_irreducible_survivor_is_the_conformal_coupling_that_turns_on_at_the_shear [OK]

sec:lock (PO-6: L-818's boundary LOCATED with a coefficient (answers 56's r2764 routing). The graviton log counterterm $= (7/40)\text{Weyl}^2 + (149/360)\text{GB} + (1/8)R^2$; GB is topological and R^2 reduces (L-818), so the ONE irreducible piece is the conformal $(7/40)\text{Weyl}^2$ — it is 0 on the shear-free FRW layer (which is WHY L-818's remainder was exactly 0 and the ledger survived) and turns on at 2nd order in the shear, where $\text{Weyl}^2=4\sigma^2$ (56 S10/r2743). The shear IS the TT-graviton tower = P10's open interacting tower; so L-818's caveat, r2743's shear and P10's open item are ONE boundary, now carrying $(7/40)\text{Weyl}^2$. *Computes:* decomposes the graviton b_4 into $\{\text{Weyl}^2,\text{GB},R^2\}$ (Weyl^2 coeff 7/40), shows only Weyl^2 is neither topological nor ledger-reducible, and computes $\text{Weyl}^2=0$ on the isotropic layer and $O(\text{shear}^2)$ on an axisymmetric anisotropic model (no linear term) (5 checks) *Bound:* NOT-A-PAPER-CLAIM — locates L-818's boundary and names the open coupling; F5 reserves the verdict; does NOT close the interacting tower (it names it); coefficients cited (t Hooft-Veltman) *Run:* python3 receipts/L819_shear_boundary_coupling/S1.the_ledgers_one_irreducible_survivor_is_the_conformal_cou

S1.the_weyl_magnitude_is_the_graviton_c_anomaly_and_the_parity_odd_partner_is_a_theta_term [OK]

sec:lock (PO-6 OWED #472: the VALUE of the one shear counterterm (Weyl^2 , c54.219) is 7/40 — the graviton's conformal-anomaly c-coefficient (b_4 in the $\{\text{Weyl}^2,\text{GB},R^2\}$ basis). And its parity-odd partner, the Pontryagin density $R\tilde{R}$ that a CIRCULAR polarisation reveals (nonzero, handedness-odd; a linear mode gives zero), is a topological theta-term at constant coefficient (Jackiw-Pi C-tensor $\text{prop nabla}(\text{coeff})=0$; no D=4 gravitational anomaly), so it does NOT enter the field equations — the dynamical count stays ONE). *Computes:* computes $R\tilde{R}$ on linear vs circular TT waves by a finite-difference Riemann (0 / +4.5 / -4.5, parity-odd), pins the b_4 Weyl^2 coefficient 7/40, and cites the topological/theta-term status (Jackiw-Pi, Alvarez-Gaume-Witten) (4 checks) *Bound:* NOT-A-PAPER-CLAIM — OWED #472, one value + one status on the fixed-background shear; the parity-odd loop coefficient is zero for the pure graviton in D=4; interacting tower untouched *Run:* python3 receipts/L821_weyl_coefficient_value/S1.the_weyl_magnitude_is_the_grav

S2.c1_and_c2_are_one_and_the_source_answers_more [OK]

sec:euclidean (C1 and C2 are ONE condition joined by so — the list is six not seven — and the same P10 passage states that the free TT sector is a harmonic oscillator whose Hamiltonian is bounded below, which PO-6 asks for). *Computes:* asserts the joined sentence verbatim, confirms PO-6s row asks whether it is bounded below and does not carry the answer, and quotes the confinement clause (11 checks) *Bound:* NOT-A-PAPER-CLAIM — corrects a count and surfaces a partial answer; joint satisfiability untested *Run:* python3 receipts/L165_defining_the_sum/S2.c1_and_c2_are_one_and_the_sou

S3.c6_is_a_theorem_not_a_condition [OK]

sec:lock (the C6/C7 tension cannot arise: the per-fibre structure is DERIVED from a commutation relation, not a requirement on a measure — so the conditions list is five plus one theorem). *Computes:* asserts D1s commutation, direct-integral and fibre-by-fibre statements verbatim, and S1s misclassification of C6 as a condition (8 checks) *Bound:* NOT-A-PAPER-CLAIM — reclassifies a list entry; joint satisfiability untested *Run:* python3 receipts/L165_defining_the_sum/S3.c6_is_a_theorem_not.

S4.the_open_half_is_the_floor [OK]

sec:lock (PO-6s open half is the FLOOR: the direct-integral decomposition needs only that both sides of the threshold are occupied, and P10 leaves whether the complete Gamma-hat is bounded below open in its own voice). *Computes:* asserts P10s uses-only clause, its leaves-open clause, the sum-of-squares form and the structural commutation verbatim, and the PO-6 rows own wording (7 checks) *Bound:* NOT-A-PAPER-CLAIM — narrows a register row and corrects r2611s caveat *Run:* python3 receipts/L165_defining_the_sum/S4.the_open_half_is_the_floor.py

D3.the_floor_does_not_survive_the_cubic [OK]

sec:lock (PO-6s floor question answered NO at the order the paper names: the cubic enters as $\pi^2 \phi$, a square times a SIGNED field, so Gamma-hat is unbounded below wherever $\phi \neq -c/g^3$). *Computes:* asserts P10s leaves-open clause, its leading-order form and its naming of the cubic verbatim, and computes the sign structure and divergence (8 checks) *Bound:* NOT-A-PAPER-CLAIM — answers a narrowed dark half; the state question is untouched *Run:* python3 receipts/L165_interacting_tower/D3.the_floor_does_not_survive_the_cubic.py

D4.the_state_fixer_has_its_own_threshold [OK]

sec:lock (the state-fixing mechanism has its own threshold: $\nu = \sqrt{\Gamma + 1/4}$ is real only for $\Gamma \geq -1/4$, below which both exponents go complex and no regular branch exists — and r2651s operator reaches -47.8). *Computes:* asserts P10s thermal-regularity sentence and its common-to-every-fibre clause verbatim, computes ν across Gammas range, and recomputes r2651s minimum (11 checks) *Bound:* NOT-A-PAPER-CLAIM — sharpens PO-6s successor question; whether the region is realised is untouched *Run:* python3 receipts/L165_interacting_tower/D4.the_state_fixer_has_its_own_threshold.py

D50.the_floor_does_survive_and_the_paper_said_so_232_revisions_earlier [OK]

sec:deparam (**D3 (r2651) HAS THE FLOOR BACKWARDS — P10 CALLS THE CUBIC'S UNBOUNDEDNESS AN "ARTEFACT OF TRUNCATION" IN THE SAME PARAGRAPH, AND HAS SINCE r2419.** The cubic is the first-order term of $\pi^2/(1+\lambda\phi)$, whose full coefficient is positive on non-degenerate metrics — so the floor survives and r2652's sub- $(-1/4)$ question has no object. Lead 'L-542'; VEIN 'L-165'). *Computes:* verifies termwise to eighth order that the cubic IS the expansion's first-order term; samples both forms on 200,000 identical points (truncation ~ 892.6 , resummed never below γ); six source checks in P10 including the affirmative verdict and its interior-only SCOPE; dates the clause by reading r2419's own copy of the file; and re-runs 'P10_gamma_hat_is_bounded_below', which passes *Bound:* LATENT — no new physics; the answer was in the paper. NOT claimed: that 'PO-6' closes (the ultraviolet definition of the tower sums is open in the same sentence), or that the floor is proved for the full interacting theory. 'D3'/'D4' and the 'PO-6' row are NOT touched — 56's band and a protected row; routed *Run:* python3 receipts/L165_interacting_tower/D50.the_floor_does_survive_and_the_paper_said_so_232_revisions_earlier.py

S50.the_counterterm_basis_is_one_dimensional_because_the_background_family_is [OK]

sec:deparam (**PO-6's UV HALF: THE COUNTERTERM BASIS IS ONE-DIMENSIONAL BECAUSE THE ADMITTED BACKGROUND FAMILY IS.** Every curvature invariant on a maximally symmetric background is a pure power of $1/\alpha^2$, and the substrates form a one-parameter family, so a divergence of any degree needs exactly one counterterm — which the framework carries. Scoped: the tower lives on the layer, whose R runs. Lead 'L-543'; VEIN 'L-165'). *Computes:* **AMENDED c54.211 ('L-544'): c54.210's SCOPE IS WITHDRAWN.** It put P15's observable rate where P10's slicing belongs; P10's own $a(T) = \alpha \cosh(T/\alpha)$ has $R = 12/\alpha^2$ CONSTANT, so the degeneracy holds on the tower's own background and the result is STRONGER than claimed. The real limit is the coupled sector, where the scale factor is quantized and there is no fixed background. Both geometries computed side by side so the error stays checkable. — computes the quadratic invariants at $D=4,5,6$ and pins their α -exponent at exactly 4; five standing-sentence checks (p_0 's invariant fact, p_0 's foliation, 'L-533's scale-only descent); computes the $\sinh^2(2/3)$ layer's Ricci scalar and asserts it NON-constant with a $12H^2$ late limit; and counts 'counterterm' in P10 from git HEAD (0) against now (7) *Bound:* LATENT, and an ASSEMBLY not a discovery. NOT claimed: that 'PO-6' closes, that the sub-leading divergences vanish, that this is renormalisability, or that the layer breaks it — **that is the question** *Run:* python3 receipts/L165_defining_the_sum/S50.the_counterterm_basis_is_one_dimensional_because_the_background_family_is.py

S7.the_successor_was_misaimed [OK]

sec:tower (L-543 INHERITS r2677s WITHDRAWN OBJECT: it asks about a RUNNING background and the free tower sits on P10s own slicing at $R = 12/\alpha^2$, CONSTANT — and the

layers late limit IS that value, so it asymptotes to it). *Computes:* computes both Ricci scalars, shows the layers late limit equals P10s constant under $H \rightarrow 1/\alpha$, and pins the successors wording and the back-reaction clause from the row (6 checks) *Bound:* NOT-A-PAPER-CLAIM — withdraws a successor this line registered; the back-reaction question stands *Run:* python3 receipts/L165_interacting_tower/S7_the_successor_was_misaimed.py

S8.the_floor_follows [OK]

sec:lock (PO-6 HAS BEEN CARRYING THE WRONG QUESTION: the complete Gamma-hat inherits its floor from the same non-degeneracy r2671 established, so boundedness is answered YES — what remains is WHERE the spectrum sits relative to $3/4$). *Computes:* asserts P10s promotion, inverse-square and Weyl-alternative clauses verbatim, quotes the straddle receipts does-not-need-a-lower-bound line, and shows the truncated coefficient goes negative where the full one stays positive (7 checks) *Bound:* NOT-A-PAPER-CLAIM — the spectrum is untouched; P10 explicitly declines the floor and the decomposition survives without it *Run:* python3 receipts/L165_interacting_tower/S8_the_floor_follows.py

S9.the_ordering_decides_it [OK]

sec:lock (PO-6s SPECTRAL QUESTION IS DECIDED BY OPERATOR ORDERING, which P10 never names: normal-ordered puts the vacuum at $1/4$ (BELOW $3/4$, freedom survives), symmetric puts $N=1$ exactly AT $3/4$ — and $3/4 = 1/4 + 1/2$, the free coefficient plus one zero-point quantum). *Computes:* computes both ordering branches against the threshold, verifies the $1/4+1/2$ identity, counts normal-order and symmetric-order at zero in the paper, and pins the both-sides-occupied clause (9 checks) *Bound:* NOT-A-PAPER-CLAIM — neither ordering is asserted correct; P10s decomposition survives either *Run:* python3 receipts/L165_interacting_tower/S9_the_ordering

Q1.the_degeneracy_is_conformal_flatness_not_maximal_symmetry_so_no_scale_factor_can_break_it

[OK]

sec:tower (PO-6's DARK HALF HAS AN OBJECT: r2677 stated TWO degeneracies as one. The quadratic trio collapses because the deficit is exactly Weyl^2 and EVERY FRW is conformally flat for EVERY $a(T)$ — not because of maximal symmetry — so no scale factor can break it and back-reaction cannot; the different-dimension half does fail. The real limit is the SHEAR). *Computes:* R , Ric^2 and Riem^2 computed from the FRW metric with $a(T)$ a free function at $k = +1, 0, -1$ (Weyl^2 identically zero); the Gauss-Bonnet density shown an exact total derivative, $\sqrt{g}(R^2 - 4\text{Ric}^2 + \text{Riem}^2) = d/dT[24(a'^3/3 + a')]$; the two relations solved to give $\text{int Ric}^2 = A/3 - \chi/2$ and $\text{int Riem}^2 = A/3 - \chi$; S50's $D=4$ row reproduced from the metric (144/36/24); P15's running layer recomputed from its own metric with the deficit zero by a second route in $u = \exp(3Ht/2)$; and Bianchi I giving $\text{Weyl}^2 = 4\sigma^2 + O(\sigma^4)$. CONTROLS: SdS gives $\text{Weyl}^2 = 48M^2/r^6$ so the identity is not vacuous; the layer's R^2 is t -dependent so the different-dimension degeneracy really does fail; and the numeric residual is a PRECISION-SCALING test (3.06e-37 at 40 digits, 2.24e-77 at 80) rather than a fitted threshold *Bound:* not that the coupled sector is renormalizable — this says which functionals a divergence can NEED, not that they are absorbable; no heat-kernel coefficient computed; Bianchi I is a HOMOGENEOUS shear, fixing the ORDER at which conformal flatness fails and not the mode-by-mode statement on the tower; and not a closure — PO-6 stays open *Run:* python3 receipts/L549_coupled_counterterms/Q1.the_degeneracy_is_conformal_flatness_not_maximal_symmetry_so

S10.the_halves_meet_at_the_shear [OK]

sec:lock (PO-6s TWO DECLARED HALVES MEET AT THE SHEAR: the counterterm half ends where conformal flatness ends ($C^2 = 4\sigma^2 + O(\sigma^4)$) and P10 names the tower as the TRANSVERSE-TRACELESS graviton modes — which ARE the shear). *Computes:* checks C^2 vanishes at $\sigma=0$, is quadratic-leading and non-zero away from it, confirms the source is trace-free, and quotes P10s tower identification twice (6 checks) *Bound:* NOT-A-PAPER-CLAIM — the second-order basis is not computed; cc54 derived the coefficient and its structure is checked here *Run:* python3 receipts/L165_interacting_tower/S10_the_halves_meet_at_the_shear.py

S11.the_ordering_question_dissolves [OK]

sec:lock (PO-6s ORDERING QUESTION DISSOLVES: P10 gives deficiency indices (1,1) INDEPENDENTLY of ordering with the coefficient $j = 1/4$ across the natural family, closes the residual via the horizons Hartle-Hawking state, and answers the coupled straddle fibre by fibre). *Computes:* pins all six of P10s own statements — the independence, the family bound, the non-uniqueness, the

Friedrichs selection, the fibre-by-fibre condition, and what remains open (6 checks) *Bound*: NOT-A-PAPER-CLAIM — the interacting tower is still open; the correction is to WHICH problem the row owes *Run*: `python3 receipts/L165_interacting_tower/S11_the_ordering_question_dissolves.py`

P10R21 `P10_ordering_selection_is_external` [OK]

sec:lock / *P7 frontier:quantum* (**PO-15 THE ORDERING EXHAUSTION: nothing internal to the construction selects the operator ordering (min Γ -hat = $1/4$ normal vs $3/4$ symmetric), and the reason is that selecting it would be solving the cosmological-constant problem.** Completes S11 (boundary CLOSURE is ordering-independent) and the r3014 thermal-state elimination (the thermal state acts downstream). ** THE PIVOT: ** the ordering is BULK-INERT (normal vs symmetric differ by the c-number zero-point $1/2 \hbar \omega_n$, a global phase under the deparametrized evolution) and BOUNDARY-PHYSICAL (the difference surfaces only in the horizon coefficient, $1/4 - i 3/4 = 1/4 + 1/2$, exactly one zero-point quantum), so "which ordering" IS "does the tower's zero-point energy gravitate at the horizon". ** THE EXHAUSTION: ** (1) the thermal state and (3) the seam's characteristic structure act DOWNSTREAM (they close the extension GIVEN Γ -hat; a regular branch $x^{(1/2+\nu)}$ exists for BOTH orderings); (2) the substrate isometry, (5) positivity, (6) the single-scale ledger and (7) covariance under configuration redefinition each respect BOTH orderings; (4) the deparametrization makes normal ordering AVAILABLE (a preferred vacuum) but not mandatory, and by SOLVING the constraint removes the anomaly-freedom lever a Wheeler-DeWitt quantization would have used. The two external naturalness heuristics DISAGREE (continuity-with-free $-i 1/4$, predictivity $-i 3/4$). ** RESULT: ** the choice is EXTERNAL because it IS the cc problem reached from inside – an epistemic gap (one datum: does vacuum energy gravitate), NOT an ontological family, so admissible under the epistemic reading.). *Computes*: the ordering difference asserted = the zero-point c-number $1/2 \hbar \omega_n$; the c-number-is-a-global-phase fact asserted numerically (identical amplitude moduli to $1e-12$, ratio = $e^{-i\pi}$); the boundary shift $1/4 - i 3/4$ asserted to be exactly one zero-point quantum with $3/4$ the threshold; the regular branch $\nu = \sqrt{\Gamma+1/4}$ asserted real for BOTH orderings (thermal closure ordering-independent); both floors asserted finite; the seven-candidate enumeration stated with a structural disqualification each and no survivor *Bound*: PO-15 answered as the row asked – the choice is genuinely external, the enumeration is the evidence, the externality is the cc problem localized. NO ordering is picked and none is claimed derivable; the interacting-tower definition (PO-6) stays separately open; builds on S11/r3014, not a re-derivation of the Friedrichs selection *Run*: `python3 receipts/P10_canonical_time/P10_ordering_selection_is_external.py`

S12.the_boundaries_coincide [OK]

sec:lock (cc54s L-818 BOUNDARY, r2743s SHEAR FINDING AND P10s OPEN ITEM ARE ONE BOUNDARY: an FRW layer is shear-free by construction, the degeneracy ends at the shear, and P10 names the transverse-traceless tower as what remains open). *Computes*: verifies cc54s Step 1 ($\text{Weyl}^2 = 0$ with R running, all three k) and pins P10s two namings of the tower (5 checks) *Bound*: NOT-A-PAPER-CLAIM — cc54s five sympy steps are not rerun; the coincidence is by identification, each link a corpus statement *Run*: `python3 receipts/L165_interacting_tower/S12_the_boundaries_coincide.py`

S13.the_ledger_does_not_reach_second_order [OK]

sec:lock (AT SECOND ORDER IN THE SHEAR THE QUADRATIC BASIS IS GENUINELY THREE-DIMENSIONAL: GB and C^2 span rank 2 of 3, C^2 vanishes identically shear-free so L-818 never had to route it, and it has NONE of L-818s three routes). *Computes*: computes the rank and nullspace of $\{GB, C^2\}$, confirms $C^2 = 0$ at all three k on the shear-free layer, and pins p0s one-constant claim as a statement about the FACES (6 checks) *Bound*: BOUNDED NEGATIVE — one order of one expansion; not a verdict on the construction, and the coefficient is not computed *Run*: `python3 receipts/L165_interacting_tower/S13_the_ledger_does_not_reach_second_order.py`

S1.the_shear_needs_exactly_one_new_counterterm [OK]

sec:tower (PO-6's OWED SHEAR CALCULATION RUN: on a propagating TT mode C^2 is $O(\hbar^2)$ and carries FOUR derivatives, but $\delta^{(1)}R = 0$ makes $\int \sqrt{g} R^2$ a shift of Einstein-Hilbert – so the shear costs exactly ONE new counterterm, Weyl-squared, and c54.215's working count of two was one too many). *Computes*: R , Ric^2 , Riem^2 and C^2 built from the corpus's own L801/N1 TT-wave-on-de-Sitter metric and expanded to $O(\epsilon^2)$; the derivative content fixed by freezing the oscillatory factors and taking the total degree in $(\hbar, k, \omega)$, giving 4 against σ^2 's 2; $\delta^{(1)}R$ shown to vanish and $\det g$ shown ϵ -independent, hence R^2 at second order equals $2 \bar{R}$ times R at second order, pointwise; the span solved from

Gauss-Bonnet plus the Weyl definition, giving B and C both in span of A and C^2 ; and the Pontryagin density computed for linear, circular and opposite-handed polarisation by a finite-difference pipeline cross-checked against a symbolic evaluation to six digits. **CONTROLS:** the linear mode returns zero for the parity-odd invariant, which is what catches the incomplete basis; the handedness flips its sign; and at $\epsilon = 0$ everything reduces to c54.215's one dimension *Bound:* no heat-kernel COEFFICIENT is computed – this says which functionals a divergence can need, not with what coefficient; the mode is a plane-wave TT on flat-sliced de Sitter and not P10's S^3 harmonics, so the COUNT is mode-independent while the COEFFICIENT is not; the Pontryagin term's total-derivative status is cited, not computed; its consequences for the corpus's chirality result are not raised; and not a closure – PO-6 stays open *Run:* python3 receipts/L553_the_shear_counterterm/S1_the_shear_needs_exactly_one_new_counterterm.py

D1.the_degeneracy_carrying_the_quartic_was_never_derived_and_its_constant_is_the_component_count [OK]

sec:tower (THE DEGENERACY CARRYING PO-6's QUARTIC WAS NEVER DERIVED: asserted in P10 and asserted-unchecked in D2. Derived from Peter-Weyl on the parallelizable S^3 it is $g(n) = 2(n-1)(n+3)$, $n \geq 2$, so the shell contribution is $2n^3$ – and the constant is the propagating-component count, which makes D2's "any tensor rank" false). *Computes:* the level-j totals of frame-valued fields on $S^3 = \text{SU}(2)$ for frame-spin 0, 1, 2, coming out as 1, 3, 5 times $(2j+1)^2$ – the component counts, which is the check the decomposition had to pass; the TT part as the two extreme summands, $2(2j+1)^2$, verified to be exactly $2/5$ of the symmetric-tracefree total; the closed form by eigenvalue with its floor $g(2) = 10$ matching P10's independently-asserted $n \geq 2$; the cumulative sum and its leading coefficient $2/3$; and Weyl's law calibrated on the scalar tower, whose degeneracy follows exactly from Peter-Weyl, with the ratio running 1.01505 to 1.00038. **CONTROLS:** the scalar tower's leading coefficient $1/3$ against the tensor tower's $2/3$, which is what refutes "any tensor rank"; and a REPORTED NEGATIVE – matching Weyl to subleading order fails to discriminate the candidate closed forms, because on a closed curved manifold that term carries the curvature through the heat kernel *Bound:* not that the eigenvalue assignment is re-derived, $\mu_n^2 = n(n+2)-2$ being P10's and taken as given; no heat-kernel coefficient, still none computed anywhere; this is the FREE spectrum's mode count and not the interacting tower; and not a closure – PO-6 stays open, this replaces an unchecked input with a derived one *Run:* python3 receipts/L554_tower_degeneracy/D1_the_degeneracy_carrying_the_quartic_was_never_derived_and_its-con

P10R22 P10_the_parity_odd_density_goes_with_the_relative_phase_of_the_two_channels [OK]

The sentence 'r6803' put into 'P10' after node 60's flag: the parity-odd density's dependence stated as a property of the configuration — vanishing a quarter-cycle apart, largest in step — rather than prescribing a helicity. Node 66 asked for a receipt so the sentence can cite one instead of standing on a reading (**Derived, and the old wording's prescription is the case that vanishes.** On the plane-wave member with an areal factor the density factorises as $\psi(u)\omega(u)$ times a **strictly positive** weight $8e^{2\psi}/(t^2 \text{rvert } t \text{rvert})$ carrying no derivative of either channel and no ω at all — so **the density's sign and zeros are exactly the product's, exactly rather than perturbatively** with both channels on one wavenumber a phase δ apart, the cycle average at second order is $\langle \psi \omega \rangle = (k^2 \epsilon^2 / 2) \cos \delta$: **maximal in step, exactly zero a quarter-cycle apart**, extremal in anti-phase, zero if either channel is silenced **and the parity cancellation is derived rather than quoted**: $\omega \text{to-} \omega$ is $\delta \text{to} \delta + \pi$ and flips the sign, so a parity-symmetric ensemble kills it as a property of the integrand, with no statement about occupation numbers). *Computes:* C1 'r6766's three facts re-run first through machinery 'exec'-imported from 'r6762' and 'r6766' so the three cannot drift — a pure plane wave gives $RR=0$ however polarised, the areal factor makes it nonzero, and silencing either channel kills it, each channel silenced by **rebuilding** the metric with it constant rather than substituting a symbol and leaving its derivatives standing the weight shown to have no zeros, so the vanishing is the product's and not the weight's *Bound:* **A torus block, not 'P10's three-sphere layer** — support for the sentence's **form**, not a computation of 'P10's own member, which is the line 'r6766' draws and this does not blur one wavenumber and equal amplitudes, which is where a relative phase is defined at all; no superposition and no unspecified ensemble average **no chiral state is supplied and 'r6770' is not reopened** — its conclusion rests on the parity argument, which is what is computed here; **what this sharpens is that balanced populations** do not by themselves kill a **cross**-correlation, so 'r6770's route

through equal populations is narrower than it reads* nothing bears on ‘PO-7’ *Run: python3 receipts/P10_canonical.time/P10_the_parity_odd_density_goes_with_the_relative_phase_of_the_two_chan*

P10R23 *P10_the_shear_breaks_conformal_flatness_but_not_the_gauss_bonnet_degeneracy_so_nothing_sees_the*
[OK]

node 66’s work order in ‘FOR_60’ (‘r6897’) — *“(1) ‘PO-52’'s second half, which the order made the whole row”. ‘r6894’ established the entry has a domain (Weyl² non-vanishing at $O(\epsilon^2)$ on the construction’s own on-shell confined wave), and the order asked the next thing: “does anything observe the coefficient?” — with the closing route named in advance, “if the honest answer is that the admitted second-order shear configurations are not ones anything observes, that closes the row and is the result”. Forbidden: importing a value for the coefficient (The row closes, and NOT by the route the order offered — the configurations are on the admitted list and the invariant is non-vanishing. What fails is the step from a non-vanishing INVARIANT to a moved OBSERVABLE, because conformal flatness was never the only degeneracy in play. (1) $C^2 = E_4 + 2(R_{ab}R^{ab} - R^2)/3$ is algebraic in four dimensions — verified on closed FLRW with $a(t)$ free (where $C^2 = 0$ and both remainders are non-zero, so it reads more than $0=0$) and on SdS (where $C^2 = 48M^2/r^6 \neq 0$) — so no order in the shear can break it (2) the Lanczos identity $H_{ab} \equiv 0$ holds on the shear configuration itself, at the amplitudes where $C^2 \neq 0$, to 1.6×10^{-30} at 30 digits: the Gauss–Bonnet half reaches no field equation (3) so the remainder is Ricci-squared, which the field equations trade for the stress tensor — and on the substrate, an exact vacuum- Λ space, $C^2 = E_4 - 8\Lambda/3$ exactly $\Rightarrow$ the subtraction-point change is a cosmological-term renormalisation — ‘PO-51’'s conclusion, reached from an identity rather than from a property of the family, so the shear cannot take it away). Computes: C1 the $\epsilon=0$ row is a control that fired: an earlier H_{ab} raised one index of one Riemann factor and not of the other and returned Hvert $H_{ab} \text{rvert} \backslash \text{Riem}^2 \sim 0.8$ on pure de Sitter, where the identity is exact — recorded because the failure was silent everywhere except on the case with a known answer every index carried down and raised only through the inverse metric, explicitly the profile re-verified on-shell against the de Sitter TT wave equation before use the Ricci residue of the linearised wave measured to scale as ϵ^2 exactly, to 10^{-6} in the ratio *Bound: ON-SHELL ONLY*, and that is the whole restriction: the reduction uses the field equations to eliminate R_{ab} , and off-shell C^2 is an independent invariant nothing here bounds a matter-sourced shear configuration gives a stress-tensor integral rather than a constant $\times \text{volume}$ — still not a new geometric constant, but not the same statement; the sharp form is the vacuum- Λ one and is labelled so the frontier’s $1/60$ is not used, not needed and not tested — the argument is coefficient-independent the linearised wave is Einstein through first order only; (2) does not need it to be a solution (an identity is an identity) and (3)’s sharp form is stated on SdS instead $\int \sqrt{g} E_4$ does shift the on-shell action by a topology-fixed constant — but at $\epsilon=0$ as much as at $\epsilon \neq 0$, so the broken degeneracy is not what produces it nothing bears on $\zeta(0)=10$ or $39/4$, which are the degenerate combination’s *Run: python3 receipts/P10_canonical.time/P10_the_shear_breaks_conformal_flatness_but_not_the_gauss_bonnet_degeneracy**

P10R24 *P10_the_degeneracy_needs_r_constant_not_the_cosh_so_the_anomaly_is_what_makes_its_own_constant*
[OK]

‘PO-23’; ‘sec:lock’'s counterterm basis, and ‘P07’'s third wall (node 66’s work order in ‘FOR_60’ (‘r6919’) — “PO-23”, the register’s last NEVER-ATTEMPTED row. The interacting tower’s ultraviolet definition: two walls are built (the operator is bounded below; its spectrum is computable branch by branch), the third untouched. The residual freedom is exactly one number, the log-scale, whose coefficient is $\zeta(0)=10 \neq 0$ — and the no-free-constant claim survives only because on this background the counterterm is degenerate with Einstein–Hilbert and Λ . The order asked the operator question: “once a is quantized, is $\int \sqrt{g} R^2$ still fixed by $\int \sqrt{g} R$ and $\int \sqrt{g}$, or independent?” — plus (2) what coupling does to $\zeta(0)$, iff (1) says observable. Guarded: “do not build the interacting theory”, and “a located wall is the attempt’s result”). Computes: INDEPENDENT — and the mechanism is the anomaly, so the degeneracy that would have hidden the one constant is switched off by exactly that constant’s non-vanishing. (1) The order’s premise is CORRECTED, and by measurement: ‘P10’'s part company as soon

as α is not the de Sitter $\cosh$ is not the operative condition. The degeneracy is equivalent to R being $\sqrt{g}$ -a.e. Λ — Λ +radiation has α nothing like the $\cosh$ and $R=4\Lambda$ exactly, and the free tower's *bare* zero-point energy is exactly radiation-like ($E=S/a$, $\rho \propto a^{-4}$, $p=\rho/3$, trace $\equiv 0$), so the naive back-reaction breaks nothing: **rank 1 to 10^{-43} on BOTH null controls** (2) what breaks it is the $\log$: the pole of Z at $s=-1$ leaves $E=\frac{1}{a}[\ln(a\mu)]$, spoiling the a^{-4} scaling, giving $\Theta=\rho p=r/(2\pi a^4)$ — μ -independent, i.e. an anomaly — hence $R=4\Lambda+4Gr/(\pi a^4)$, non-constant **iff $r \neq 0$ **, and $r=\frac{39}{4}$ (3) rank 3 at $s_3/s_1=9.06 \times 10^{-8}$, 35.5 decades above the MEASURED floor; the exact identity is $I_{2L0-L_1^2}=I_0^2 \mathrm{Var}\{\sqrt{g}(R)\}$, so the gap is **second order in r ** (r^2 constant to 7×10^{-4} over three decades — the same order ‘P10’ found for the shear) (4) (2) answers as a THEOREM: $\zeta(0)=10+\frac{12B_3}{3}(0)$ exactly, so only the m^{-1} coefficient can move it, through ζ_R 's pole at 1 — **immune to every constant (mass-like) shift, every multiplicative rescaling** (containing ‘r6411’s’ α factors out” as one member) **and every even $1/m^{2k}$ shift**; it moves only on an ODD power of the mode label ($\delta=cm \rightarrow c^3/3$; $\delta=c/m \rightarrow -c$), and local operators give only even powers. $\Rightarrow$ **the whole even sector is foreclosed** *Bound:* **The wall, located rather than hit.** The back-reaction is **SEMICLASSICAL** — the free tower's regularized stress tensor in the classical constraint — and the interacting theory is **not built**, as ordered. What higher orders could do is the wall, and it is narrow: restoration needs the TOTAL trace exactly constant, i.e. a pure Λ , and an a^{-4} anomaly is not that. **One premise taken, not proved: that the coupled sector admits states without a definite α — the order's own words and ‘P10’s; if the physical Hilbert space selected one background the degeneracy would survive, and that is the single way (1) reverses. No value for the coupled $\zeta(0)$. **No detectable signal** — the gap is $O(r^2)$; structural independence is not laboratory observability. **This is NOT ‘PO-43’: that entry is a NEW invariant (C^2), this is the SAME invariant becoming an independent functional. Nothing touches the ordering ambiguity or ‘prop:flat’ *Run: python3 receipts/P10_canonical_time/P10_the_degeneracy_needs_r_constant_not_the_cosh_so_the_anomaly_is_what*

P10R25 P10_the_premise_is_not_load_bearing_because_superselection_forbids_coherence_and_not_the_ensemble [OK]

‘PO-23’; ‘sec:lock’s direct integral over $\mathrm{spec}\{\hat{\Gamma}\}$, and $\kappa=1/\alpha$ *common to every fibre* (node 66’s ‘r6925’ order, on the one premise ‘r6920’ took and did not prove: “Does that Hilbert space carry a superselection rule in α ? Equivalently: is α — or whatever operator carries the background’s scale — central in the algebra the true Hamiltonian and the tower generate, or does something in that algebra fail to commute with it?”) with the three outcomes costed: *superselected* reverses ‘r6920’s first item and the degeneracy survives; *not superselected* leaves the finding standing on something proved; *neither* is honest if the algebra is not determined until the coupling is, and then “say which step needs it”. And the guard is this line’s own returned: “Whatever you build here, build its controls first — a case where the answer is known to be superselected and one where it is known not to be — because a superselection statement with no measured floor is exactly the shape of an unfalsifiable one”. Not asked: an interacting theory, a value for the coupled $\zeta(0)$, or a detectable signal). *Computes:* **THE THIRD BRANCH, AND FOR A DIFFERENT REASON THAN THE ORDER OFFERED IT: THE TWO BRANCHES ARE NOT ALTERNATIVES.** (1) Superselecting α does NOT restore the degeneracy, so the first branch reverses nothing. A superselection rule forbids *coherence* between sectors and not an *ensemble* over them; $\int \sqrt{g} R^k$ is read off the ensemble **linearly**; so an ensemble over three distinct α reaches rank 3 at $s_3/s_1=1.60 \times 10^{-4}$, **39 decades above the measured floor, INSIDE a superselected theory** — and the ensemble is exhibited as a legitimate state ($\rho \geq 0$, $\mathrm{tr} \rho=1$, commuting with the label) (2) and ‘r6920’ took neither route, so the premise was never a dependency. Two exact rank lemmas separate them: R constant on ONE history gives $M[i,k]=R_0^k V[i]$, **rank 1 for any number of regions**; an ensemble over N distinct α is a sum of N outer products, **rank exactly $\min(N,3)$ ** — confirmed at $N=2$, where the third singular value sits at 8.2×10^{-72} , *below* the quadrature floor. ‘r6920’ reached rank 3 at $N=1$, where mixing caps the rank at 1, and rank 1 is exactly what its $r=0$ control measured (3) and the question as put to α has no object: $\alpha=\sqrt{3/\Lambda}$

is a **coefficient** of $\mathbb{H}^{\text{phys}}$, so $\hat{\alpha} = \alpha \cdot \mathbf{1}$ is central ***with a ONE-point spectrum***, against the control's three — and a central operator with one point decomposes nothing, so there is no rule for the degeneracy to survive on; the enlargement that would supply one is a direct sum over **theories**, which solving the constraint does not return *** (4) *** the operator R actually depends on, $\hat{a}$, is central in nothing: $[\hat{a}, \mathbb{H}^{\text{phys}}] = [\hat{a}, \hat{p} a^2]$ ***exactly and at every order***, the residual 7.8×10^{-13} sitting below a floor of 7.7×10^{-12} measured on eighteen members of the $f(\hat{a}) \otimes T$ family, because every coupling term $S^{\text{'lock'}}$ names is $f(\hat{a}) \otimes (\text{tower operator})$ and functions of $\hat{a}$ commute with $\hat{a}$ *** (5) the theory's ONE genuine superselection label is $\hat{\Gamma}$ — central in the radial algebra (5.3×10^{-18}) and NOT in the algebra the order names (2.7×10^{-2}), with 7 spectral points — **and it labels the boundary condition, not the background scale***, which is the two-sided control showing that centrality is relative to an algebra *Bound: **AND IT CORRECTS THIS LINE'S OWN LANDED WORK.*** 'r6920' S^E lists as a wall **\{}* "if the physical Hilbert space selected a single background, the degeneracy would survive *\{* " ***that clause is false*** — with a single background the rank is 3 at 9.06×10^{-8} , 'r6920's own headline number, computed at a single Λ . The wall list is shorter by one item, not longer. ***The step that DOES need the coupling is a different question from the one the order posed***: the criterion is $\text{Var}_{\sqrt{g}}(R) = 0$, so its operator form asks whether any physical state makes $\hat{R} = 4\Lambda + 8\pi G \hat{\Theta}$ ***sharp*** — an eigenvector of $\hat{\Theta}$, whose ultraviolet definition $S^{\text{'lock'}}$ names as the open frontier. A conditional is stated as one: if $\text{spec}(\hat{\Theta})$ is purely continuous, no state makes $\hat{R}$ sharp either. ***This receipt does not compute that spectrum.** **The truncated models are models of the ALGEBRA, not of the spectrum*** — they settle commutators and spectral-point counts and settle nothing about the half-line operator's spectrum or its self-adjoint extensions, which 'P10' treats and this does not touch No interacting theory, no coupled $\zeta(0)$, no signal; S^D 's finding does not enlarge the gap, which is still $O(r^2)$ on one history, the larger ensemble number being an effect a single history does not have. Nothing here touches the ordering ambiguity or 'prop:flat' *Run: python3 receipts/P10_canonical_time/P10_the_premise_is_not_load_bearing_because_superselection_forbids_coh*

P10R26 *P10_no_state_makes_the_curvature_sharp_and_the_obstruction_is_the_anomaly_coefficient_itself* [OK]

'PO-23'; $S^{\text{'lock'}}$'s open frontier, and the sentence **\{* "the one ultraviolet constant becomes observable once the scale factor is quantized *\{* " *** (node 66's 'r6929' order, the step this line itself named at 'r6928': **\{* "whether any physical state makes $\hat{R}$ sharp — whether $\text{operatorname{spec}}(\hat{\Theta})$ has a point spectrum *\{* " *****, with the two readings costed: **only continuous** $\Rightarrow$ every state carries a spread in R and the independence is ***\{* " a statement about the ensemble the state already is — which is stronger than it sounds and needs saying in those terms *\{* " *****; **point states** $\Rightarrow$ the independence is realized on individual states and the counterterm is observable in the ordinary sense. Plus: ****say which of the two 'P10' is entitled to claim now**** — and ***\{* "if the answer changes what that sentence may assert, that is a correction to a paper and it comes to me — do not edit 'P10' *\{* " *****. Controls demanded by name: ***a case with known point spectrum and one with known continuous spectrum, run through the same arithmetic*** Not asked: an interacting theory, a coupled $\zeta(0)$, a detectable signal; and 'PO-23' is not expected to close). *Computes: **THE FIRST READING, AND AS A THEOREM RATHER THAN A MEASUREMENT: NO PHYSICAL STATE MAKES $\hat{R}$ SHARP, AND THE OBSTRUCTION IS THE ANOMALY COEFFICIENT ITSELF.** ** (1) $\hat{\Theta}$ is a multiple of the identity on the tower at this order***, which is why the question is answerable without the coupling: $\omega_n = \mu_n/a$ makes the excitation energy $\bar{S}/a$ with $\bar{S} = \sum_n \mu_n \hat{N}_n$ free of a , so $p = \rho/3$ and ***the excitation trace vanishes IDENTICALLY as an operator*** — in every state, not merely the vacuum — while the whole anomaly sits in the renormalized zero point, a c-number. $\Rightarrow \hat{\Theta} = (r/2\pi^2) a^4 \mathbf{1}$ and $\hat{R} = 4\Lambda + \kappa \hat{a}^{-4}$ is a function of $\hat{a}$ ***alone*** *** (2) so the spectral question becomes one about multiplication operators, and there it is exact***: $R' = -4\kappa a^{-5}$ on the half-line, so R is strictly monotonic hence injective, so every level set above 4Λ is ONE point (and below it, empty) — and multiplication by f has an eigenvector at λ ***iff*** $f = \lambda$ carries positive measure. $\Rightarrow \hat{\Theta}$

R has NO eigenvectors; purely continuous spectrum, the range $[4\Lambda, \infty)$ with 4Λ itself not attained** (3)** and the instrument agrees, against controls built first: two refinement exponents (median eigenvalue gap, eigenvector participation length) sit at $\mathbf{-1.00}$ for $\hat{R}$, matching multiplication-by- x (-1.00) and **a full decade of exponent away from the harmonic oscillator's 0.00 ** (4) and the obstruction is QUANTITATIVE and exact**: $[R(\hat{a}), \hat{p}_a] = i\hbar R'(\hat{a})$ gives $\operatorname{Var}(\hat{R}) \operatorname{Var}(\hat{p}_a) \geq \frac{\hbar^2}{4} \langle R'(\hat{a}) \rangle^2$, **saturated to 0.3% ** by narrow admissible states (5) and the right-hand side vanishes iff $r=0$ — where $\hat{R}=4\Lambda \mathbf{1}$ and the variance sits at the measured floor 1.9×10^{-12} . $\Rightarrow$ **the same residue that makes the counterterm observable is what forbids any state from resolving the curvature it is read off** — the row's shape a third time, from a third direction (6)** and a consistency that was not put in by hand: $\langle \hat{a}^{-4} \rangle$ converges at the origin **iff $\nu \geq \frac{13}{4}$ **, and the horizon's own thermal condition gives $\nu = \sqrt{\gamma + \frac{1}{4}} \geq \sqrt{\frac{1}{2}}$ — **the boundary condition that closes the deficiency is the same one that makes $\hat{R}$'s expectation exist** Bound: **P10' IS NOT EDITED, AS ORDERED — the verdict is routed.** The sentence $\langle \rangle$ becomes observable once the scale factor is quantized** **stands**, and the mode is now fixed as **distributional, not sharp-valued**: no state assigns R a definite value, so the third integral's independence is a statement about the spread every state already carries — *unconditional rather than weakened, since there are no special states to except.* What the paper is **not** entitled to is the ordinary-observable reading, a measurement returning a sharp R carrying the counterterm as a definite number; **that is excluded by a theorem, not by a missing calculation** **The step that still needs the coupling, and it is STRICTLY SMALLER than the one it replaces**: $\hat{\Theta}$ is a c-number on the tower *at this order*; cubic and higher terms put genuine tower operators into it, and then $\hat{R}$ is not a function of $\hat{a}$ alone, so the level-set argument — which uses injectivity of a function of ONE variable — does not survive the promotion. Whether a correlated interacting state can make $\hat{R}$ sharp is **not settled here** *One direction flagged and not computed: the D bound needs only the commutator, so any interacting $\hat{\Theta}$ still containing a non-constant function of $\hat{a}$ inherits a bound of the same form from that piece alone — removing the obstruction would need the a -dependence to CANCEL, not merely to be accompanied* **The grid instrument is an instrument, not a proof**: a finite matrix has pure point spectrum by construction, which is exactly why the statistic is a REFINEMENT EXPONENT and not a spectrum; the verdict rests on the level-set theorem **And one methodological catch recorded because it read as a refutation**: the commutator identity integrates by parts, so trial states with amplitude at the grid boundary break it — two of seven widths, and those two are exactly the ones whose ratio fell below 1 (to 0.26). *The domain of the identity is part of the statement.* Back-reaction still SEMICLASSICAL; no interacting theory, no coupled $\zeta(0)$, no signal; gap still $O(r^2)$; nothing touches the ordering ambiguity or 'prop:flat' Run: python3 receipts/P10_canonical_time/P10_no_state_makes_the_curvature_sharp_and_the_obstruction_is_the_anoma

P10R27 P10_the_commutator_bound_survives_the_cubic.because.the_scaling_that_could_cancel_it_carries_no [OK]

'PO-23'; the promoted $\hat{\Theta}$, and S' 's ultraviolet frontier (node 66's 'r6933' order, on the one step 'r6930' left: the level-set argument dies at the cubic **but the commutator may not**, since ** $\langle \rangle$ ** a commutator is algebra and not dynamics**). $\Rightarrow$ ** $\langle \rangle$ ** Compute $[\hat{R}, \hat{p}_a]$ at that order and report whether the variance bound survives with the residue as its coefficient. $\langle \rangle$ *** Three outcomes coded: correction $\hat{a}$ -independent or carrying the residue as a common factor $\Rightarrow$ **the bound survives verbatim and the theorem generalizes**; the correction able to **cancel** the residue at some tower state $\Rightarrow$ **the question needs the interacting theory, but with the cancellation condition in hand**; or the promotion not well posed without a choice $\Rightarrow$ ** $\langle \rangle$ ** say which choice... if this lands on the ordering, that is a different row's business and you should say so rather than pick. $\langle \rangle$ *** Guard, this line's own catch returned: ** $\langle \rangle$ ** an inadmissible case shown with its arithmetic is evidence about the instrument; the same case dropped is a filtered result. Whatever you build for (1), carry that. $\langle \rangle$ *** Not asked: an interacting theory, a coupled $\zeta(0)$, a signal; and 'PO-23' is not expected to close). Computes: **THE BOUND SURVIVES — AND THE REASON IS THE ROW'S OWN SHAPE A FOURTH TIME: THE ONLY SCALING THAT COULD CANCEL IT IS THE SCALING THAT CARRIES NO TRACE.** (1) The bound survives verba-

tim in FORM at every order, ordering-blind.** $[f(\hat{a}) \otimes \hat{T}, p_a] = i \hbar f'(\hat{a}) \otimes \hat{T}$ for *any* $\hat{T}$ — commuting or not, ordered any way — because p_a differentiates only the a -function and $\hat{T}$ rides through; exact symbolically, and on a grid the residual falls at *second order* ($\$-1.99\$$ to $\$-2.00\$$ across four powers with an arbitrary non-commuting $\hat{T}$), i.e. it is the central difference and not the identity *(2)* And pointwise cancellation is IMPOSSIBLE, by a theorem.** An energy term $h(a) \otimes \hat{T}$ contributes $\hat{\Theta} = (h+ah') \hat{T} / 2 \pi^2 a^3$ — *exact* for any h — so a term is traceless *iff* $h \propto 1/a$; a power law $c a^{-n}$ gives $\partial_a \hat{\Theta} \propto (n-1)(n+3) a^{-n-4}$, and matching the anomaly's a^{-5} needs $n=1$, *where* the coefficient $(1-n)$ is zero and the term contributes nothing to $\hat{\Theta}$ at all. $\Rightarrow$ *The anomaly exists because $\ln a$ breaks the $1/a$ scaling that made the bare sum traceless; cancelling it would need exactly the scaling it broke* *(3)* The expectation CAN be tuned to zero — and that is not sharpness.** At B^* the coefficient $\langle \partial_a R \rangle$ vanishes to $\$-1.4 \times 10^{-17}\$$ and the bound collapses to $\$1.2 \times 10^{-35}\$, *vacuous* — but $\langle \text{Var}(\hat{R}) \rangle = 1.56 \times 10^{-6}$ there, *5.9 decades above the measured floor* $\$1.9 \times 10^{-12}\$. *A vacuous bound only stops constraining* *(4)* And the level-set argument survives FURTHER than the order expected.** On every fibre where the cubic's tower operators are simultaneously diagonal — all product states, any commuting family — $\hat{R}$ is multiplication by $R_{\text{eff}} = 4 \sum_j c_j a^{-n_j}$, and $\partial_a R_{\text{eff}} \equiv 0$ has *no* solution at all *while* $R \neq 0$; a level set clears to a degree-6 polynomial, so at most 6 roots, measure zero $\Rightarrow$ *no eigenvector*; and a direct sum over fibres keeps the refinement exponent at $\$-1\$$ ($\$-1.01\$, \$-1.02\$$), so stacking fibres does not manufacture a point spectrum *(5)* So the residue is NOT the coupling — it is non-commutativity, measured: the cubic's own factors π^2 vs ϕ^2 at $\$9.8 \times 10^{-2}\$$ and two cubic terms at $\$2.8 \times 10^{-1}\$, against a commuting family's $\$5.2 \times 10^{-17}\$ *Bound*: *The order's third branch does NOT fire, and the choice is named rather than made.* The commutator needs *no* ordering choice — B differentiates only the a -function — so (1) answers without reaching the ordering. Where the ordering *does* enter is one place: it sets $\langle \hat{T}_j \rangle$, hence B^* , hence whether D 's tuned cancellation is reachable on a physical state. *P10* calls the ordering selection external and localizes it as one computable datum (whether the tower's zero-point energy gravitates at the horizon); this receipt does not pick it *What is still open, narrowed rather than restated: entangled states over a NON-COMMUTING cubic tower family.* Product states and commuting families are covered by (4); the measured non-commutativity is what blocks a common fibre basis in general. Such a state would have to sharpen R by a mechanism neither (2) nor (3) touches — *not settled here* *The power-law form in (2) is an assumption about the cubic and is stated as one*: the theorem is exact for any finite sum of powers, but a cubic term carrying a *second logarithm* would need its own line — and that is a question about the ultraviolet definition, S 'lock's open frontier, not this receipt's. *Flagged, not computed* *The grid is an instrument, not a proof*: B 's verdict rests on the symbolic identity and the exponent is the falsifiable check; comparing MATRICES on a finite-difference grid fails outright, since the commutator sits off-diagonal there — recorded, because it read as the identity failing at a residual of $\$1.2\$$ No interacting theory, no coupled $\zeta(0)$, no signal; back-reaction still semiclassical; nothing touches the ordering ambiguity or 'prop:flat' *Run: python3 receipts/P10_canonical_time/P10_the_commutator_bound_survives_the_cubic_because_the_scaling_that-co*$$$$

P10R28 `P10_the_second_logarithm_holes_the_mechanism_and_not_the_conclusion_because_the_leading_log_is_`
[OK]

'PO-23'; S 'lock's ultraviolet frontier, and 'r6934' S C's mechanism sentence (node 66's 'r6939'/'r6941' order, on the one thing 'r6934' flagged and did not compute: **** $\setminus$ "Generalise the trace theorem to $h(a)=c a^{-n} \ln^m a$ and report $\hat{\Theta}$ and $\partial_a \hat{\Theta}$ for general $(n,m) \setminus$ "*** $((1)^a)$, **** $\setminus$ "does the tower sum at cubic order generate $m=2 \setminus$ "*** $((1)^b)$; and **** $\setminus$ "report whether the bound's right-hand side can be cancelled at any (n,m) the cubic actually generates, as against the $m=0$ answer which is no $\setminus$ "*** $((1)^c)$. With 66's *own hypothesis offered to be refuted*: $\setminus$ " at $n=1$ the $\ln^m$ term vanishes and the $\ln^{m-1}$ term does not ... the mechanism (2) rests on has a hole at exactly $m \geq 1$, which is exactly where the anomaly lives $\setminus$ "*, flagged as $\setminus$ "a hypothesis with a poor pedigree $\setminus$ ". Three guards: *never compare the identity as matrices* (this line's own 'r6934' note, made the standing instrument warning); *the ordering stays named and un-****

picked** ('PO-15's); and **any new cancellation gets the second measurement** against the variance floor π or it will be read as sharpness by the next reader π . Not asked: an interacting theory, an ordering choice, or any corpus edit). *Computes:* **66's HYPOTHESIS IS CORRECT AND 'r6934's MECHANISM IS HOLED — BUT THE CONCLUSION SURVIVES BY A STRICTLY BETTER ARGUMENT THAT DOES NOT DEPEND ON (1)^b.**

(1) ^a exactly**: $2\pi^2 \hat{\Theta}[c, a^{-n}] \ln^m a = c, a^{-(n+3)} [(1-n) \ln^m a + m \ln^{m-1} a]$, confirming the order's algebra. At $n=1$ the first term dies and the second does not **(2)** and the witness is the ANOMALY ITSELF**: $r/(2\pi^2 a^4)$ is the $(n=1, m=1)$ entry of that very formula, so 'r6920's trace was always a member of the family 'r6934' assumed excluded. $\Rightarrow$ 'r6934' $\setminus S$ C's $\setminus$ the only scaling that could cancel carries no trace $\setminus$ is true for $m=0$ and false for $m \geq 1$ **(3)** ^c — AND YET CANCELLATION IS STILL IMPOSSIBLE, at ANY (n, m) : the LEADING LOGARITHM IS UNPARTNERED. From an $(1, m)$ term the highest power in $\partial_a \hat{\Theta}$ is $\ln^{m-1} a$ with coefficient **exactly** $-4mc$ (checked $m=1.4$); terms chain downward and never upward, so the coefficient system is **lower triangular with diagonal** $-4m$ and $\partial_a \hat{\Theta} \equiv 0$ forces every $c=0$ — sympy returns only the trivial solution **(4)** and this is UNCONDITIONAL: whatever m the cubic generates, its own top logarithm has no partner, **so** ^c does not wait on ^b **(5)** but PARTIAL cancellation is real, and is a third answer: an $(1, 2)$ term at exactly $c_2=2r$ kills the $\ln^0$ coefficient the anomaly sits in and leaves $\ln^1$ at $-16r$ — the obstruction RELOCATES one logarithm up rather than being removed, neither $\setminus$ cancels $\setminus$ nor $\setminus$ adds $\setminus$. And the second measurement: $\operatorname{Var}(R)=5.1 \times 10^{-3}$ there, **11.2 decades above the floor** 3.6×10^{-14} **(6)** ^b — a second logarithm needs a NESTED SUM: m is the pole order, and a double pole needs a $\ln$ of the mode label *in the summand*, since the Mellin transform of a log-free power asymptotic has only simple poles. The free tower's $d(m) \mu(m) = 2(m^2-4) \sqrt{m^2-3}$ expands at large m as a **pure Laurent series** — log-free, **exactly** — growing as $2m^3$, which is the power putting the pole at $s=-1$. So the free sum yields $m=1$ and **never** $m=2$ *Bound*: **This CORRECTS** this line's own landed 'r6934' — the mechanism, not the conclusion, and the correction came from a limitation that receipt itself recorded. *Sixth self-correction of landed work, second of that kind.* The conclusion 'r6934' reached is re-established on a stronger footing; its stated reason is not general **(1)^b** is NOT computed, and the datum is named rather than handed off: it turns on **one property** — whether the cubic vertex's large-mode-label asymptotic carries a $\ln m$. If it does, the cubic reaches $m=2$ and ^c's partial cancellation becomes physically available at $c_2=2r$; if not, the cubic stays at $m=1$. **Either way (3) holds.** *The order's point that this row IS the ultraviolet definition is taken — which is why the answer is a named property of the sum and not a hand-off* **The ordering stayed named and unpicked, and this order's answer does not depend on it**: nothing in the trace formula or the leading-log argument uses an ordering choice, so there was nothing to stop for **Guard 1** carried live, not recited: the identity's convergence order is re-measured at -1.99 on a smooth state, so a flat residual would be a representation error. And $\mathcal{D}$'s rejected width is **printed with its arithmetic**, per the standing rule No interacting theory, no coupled $\zeta(0)$, no signal, **no corpus edit** — the site in 'P10' and what it should say are routed; back-reaction still semiclassical; nothing touches 'prop:flat', 'PO-31' or 'PO-15' *Run: python3 receipts/P10_canonical.time/P10_the_second_logarithm_holes_the_mechanism_and_not_the_conclusion_bec*

P10R29 *P10_the_second_logarithm_line_closes_because_the_interaction_cannot_reach_the_anomalys_own_power*
[OK]

'PO-23'; $\setminus S$ lock's ultraviolet frontier, and whether 'r6942's relocation is physically available (node 66's 'r6945' order, on the one property 'r6942' reduced the second-logarithm question to and declined to guess: **the vertex's large-label asymptotic, to the order that decides the pole structure ... report whether the inner sum produces one and at what order in the label, with the expansion printed rather than characterised** **((1)^a)**; and **whichever way it falls, say what it does to the relocation** **((1)^b)**. With the order's own statement of the stake — **(1)^c** is unconditional, so this does not gate the verdict. **What it gates is whether the relocation at $c_2=2r$ is PHYSICALLY AVAILABLE** — and its costing that **either outcome closes a branch rather than opening one**. Three guards: the convergence-order measurement **stays live rather than recited**; the ordering **stays named and unpicked**, with the addition that **if the vertex asymptotic turns out to depend on an ordering choice, that is a finding and the move is to name the datum and stop**; and — NEW, about the shape of the

answer — $\{\}$ a second completed argument is worth more than three narrowings, so if the vertex computation closes the second-logarithm line outright, say so in those terms $\{\}$ ***, with $\{\}$ not computed $\{\}$ * and $\{\}$ computed and inconclusive $\{\}$ * named as different states. Not asked: an interacting theory beyond the vertex's own asymptotic, an ordering choice, or any corpus edit). *Computes:* **THE SECOND-LOGARITHM LINE IS CLOSED — TWICE, AND THE STRONGER CLOSURE NEEDS NO PROPERTY OF THE VERTEX AT ALL.**

*(1) The relocation lives at a power of a the interaction cannot reach. 'r6942's tuned term is $c_2 a^{-1} \ln^2 a$ — an $(n=1, m=2)$ term, because $n=1$ is where the anomaly sits. The cubic's second-order vacuum shift is an $n=3$ term: in closed form $E^{(2)} = (\lambda^2 \kappa / 32 a^3) \sum G_{123} \mu_1 \mu_2 \mu_3 \mu_1$ — matched against direct diagonalization to 2.5×10^{-9} relative (both give $-1/192$ at $a=1$), with the measured log-log exponent -2.999992 against the free tower's -1.000000 , truncation-stable to 4×10^{-6} (2) And the counting makes that general rather than a feature of the cubic: the only scales are a and ℓ_P , so a term carrying ℓ_P^j carries $a^{-(1+j)}$ — $n=1+j$, and $n=1$ is populated at $j=0$ alone, by the FREE tower, whose summand is log-free exactly ('r6942'). $\Rightarrow$ At $n=1$ the logarithm is capped at $m=1$ at every order of the coupling, so there is no $(n=1, m=2)$ coefficient in this theory to tune; and sympy returns *no solution* when the two families are asked to cancel at two values of a (3) (1)^a answered on its own terms, and the answer is also no. The S^3 triple-harmonic overlap has an exact multiplet sum rule $\sum \alpha_{\text{vert}} C_{123} = n_1 n_2 n_3 / 2 \pi^2$ on the selection-rule set (triangle inequality and $n_1 + n_2 + n_3$ odd) — the integral is $\pi/2$ on that set and 0 off it over all 343 triples against quadrature, and the rule reproduces the $n_3=1$ constant-harmonic case exactly. In both soft corners the internal-label dependence is $j(j-3)^p$, $p \ln \frac{12}{\mathbb{Z}}$: an odd polynomial terminating at $j+1$ for integer p , an even-power series for half-integer $p \Rightarrow$ the coefficient of $1/j$ is exactly zero at every order in both corners — no harmonic number, no $\ln$ of the label, only simple poles, $m=1$ at the cubic's own power of a too (4) And the condition is sharp rather than lucky: give the weight one inverse power and the integer- p branch stops terminating, producing a $1/j$ coefficient of exactly $c(-3)^p$ at every odd k — the instrument returns a harmonic number when one is there, so the zero is a measurement (5) The second measurement carried anyway, because a relocation is not a sharpening: $\text{operatorname{Var}}(R) = 5.1 \times 10^{-3}$ at $c_2 = 2r$, 11.2 decades above the floor 3.6×10^{-14} Bound: This DEMOTES 'r6942's relocation rather than correcting it. 'r6942' reported it as $\{\}$ a fact about the formula $\{\}$ and asked whether it was available; it is not — the tuned state $c_2 = 2r$ exists in the trace formula's closure and not in this theory's spectrum. Nothing in 'r6942's verdict changes: its S C was already unconditional, and it neither asserted the relocation was reachable nor guessed its sign ONE STEP NAMED, and it is not load-bearing. The scalar sum rule's weight $n_1 n_2 n_3$ is a polynomial exactly, so (3)'s zero is exact there; the transverse-traceless vertex differs by index contractions whose multiplet-summed weight is a rational function of the labels, and whether that function is polynomial in each label is the one step at which (3) could differ. The TT zonal kernel is not computed here and not guessed — and it decides nothing about (1)–(2), which is where the verdict is The ordering stayed named and unpicked, and the new guard is answered rather than deferred: the cubic's reordering difference is measured to be exactly iP — linear in the momenta and supported only on coincident labels (residual below 10^{-15} across three truncations), so it is not a cubic term and does not enter the asymptotic Guard 1 carried live: the identity's convergence order re-measured at -1.99 on a smooth state this revision, so a flat residual would still be a representation error. The inadmissible width is printed with its arithmetic The $n=3$ scaling is measured on a three-mode truncation of the reduced system, not on the full tower; the closed form and the ℓ_P counting are what carry the claim, and the diagonalization is the falsifiable check on them No interacting theory beyond this vertex's own asymptotic, no coupled $\zeta(0)$, no signal, no corpus edit — the 'P10' site and its sentence are routed; nothing touches 'prop:flat', 'PO-31' or 'PO-15' Run: python3 receipts/P10_canonical.time/P10_the.second.logarithm.line.closes.because.the.interaction.cannot.re

P10R30 P10_the_entangled_case_is_a_singular_pencil_question_and_the_pencil_is_non_singular_at_every_tr
[OK]

'PO-23'; the entangled case — 'r6934's remainder, which has been this row's whole wall for four

revisions (node 66's 'r6949' order, reformulating that wall from a category into a computation and offering the reformulation **as a hypothesis to be broken** ($\{\}$ my record on this row is three withdrawn clauses, one of which you withdrew last revision, so weigh it accordingly $\{\}$): (1)^a $\{\}$ first, the structural claim, which everything after it rests on: $\{\}\hat{R}$ commutes with $\{\}\hat{a}$... check it rather than assume it — the conformal factor, the measure, and whatever operator ordering the momentum constraint imposes are all places it could fail, and if it fails the rest of this order is void and that is the finding $\{\}$; (1)^b $\{\}$ does the family $\{\}\hat{R}(a)$ have an eigenvalue branch that is CONSTANT in a on a set of positive measure? $\{\}$ — **a question about a matrix family rather than about states**; (1)^c $\{\}$ solve it at the smallest truncations that carry two non-commuting tower operators and report whether a solution exists $\{\}$, the determinant being a polynomial in the powers of $1/a$ hence analytic, so vanishing on positive measure means vanishing identically. Guards: **if (1)^a fails, stop there**; $\{\}$ do not let the truncation answer for the tower ... if the polynomial system's solvability changes with N , that dependence IS the result $\{\}$; the ordering stays named and unpicked, **and if (1)^a depends on it that is a third surfacing and worth flagging as such**; and $\{\}$ reformulated and open is progress and not attempted is not $\{\}$. *Computes*: **THE PREMISE HOLDS, THE ENTANGLED CASE IS NOW A SINGULAR-PENCIL QUESTION, AND THE PENCIL IS NON-SINGULAR AT EVERY TRUNCATION TESTED.** $\{\}$ (1)^a is TRUE, and true for a reason rather than by construction. $\{\}\hat{R}$ is **algebraic** in the matter trace—the trace of the field equation gives $R=4\{\}\Lambda+\{\}\kappa\{\}\Theta$ with no derivative of a in it—and 'r6934's trace formula makes every term a function of $\{\}\hat{a}$ times a tower operator on the other factor, so $\{\}\hat{R},\{\}\hat{a}$ is **exactly zero**. $\Rightarrow$ **And the control shows it is not vacuous: the variant that keeps a $\{\}\hat{p}_a$ —what taking R from the kinetic form would do—measures $8.4\{\}\times 10^{-4}$ against that exact zero.** The **measure** does not touch it (self-adjointness residual 10^{-16} in a weighted product), and (1)^a is **ordering-blind** across three orderings of the tower factor $\Rightarrow$ **the ordering does NOT surface a third time** $\{\}$ (2) (1)^b the branch value is FORCED before any determinant is taken. Every power in $\{\}\hat{R}(a)=4\{\}\Lambda+\{\}\sum_k\{\}\kappa_k a^{-m_k}\{\}\hat{T}_k$ is negative, so $\{\}\hat{R}(a)\{\}\rightarrow 4\{\}\Lambda$ and a constant branch must equal $4\{\}\Lambda$ exactly $\Rightarrow$ **the question collapses to: is $M(a)=\{\}\sum_k\{\}\kappa_k a^{-m_k}\{\}\hat{T}_k$ singular for almost every a ?** **No truncation in that step** (3) (1)^c and at finite N that is EXACTLY a singular-pencil question, which is a -free and coupling-free. $\{\}\det(\{\}\sum x_k\{\}\hat{T}_k)$ is a degree- N form and for two distinct powers each monomial lands on its own power of $1/a$ (checked: six monomials, six powers at $N=5$), so $\{\}\det M(a)\{\}\equiv 0\{\}\iff$ **the pencil is singular**, with extreme coefficients $\{\}\kappa_k^N\{\}\det\{\}\hat{T}_k$. $\Rightarrow$ For the cubic's own **non-commuting** factors $\{\}\hat{T}_1=\{\}\hat{\pi}^2$, $\{\}\hat{T}_2=\{\}\frac{12}{\{\}\hat{\pi}^2}\{\}\hat{\pi}^2\{\}\hat{\pi}$ the pencil is **NON-SINGULAR** at $N=2\{\}\dots 8$, in exact integer arithmetic (4) At three operators the powers COLLIDE ($6\{+ \}10=8\{+ \}8$), so the pencil equivalence does not carry and the curve determinant is computed directly—non-zero at $N=3\{\}\dots 6$ with symbolic couplings, and the full coupling system has **no solution** with all three couplings non-zero at $N=3,4,5$ (5) **AND THE ONE N -DEPENDENCE IS THE ORDER'S WARNING MADE CONCRETE.** At **odd** N a ladder matrix is singular, so $\{\}\det\{\}\hat{T}_k=0$ and the system **does** acquire non-trivial solutions—**every one of which switches off** at least two of the three couplings, collapsing to a single-operator model that is singular for that reason alone. $\Rightarrow$ **The necessary conditions are N -parity dependent; the verdict is not.** And the bias runs the helpful way: an odd truncation MANUFACTURES a null vector the tower has not got, so the instrument is tilted toward the affirmative and still returns none (6) The instrument is **live**: two operators sharing one null vector give a **singular** pencil at $N=4,5,6$, so the zeros are measurements. And the measure-zero refinement: a non-zero Laurent polynomial has finitely many positive roots (2 at $N=4$), so even where the determinant vanishes it vanishes on measure zero and $\{\}$ for almost every $a\{\}$ fails *Bound*: **THE ONE NAMED STEP IS THE TOWER LIMIT OF THE DETERMINANT CRITERION.** At finite N , singularity is $\{\}\det=0$; in the tower it is a **null eigenvector** for almost every a , and the odd- N parity is exactly where the two criteria part company. **What survives dimension is stated separately and is three things:** $\{\}\lambda=4\{\}\Lambda$ is forced; the condition is a null eigenvector of $M(a)$; and $\{\}\hat{\pi}, \{\}\hat{\pi}^2$ have purely continuous spectrum so neither has the null eigenvector the odd- N determinant reports—

making the finite- N proxy strictly more permissive, so a finite- N “no” is the stronger of the two statements. $\Rightarrow$ So this is reformulated and ANSWERED AT EVERY TRUNCATION with the limit named, not a closed statement about the untruncated tower — the wall is a different object now, “the tower limit of a singular-pencil condition” rather than “entangled states over a non-commuting family”. The momentum constraint’s algebra is NOT derived; it can only bear on (1)^a by putting a $\hat{p}_a$ into $\hat{R}$, which is what the control measures, and transverse-traceless modes satisfy it identically AND A HABIT ADOPTED FROM ‘r6947’, which is why this receipt has almost no floats in it. That revision found ‘r6946’'s one tolerance was the authoring machine’s round-off floor. The repair for a claim of this shape is not a better tolerance but NO tolerance: the ladder matrices are carried in the basis $D = \text{diag}(1/\sqrt{n!})$ where both are integer matrices—a diagonal similarity, so spectra and determinant-vanishing are untouched—and every load-bearing determinant, Laurent coefficient and coupling system is computed over the integers. The two remaining floats are the commutator measurements, each reported against a control rather than a threshold No ordering choice, no interacting theory beyond the cubic’s own two tower factors, no signal, “no corpus edit”; nothing touches ‘prop:flat’, ‘PO-31’ or ‘PO-15’ Run: python3 receipts/P10_canonical_time/P10_the_entangled_case_is_a_singular-pencil-question-and-the-pencil-is-

P10R31 P10_the_tower_limit_closes_at_first_order_and_the_wall_moves_into_a_boundary_condition_at_second_order [OK]

‘PO-23’; the tower limit — the one step ‘r6950’ named, and the boundary condition it turns out to hide (node 66’s ‘r6953’ order, handing the named step over as a hypothesis they had run rather than guessed: in the momentum representation $M(a)\psi=0$ is a first-order ODE with closed form $\psi=Ce^{\int \mathrm{i}c(a)p/p}$, non-normalizable because of the $1/p$ at the origin, “and that is not normalizable for a reason independent of both couplings and of the scale”. (1)^a “Confirm or refute the reduction itself ... is the representation I have assumed the one it uses? If the measure is not $\mathrm{d}p$, the $1/p$ verdict changes”; (1)^b “then the domain question, which is the load-bearing one. Is $M(a)$ essentially self-adjoint on the natural domain, or does it admit a family of extensions?” — with the conditional that “if the extension selection is the datum ‘PO-15’ carries, that is the ordering surfacing a third time”; (1)^c “and whether more than two operators enter ... report the order of the equation the actual operator content gives, since that decides whether (1)^a’s closed form is the answer or the first term of one”. Guards: “do not take my ODE on trust” — four named places it can fail (representation, measure, domain, content), “and the last three are exactly where this row’s earlier results came from”; if it closes, say the row has a “third completed argument” and the wall is gone as an object; “and if the extension family is non-trivial, say the wall has MOVED into the extension choice and name it as a different open thing”; and keep the arithmetic exact). Computes: THE REDUCTION IS CORRECT; THE OPERATOR IS NOT ESSENTIALLY SELF-ADJOINT; THE VERDICT IS UNIFORM OVER THE EXTENSION FAMILY ANYWAY — AND (1)^c IS NOT A FOOTNOTE, BECAUSE AT SECOND ORDER THE ANSWER REVERSES. (1) (1)^a confirmed, and in TWO representations. With $\hat{\varphi} \rightarrow \hat{\pi}$ to p , $\hat{\varphi} \rightarrow \hat{\varphi}$ the symmetrised vertex is exactly $\hat{T}_2 = \int \mathrm{i}p \cdot \frac{\mathrm{d}}{\mathrm{d}p} \frac{\mathrm{d}}{\mathrm{d}p}$ — which is where the $1/p$ comes from: the symmetrisation’s own inner derivative — it is symmetric with the difference an exact total derivative $\frac{\mathrm{d}}{\mathrm{d}p} p^2 \bar{f}g$, and sympy returns 66’s closed form unprompted. The measure IS $\mathrm{d}p$ (Plancherel), and the verdict is “scale-invariant”: a dilation carries $1/p$ to $1/\lambda$ p , so no oscillator mass or frequency rescues it. $\Rightarrow$ And the position representation is an independent route to the same place: $M = -(A+Bx)\psi'$, whose $z=0$ solutions are $c_1 + c_2 \ln \left| \frac{A+Bx}{A+Bx} \right|$ — “neither in L^2 , and no numerics in either route” (2) (1)^b 66’s worry is JUSTIFIED: the operator is NOT essentially self-adjoint. At general z , verified by substitution, $\psi_z = \frac{C}{p} \exp(\int \mathrm{i} \frac{A}{B} p)$ at $z = \int \mathrm{i} \sigma$ the second exponential is “real”, and the near-origin integrals are exact — $\int_0^\infty e^{-k/p} p^{-2} \mathrm{d}p = e^{-k/k}$ finite, $\int_0^\infty e^{+k/p} p^{-2} \mathrm{d}p = \infty$ — so each half-line carries an L^2 solution for one sign of $\operatorname{Im} z$ and none for the other: “deficiency indices $(1,1)$, a $U(1)$ family of extensions” (3) AND YET THE QUESTION IS SETTLED UNIFORMLY OVER THAT FAMILY. At $z=0$ the exponential is a “pure phase”, $\left| \psi_0 \right|^2 = \left| \psi_0 \right|^2$

$C \backslash \backslash \backslash \text{rvert}^2 / p^2$, and $\$ \backslash \backslash \backslash \int_0^1 \backslash \backslash \backslash \mathrm{d} p / p^2 = \backslash \backslash \backslash \text{infy}$ — ****no L^2 solution on EITHER side****. A boundary condition selects among L^2 solutions and there are none to select from $\Rightarrow$ *****\$0\$ is not an eigenvalue of ANY extension: the first-order tower criterion closes WITHOUT choosing one*****, which is stronger than closing it by choosing the right one. $\Rightarrow$ ****So the order's conditional answers in the negative — the selection is never needed, so it cannot be 'PO-15's datum surfacing** (4) (1)^c AND AT SECOND ORDER THE ANSWER REVERSES.**** A $\$ \backslash \backslash \backslash \text{hat} \backslash \backslash \backslash \text{varphi}^2$ structure makes the equation second order; then $\$ \backslash \backslash \backslash \text{psi} = \backslash \backslash \backslash \text{alpha} + \backslash \backslash \backslash \text{beta} p$ solves it to $\$O(p)$ at the momentum origin, so ****both solutions are regular there and the $1/p$ obstruction is GONE**** — the failure mode the order flagged, confirmed. The constraint moves to large scale, where $\$ \backslash \backslash \backslash \text{psi}'' + \backslash \backslash \backslash \text{psi}' / x - (C/B)x \backslash \backslash \backslash \text{psi} = 0$ is ****exactly Bessel**** in $\$ \backslash \backslash \backslash \text{xi} = \backslash \backslash \backslash \text{frac} 23 \backslash \backslash \backslash \text{sqrt} [C/B] \backslash \backslash \backslash \text{lvert} x \backslash \backslash \backslash \text{rvert}^{3/2}$: exponentially decaying at one end, oscillatory with amplitude $\$ \backslash \backslash \backslash \text{lvert} x \backslash \backslash \backslash \text{rvert}^{-3/4}$ at the other — the full equation measured at $\$-0.7511$ against that ****exact**** value — so $\$ \backslash \backslash \backslash \text{lvert} \backslash \backslash \backslash \text{psi} \backslash \backslash \backslash \text{rvert}^2 \backslash \backslash \backslash \text{sim} \backslash \backslash \backslash \text{lvert} x \backslash \backslash \backslash \text{rvert}^{-3/2}$ is integrable $\Rightarrow$ *****both ends admit L^2 , a normalizable null vector is NOT excluded, and everything turns on the boundary condition at the degenerate point $x_0 = -A/B$ where $A+Bx$ vanishes and the solutions are a constant and a logarithm (limit-circle)***** *Bound:* ****NOT a third completed argument — it is one AT A STATED OPERATOR CONTENT****, and the row's status is said in those terms rather than banked. **Closed for the content the cubic's kinetic vertex gives $(\$ \backslash \backslash \backslash \text{hat} \backslash \backslash \backslash \text{pi}^2) \backslash \backslash \backslash \text{hat} \backslash \backslash \backslash \text{varphi}$, one power of $\$ \backslash \backslash \backslash \text{hat} \backslash \backslash \backslash \text{varphi}$, hence first order); the first term of an answer otherwise.* ****AND THE WALL HAS MOVED RATHER THAN VANISHED, named as the order asked****: at second order it is ****the boundary condition at the degenerate point****, which is a **different** open object from a limit of determinants and must not be carried as the same item. **It is also the same KIND of object $\backslash \backslash \backslash \text{S'lock}$ already closes at $a=0$ with the horizon's own thermal state — the one concrete suggestion this receipt makes about where to look** ****The operator content is the deciding datum and is NOT established here.**** What is established: the FREE tower contributes no tower operator at all ('r6930's traceless excitation, at fixed occupation), so the content begins at the cubic. ****Whether a $\$ \backslash \backslash \backslash \text{hat} \backslash \backslash \backslash \text{varphi}^2$ structure enters $\$ \backslash \backslash \backslash \text{hat} R$ alongside $\$ \backslash \backslash \backslash \text{hat} \backslash \backslash \backslash \text{pi}^2 \backslash \backslash \backslash \text{hat} \backslash \backslash \backslash \text{varphi}$ is not computed and not guessed**** ****Arithmetic discipline carried from 'r6950'/'r6947****: every load-bearing step is closed form — the reduction, the general- z solution (verified by substitution, not by form-matching), the deficiency integrals, the Bessel reduction. **The single float is the amplitude exponent, reported against the exact Bessel value rather than a threshold** No ordering choice ($\$C$ never reaches one), no interacting theory beyond the two cubic operators and the hypothetical $\$ \backslash \backslash \backslash \text{hat} \backslash \backslash \backslash \text{varphi}^2$, ****no corpus edit****; the TT weight stays where 'r6946' left it; nothing touches 'prop:flat', 'PO-31' or 'PO-15'*

Run: python3 receipts/P10_canonical_time/P10_the_tower_limit_closes_at_first_order_and_the_wall_moved

P10R32 P10_the_field_squared_is_already_in_the_trace_at_quadratic_order_and_the_degenerate_point_is_reduced [OK]

'PO-23'; the one datum the row was reduced to — whether a $\$ \backslash \backslash \backslash \text{hat} \backslash \backslash \backslash \text{varphi}^2$ structure enters $\$ \backslash \backslash \backslash \text{hat} R$ alongside the kinetic vertex's $\$ \backslash \backslash \backslash \text{hat} \backslash \backslash \backslash \text{pi}^2 \backslash \backslash \backslash \text{hat} \backslash \backslash \backslash \text{varphi}$ (node 66's 'r6959' order, making that datum the whole row and naming the two outcomes as ****not symmetric****: ***if it does not enter, 'r6954's first-order argument is the answer at the theory's actual content ... and **the wall is gone as an object** rather than moved ... that is the third completed argument, and it should be reported in those words***; ***if it does enter, the obstruction at the momentum origin is gone, both ends admit square-integrable behaviour, and **the wall sits in a boundary condition at $x_0 = -A/B$ ****. (2) **only if (1) says yes**: ****report the deficiency indices there and the form the family takes****, say whether ***the same argument that closed the first-order case — no square-integrable solution to select among — has any purchase at all***, and whether the $a=0$ mechanism reaches that point. The direction, named and not claimed: ***the curvature enters through the trace, so the question is which structures the stress trace carries at cubic order ... a $\$ \backslash \backslash \backslash \text{hat} \backslash \backslash \backslash \text{varphi}^2$ in $\$ \backslash \backslash \backslash \text{hat} \backslash \backslash \backslash \text{Theta}$ is a potential-like term, and the cubic vertex you used is kinetic***, flagged with its pedigree of four withdrawn clauses. Guards: ***computed and inconclusive with the step named is progress; not attempted is not***; *****do not let the kinetic vertex answer for the content*****, and ***if the content turns out to be larger than two structures, say what the equation's order becomes and stop rather than extrapolating***; keep the arithmetic exact and report any float against an exact predicted value; the ordering stays named and unpicked). *Computes:* **** (1) IS YES — AND YES EARLIER IN THE EXPAN-**

SION THAN THE ORDER'S DIRECTION LOOKED. THE DIRECTION IS RIGHT AND THE ORDER IS WRONG: $\hat{\varphi}^2$ DOES NOT WAIT FOR THE CUBIC.** (1) The answer, with three independent derivations.** 'sec:lock's own tower Hamiltonian $H = \sum_n [\hat{\pi}_n^2 / 2a^3 + \frac{1}{2} \mu_n^2 \hat{\varphi}_n^2]$ under 'r6934's trace formula $\text{Tr} \mapsto (h+ah') T / 2 \pi^2 a^3$ gives $2 \pi^2 \hat{\Theta}_a \hat{\pi}_n^2 / a^6 + \mu_n^2 \hat{\varphi}_n^2 / a^2$ — so $\hat{\varphi}^2$ enters R with coefficient $\kappa \mu_n^2 / 2 \pi^2$ at the power a^{-2} **, from the tower's own **gradient** energy, at $j=0$ and every order above. *Route 2, taking no $\partial \partial V$ at all*: $\rho_3 = (\partial \varphi)^2 = -\dot{\varphi}^2 + \mu^2 \varphi^2$ with $\dot{\varphi} = \pi/a^3$, term for term the same. *Route 3, 'r6930's own instrument*: its 'trace_of' on $E = S(a)/a$ is exactly $S'(a)/2 \pi^2 a^3$, and with $S(a) = \mu(\hat{\varphi}^2 + \frac{1}{2} \pi^2)$ $N + \frac{1}{2} = (a^2 \mu^2 \hat{\varphi}^2 + \pi^2)/2$ in the **fixed** field operators it reproduces both. $\mu_n^2 = m^2 - 3 \geq 6$, so the coefficient never vanishes; and the structure is **degree two** in the fields, not a cubic-order effect ** (2) WHERE THE ROW LOST IT, AND IT IS A SIXTH FACE OF THE STANDING LESSON: THE DOMAIN OF A VANISHING.** 'r6930's 'trace_of' docstring scopes itself **"at fixed occupation numbers"**; its verdict line drops the scope — **"traceless IDENTICALLY**, as an operator ... whatever the occupation numbers". Since $M \omega = a^2 \mu$, $\hat{a}_n$ depends on a : S is free of a for the c-number zero point $\sum \mu_n / 2$ and along fixed occupation, and **not** for the operator**. In the adiabatic basis the same operator is exactly $(\mu/a^4)(\hat{a}^2 + \hat{a}^{\dagger 2}) = \hat{X}^2 - \hat{P}^2$ with $[\hat{X}, \hat{P}] = i \mu/a^4$, whose **diagonal** is exactly zero at every truncation — which **is** the harmonic virial theorem $\langle n | \hat{K} | n \rangle = \langle n | \hat{V} | n \rangle$, verified exactly, and it holds in every mixture of number states too. **But** on the squeezed state $|0 \rangle_t |2 \rangle_r$ the expectation is $2 \sqrt{2} \mu t/a^4 (1+t^2)^{-1/2}$, and normal ordering cannot remove it, $\hat{a}^2 + \hat{a}^{\dagger 2}$ carrying no contraction ** (3) THE LANDED CONCLUSION SURVIVES, ON STRONGER GROUND.** The residual is an **inverted oscillator**, point spectrum empty: the null equation $\psi'' + p^2 \psi/a^4 = 0$ is solved by substitution by $\sqrt{p} J_{-1/4}(p^2/2 \mu a^2)$, **both** regular at the momentum origin and neither square-integrable at infinity ($\psi \sim 1/p$, $\int_1^\infty dp/p = \infty$; envelope exponent -0.50000 against the exact $-\frac{1}{2}$, and the $-\frac{1}{2}$ exact at every real z since the WKB amplitude is $(p^2+z)^{-1/4}$). $\Rightarrow$ **R has no eigenvector at quadratic order** — not because it is a function of $\hat{a}$ alone, which it is not, but because an inverted oscillator has no normalizable eigenvector at any real eigenvalue**, which is a warrant that does not depend on a term being absent ** (4) (2) AND THE SECOND BRANCH'S OBJECT DOES NOT EXIST: THE DEGENERATE POINT IS A FEATURE OF ONE REPRESENTATION.** One abstract operator $\hat{M} = c_1 \hat{\pi}^2 + c_2 \frac{1}{2} \hat{\pi}^2 \hat{\varphi}^2$ verified to be one operator by its exact Hermite matrix against the abstract ladder matrix: in **position** it is $-(c_1 + c_2 x) \psi'' - c_2 \psi' + c_3 x^2 \psi$, leading coefficient vanishing at $x_0 = -c_1/c_2$; in **momentum** it is $-c_3 \psi'' + i c_2 (p^2 \psi' + p \psi) + c_1 p^2 \psi$, **leading** coefficient the constant $-c_3$, vanishing nowhere**. At large p the two behaviours are $\psi_1 \sim p^{-1} e^{i c_2 p^3 / 3 c_3 - i c_1 p / c_2}$ and $\psi_2 \sim e^{i c_1 p / c_2}$, each verified by an exact limit on its relative residual and both exponents purely imaginary for real couplings, so $\psi_1 = 1/p$ and $\psi_2 = 1$ exactly $\Rightarrow$ **exactly one L^2 solution at each end**, both ends limit-point, deficiency indices $(0,0)$, essentially self-adjoint — the family is a single point and there is nothing to select.*** And what distinguishes x_0 from $a=0$: $a=0$ is an **endpoint** of the physical configuration space** and a physical locus (the horizon, $\kappa=1/\alpha$), so the freedom is genuine and the thermal state closes it; x_0 is an interior point in one representation and **no point at all** in the other, carrying no locus — **the mechanism does not reach it** because there is nothing there to reach* **Bound**: **NOT** the third completed argument, and two of the order's own guards are why.*** **First**, the closure at $c_2=0$ **does not extend by continuity**: at $c_2=0$ neither solution is L^2 , at $c_2 \neq 0$ exactly one is at each end, so switching on the kinetic vertex **CREATES** a square-integrable solution where there was none. **What remains is a CONNECTION condition** — one analytic equation asking the two ends' one-dimensional subspaces to coincide — **and not a boundary condition**. The scaling $p \rightarrow \lambda q$ leaves one dimensionless parameter $v = c_1 c_3^{-1/3} c_2^{-4/3}$, constant in a only if the cu-

bic's trace power is exactly π^2 and the corpus's own π^2 in kind" puts it at π^2 ; the π^2 endpoint says the condition is not identically satisfied. *A shape, not a verdict, and not claimed either way* *Second*, **the content may be larger than three structures.** The kinetic expansion $\pi^2/(1+\lambda\varphi)$ always carries two powers of momentum; the **potential** sector — the $\frac{1}{2}\mu^2\varphi^2$ term (1) came from, $\sqrt{h}\varphi^3$ expanded in the transverse-traceless perturbation — gives φ^2 at quadratic order and φ^3 at cubic, with no momenta at all. 'sec:lock' named only the kinetic vertex because that is the one singular at $a=0$; the potential cubic is regular there and so was never in view, **but the trace does not care which is singular**. $\Rightarrow$ **Order of the null equation reported and stopped, per the guard: 2 at the content established here; 3 if $\sqrt{h}\varphi^3$'s cubic enters** **AND ONE LANDED CLAIM DOES NOT SURVIVE AS STATED**: "the spectrum is purely continuous, the half-line above Λ with Λ itself not attained" rested on the same absence, and $X^2-\varphi^2$ has spectrum all of $\mathbb{R}$, so the restored content puts spectrum **below** Λ too. *Purely continuous survives; two-sided rather than a half-line.* Claimed at quadratic content only, flagged, routed, not extrapolated **The ordering did not surface**: the quadratic operator has no ordering ambiguity at all **One float in the receipt** — the envelope exponent, against an exact predicted value; an earlier draft measured the two φ^2 exponents by integration and the exact limit replaced it, the recessive branch being ill-conditioned outward with a phase integral growing as φ^3 No ordering choice, no interacting theory beyond the trace's content at cubic order, **no corpus edit** — three 'P10' sites and their replacement sentences are routed in 'FOR.66_FROM.60.md'; the TT weight stays where 'r6946' left it; nothing touches 'prop:flat', 'PO-31' or 'PO-15' *Run: python3 receipts/P10_canonical_time/P10_the_field_squared_is_already_in_the_trace_at_quadratic_order_and.th*

P10R33 *P10_the_operator_is_an_inverted_quartic_so_the_deficiency_is_two_two_and_the_measure_closes_it.*
[OK]

'PO-23'; the row's last two data — the potential sector's cubic, and the connection coefficient 'r6962' reduced the criterion to (node 66's 'r6965' order, taking the two data in the order that fixes tractability. (1)^a ***"Expand $\sqrt{h}\varphi^3$ to cubic order in the perturbation and report whether a φ^3 structure survives into Θ , with the coefficient if it does"*** — "the trace formula is in hand and the quadratic case is worked, so this is the same computation one order up, and the answer is a yes or a no about one structure, not a range"; (1)^b ***"and say what the order of the null equation then is"***. (2)^a ***"Compute the connection coefficient as a function of v and report where it vanishes"*** — with the offer that if it has no zero, "the criterion closes at the three-structure content and the wall is gone as an object at that content ... report it in those words only if it holds at the content (1) establishes"; (2)^b ***"and if it does have a zero, say what it means that v is not constant in the scale factor"*** — "an eigenvector needs the condition to hold on a set of positive measure in a , which is 'r6950's own criterion returning one level up". Guards: **if (1) says the potential cubic enters, take (2) at the three-structure content only and say so**, "do not extrapolate across the order the way the row just extrapolated across the diagonal"; exact arithmetic, or a float against an exact prediction; the ordering stays named and unpicked; and the one that fires — ***"the sixth face applies to your own new results ... for each, say what its scope is in the same sentence that states it, since the defect this revision found was exactly a scope that lived in one place and a claim that lived in another"***). *Computes: **THE GUARD FIRED ON ITS OWN AUTHOR, AND THE WITHDRAWAL IS THE FIRST THING HERE.** ** (1) 'r6962's DEFICIENCY COUNT IS WRONG AND IS WITHDRAWN. It reported "exactly one L^2 solution at each end, both ends limit-point, deficiency indices $(0,0)$, essentially self-adjoint, nothing to select" — **all four clauses**. The test was an exact limit on an ansatz's *relative residual*, and for the dominant branch that residual vanishes **at order $1/p$ **: e^{ic_{1p}/c_2} and $p^{-1}e^{\mathrm{ic}_{1p}/c_2}$ **both pass** ($p \cdot \text{residual}$ to β and φ^2 respectively), so ***the check's discriminating order equalled the effect's order*** and the modulus claim — hence the L^2 count, hence the deficiency, hence "nothing to select" — lived outside what it could see ** (2) AND THE REPAIR IS A CLOSED FORM THAT SETTLES EVERYTHING AT ONCE. The **unimodular** gauge $\psi = \exp(\mathrm{ic}_{2p}^3/6c_3)\chi$, modulus exactly 1 hence unitary on $L^2(\mathbb{R}, \mathrm{dp})$, removes the first-order term with residual **exactly zero**: $M = c_1\pi^2 + c_2\varphi^2 + c_3\mathrm{sym}(\pi^2\varphi) + c_3\hat{\varphi}$*

$c_3 \partial_p^2 + c_1 p^2 - \frac{c_2^2}{4c_3} p^4$, and the quartic coefficient is negative for $c_3 \neq 0$ — whatever the sign of the cubic vertex — the three-structure operator is an inverted quartic. $\Rightarrow$ At every real z both solutions have WKB amplitude $\sqrt[4]{V}$ (measured -0.9985 against the exact -1), so $\sqrt[4]{V}$ is integrable at each end: limit circle at both ends, deficiency $(2,2)$, NOT essentially self-adjoint. And the dichotomy is exact at $c_2=0$: $\int_1^\infty \frac{dp}{\sqrt{p^4}}=1$ converges while $\int_1^\infty \frac{dp}{\sqrt{p^2}}=\infty$ diverges — so ‘r6962’'s quadratic-content closure stands, and the discontinuity it flagged is real and sharper than it said. (3) (1)^a THE TERM IS IN THE TRACE, AND THEN THE CONTROL SPLIT THE QUESTION. Exactly: $\text{tr}(\epsilon^3)=3\det \epsilon$ for every traceless symmetric 3×3 matrix; $\sqrt{\det(\gamma+\epsilon)}=1-\frac{14}{15}\text{tr}(\epsilon^2)+\frac{16}{15}\text{tr}(\epsilon^3)+O(\epsilon^4)$ for $R^{(1)}=0$ for transverse-traceless on a maximally symmetric section, so the cross term dies and the cubic order of $\sqrt{h}R^3$ is $a\sqrt{\gamma}R^3+\text{tr}(\epsilon^3)$ — the second piece being the volume element’s own, against the UNPERTURBED $6/a^2$, needing no curvature perturbation. And the trace formula annihilates exactly one power, $h \propto 1/a$ by ‘dsolve’, while the potential sector’s $h \propto a^{+1}$ gives $h+ah'=2h$, so the term reaches $\hat{\theta}$ at a^{-2} , one power of ℓ_P . (4) BUT THE SINGLE-MODE DIAGONAL SITS EXACTLY ON THE DEGENERATE SET, AND THAT IS COMPUTED RATHER THAN FEARED. Asked for $R^{(3)}$ on a flat section with an explicit transverse-traceless plane wave, the answer is exactly zero — and the reason is algebraic: for a single wave vector transversality is $k^j \epsilon_{ij}=0$, making k a null eigenvector, so the whole polarisation space $[[A,B,0],[B,-A,0],[0,0,0]]$ has $\det=0$ and $\text{tr}(\epsilon^3)=0$ identically. Two wave vectors lift it off: $\det=f_1 f_2 (f_1-f_2)$. $\Rightarrow$ So the cubic potential is intrinsically a MULTI-MODE term at flat order, and what would put a $\hat{\varphi}^3$ on ONE excitation factor is the one integral $\int \sqrt{\gamma} \det e_{(n)}$ on S^3 , where transversality is differential and does not force a pointwise null eigenvector. (5) (1)^b SO THE ORDER IS TWO AT THE ESTABLISHED CONTENT, three only if that integral is non-zero. (6) (2)^a THE CONNECTION COEFFICIENT DOES NOT EXIST AS PUT, AND 0 IS AN EIGENVALUE OF SOME REALISATION. The question asked two one-dimensional subspaces to coincide; the deficiency space at $z=0$ is two-dimensional at each end and globally, the momentum-side operator being regular everywhere. The gauge-transformed equation has real coefficients, so it carries a real L^2 null solution (tail exponent -0.9985 against the exact -1), and $\hat{M}_{\min} + \text{span}\{\chi_0\}$ is symmetric and extends. $\Rightarrow$ ‘r6954’'s uniform argument has no purchase here: there are two solutions to select among, not none. (7) AND YET (2)^b CLOSES IT, FOR EVERY REALISATION HELD FIXED ACROSS FIBRES. The rescaling $\lambda^6=4c_3^2/c_2^2$ gives the canonical $\hat{H}(w)=-\partial_q^2+wq^2-q^4$ with $w=2^{4/3}c_1c_3^{1/3}\sqrt[4]{c_2}$ and $w \propto a^{4m/3-20/3}$, constant only at $m=5$ where the corpus’s own ‘in kind’ puts $m=6$, so $w \propto a^{4/3}$. Limit circle at both ends makes every realisation’s resolvent compact, hence its spectrum purely discrete; and Hellmann–Feynman gives $d\lambda_n/dw=\langle \psi_n | \partial_q^2 | \psi_n \rangle$ STRICTLY for every realisation — measured against that exact value at two different wall positions, worst relative departure 8.2×10^{-9} and 1.4×10^{-8} , every derivative positive, and the first six eigenvalues all rising from $w=-4$ to $w=+4$. $\Rightarrow$ Each branch crosses zero at most once, so $\{w:0 \in \text{spec}\}$ is discrete, so $\{a\}$ is discrete, so the condition cannot hold on a set of positive measure: NO EIGENVECTOR at the three-structure content — which is ‘r6950’'s criterion returning one level up, exactly as the order predicted *Bound*: NOT the row’s “the wall is gone as an object”, for the third time — and this time not because the argument failed to reach but because a claim of mine had to be withdrawn and what replaces it is an extension family rather than a closure. THE ONE LOOPHOLE, NAMED AND NOT CLOSED, AND IT LANDS ON AN ITEM ALREADY OPEN: the measure argument fixes ONE realisation across fibres, and a realisation chosen fibre by fibre could track the zero and would supply an eigenvector. But the extension sits at $p \rightarrow \infty$, the tower’s ultraviolet, and ‘S’lock’ already carries “the ultraviolet definition of the tower sums” as its open frontier — so the wall lands on the item the paper already names rather than a new one, and is a different kind of object from the $a=0$ extension, which is at a

physical boundary and is already closed by the horizon's thermal state **NOT the VALUE** of the S^3 diagonal overlap — only that its flat analogue vanishes for an algebraic reason and that the section's curvature is the only thing that could lift it; 'r6946's selection rule is cited and is **not** used to infer a value **(2)** is at the **THREE**-structure content and is not extrapolated, per the guard: a third-order operator is not a Schrödinger operator and the gauge argument does not reach it **WHAT OF 'r6962' IS NOT WITHDRAWN**, and why, one by one: that $\hat{\varphi}^2$ enters $\hat{\Theta}$ at quadratic order (three independent exact derivations, none asymptotic); the sixth face on 'r6930' (an exactly-zero diagonal against a non-zero matrix element); the quadratic-content inverted **oscillator** and its empty point spectrum (the $c_2=0$ case, where the limit-circle integral **diverges**, so there neither solution is L^2); and the degenerate point's representational status — **strengthened**, the gauge showing the momentum-side operator is unitarily a plain Schrödinger operator with **no singular point at all** **Two floats**, each against an exact predicted value: the envelope exponent against -1 , and $d\lambda_n/dw$ against the Hellmann–Feynman expectation computed in the same arithmetic No ordering choice — the potential cubic is a product of configuration variables and carries none; no realisation chosen; **no corpus edit**, the withdrawn sentences and their replacements routed in 'FOR.66_FROM.60.md'; the TT weight stays where 'r6946' left it; nothing touches 'prop:flat', 'PO-31' or 'PO-15' *Run: python3 receipts/P10_canonical_time/P10_the_operator_is_an_inverted_quartic_so_the_deficiency_is_two_two_and*

Appendix L The Knowledge Ledgers

Each claim marked ^L in the text rests on a field bake or the figure–theorem bake recorded in the named ledger, which ships with this work and carries the probes, the receipts, and the three registers kept apart — what bit, what bounced, and where the boundary is. A ledger is working apparatus, not a publication, and is cited here rather than in the bibliography. This appendix is generated from the ledger index, not maintained by hand.

- L10** `FUNCTIONAL_ANALYSIS_LEDGER.md` [field bake]
The functional-analysis / unitarity field-bake ledger — the field that bounced, and the one routing fact it returned. Third of the four fields ‘L-272’'s re-survey left outstanding. ‘OWED’ 622
- L17** `QUADRIC_GEOMETRY_LEDGER.md` [field bake]
projective geometry of quadrics against the CR substrate — the CK metric identification, and the ladder gap it exposes
- L20** `VARIATIONAL_LEDGER.md` [field bake]
variational / action against CR — the field the corpus uses and never names
- L14** `INTEGRABLE_SYSTEMS_LEDGER.md` [field bake]
The integrable-systems field-bake ledger — what bit, what bounced, and where the boundary is